

arXiv:2210.03629v3 [cs.CL] 10 Mar 2023

REACT: SYNERGIZING REASONING AND ACTING IN LANGUAGE MODELS

Shunyu Yao^{*1}, Jeffrey Zhao², Dian Yu², Nan Du², Izhak Shafran², Karthik Narasimhan¹, Yuan Cao²

¹Department of Computer Science, Princeton University

²Google Research, Brain team

¹{shunyuy, karthikn}@princeton.edu

²{jeffreyzhao, dianyu, dunan, izhak, yuanco}@google.com

ABSTRACT

While large language models (LLMs) have demonstrated impressive performance across tasks in language understanding and interactive decision making, their abilities for reasoning (e.g. chain-of-thought prompting) and acting (e.g. action plan generation) have primarily been studied as separate topics. In this paper, we explore the use of LLMs to generate both reasoning traces and task-specific actions in an interleaved manner, allowing for greater synergy between the two: reasoning traces help the model induce, track, and update action plans as well as handle exceptions, while actions allow it to interface with and gather additional information from external sources such as knowledge bases or environments. We apply our approach, named ReAct, to a diverse set of language and decision making tasks and demonstrate its effectiveness over state-of-the-art baselines in addition to improved human interpretability and trustworthiness. Concretely, on question answering (HotpotQA) and fact verification (Fever), ReAct overcomes prevalent issues of hallucination and error propagation in chain-of-thought reasoning by interacting with a simple Wikipedia API, and generating human-like task-solving trajectories that are more interpretable than baselines without reasoning traces. Furthermore, on two interactive decision making benchmarks (ALFWorld and WebShop), ReAct outperforms imitation and reinforcement learning methods by an absolute success rate of 34% and 10% respectively, while being prompted with only one or two in-context examples.

1 INTRODUCTION

A unique feature of human intelligence is the ability to seamlessly combine task-oriented actions with verbal reasoning (or inner speech, Alderson-Day & Fernyhough, 2015), which has been theorized to play an important role in human cognition for enabling self-regulation or strategization (Vygotsky, 1987; Luria, 1965; Fernyhough, 2010) and maintaining a working memory (Baddeley, 1992). Consider the example of cooking up a dish in the kitchen. Between any two specific actions, we may reason in language in order to track progress (“now that everything is cut, I should heat up the pot of water”), to handle exceptions or adjust the plan according to the situation (“I don’t have salt, so let me use soy sauce and pepper instead”), and to realize when external information is needed (“how do I prepare dough? Let me search on the Internet”). We may also act (open a cookbook to read the recipe, open the fridge, check ingredients) to support the reasoning and to answer questions (“What dish can I make right now?”). This tight synergy between “acting” and “reasoning” allows humans to learn new tasks quickly and perform robust decision making or reasoning, even under previously unseen circumstances or facing information uncertainties.

Recent results have hinted at the possibility of combining verbal reasoning with interactive decision making in autonomous systems. On one hand, properly prompted large language models (LLMs) have demonstrated emergent capabilities to carry out several steps of reasoning traces to derive

^{*}Work during Google internship. Projet page with code: <https://react-lm.github.io/>.

REACT: 促使推理与行动协同的语言模型

Shunyu Yao^{*1}, Jeffrey Zhao², Dian Yu², Nan Du², Izhak Shafran², Karthik Narasimhan¹, Yuan Cao²

普林斯顿大学计算机科学系 Google Research, Brain团队

{shunyuy, karthikn}@princeton.edu

{jeffreyzhao, dianyu, dunan, izhak, yuanco}@google.com

摘要

虽然大型语言模型 (LLMs) 在语言理解和交互式决策任务中展现出令人印象深刻的性能, 但其推理 (例如链式思维提示) 和行动 (例如行动计划生成) 能力主要作为独立主题进行研究。在本文中, 我们探索了使用 LLMs以交错方式生成推理轨迹和任务特定行动的方法, 从而实现两者之间的更大协同: 推理轨迹帮助模型推导、跟踪和更新行动计划以及处理异常, 而行动则使其能够与外部资源 (如知识库或环境) 进行接口并收集额外信息。我们将该方法命名为ReAct, 并将其应用于多样的语言和决策任务, 证明其在优于最先进基线的时候, 还提升了人类的可解释性和可信用度。具体而言, 在问答任务 (HotpotQA) 和事实验证 (Fever) 中, ReAct通过与简单的维基百科API交互, 克服了链式思维推理中常见的幻觉和错误传播问题, 并生成比无推理轨迹基线更具可解释性的类人任务解决轨迹。此外, 在两个交互式决策基准 (ALFWorld和WebShop) 中, ReAct以仅一到两个上下文示例的提示, 分别在绝对成功率上超越模仿学习和强化学习方法, 成功率分别提高34%和10%。

1 引言

人类智能的一个独特特征是能够无缝结合任务导向的行动与言语推理 (或内心独白, Alderson-Day & Fernyhough, 2015), 这一能力被认为在人体认知中发挥着重要作用, 促使自我调节或策略制定 (Vygotsky, 1987; Luria, 1965; Fernyhough, 2010) 以及维持工作记忆 (Baddeley, 1992)。以厨房烹饪一道菜为例, 在任何两个具体行动之间, 我们可能会用语言进行推理, 以跟踪进展 (“既然都切好材料了, 我应该加热水锅”), 处理例外情况或根据情境调整计划 (“我没有盐, 那就用酱油和胡椒代替”), 以及意识到何时需要外部信息 (“我怎么准备面团? 让我在网上搜索”)。我们还可能采取行动 (翻开食谱书查阅食谱, 打开冰箱检查食材) 以支持推理并回答问题 (“我现在能做什么菜?”)。这种 “行动” 与 “推理” 之间的紧密协同, 使人类能够快速学习新任务, 即使在之前未曾遇到的情境或面对信息不确定性时, 也能进行稳健的决策或推理。

最近的研究结果暗示了在自主系统中结合语言推理与交互式决策的可能性。一方面, 经过适当提示的大型语言模型 (LLMs) 已展现出在推理过程中进行多步骤推导的涌现能力, 以得出

^{*}在 Google 实习期间的工作。项目页面及代码: <https://react-lm.github.io/>。

arXiv:2210.03629v3 [cs.CL] 10 Mar 2023

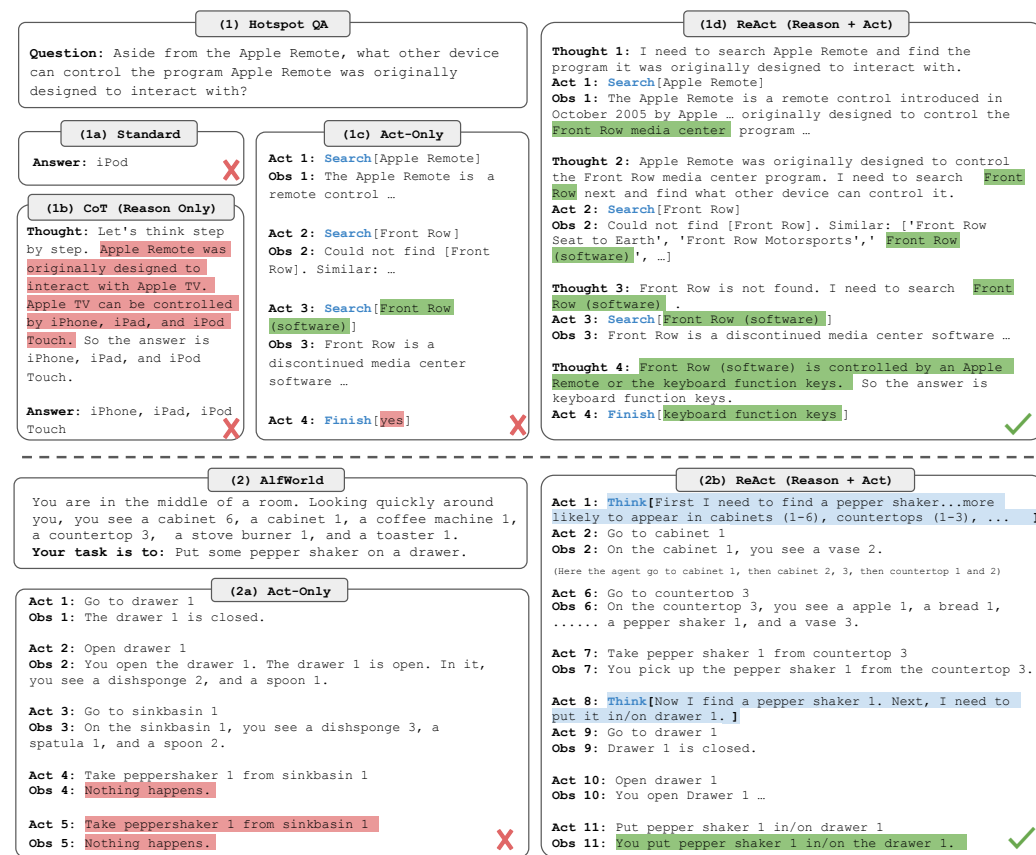


Figure 1: (1) Comparison of 4 prompting methods, (a) Standard, (b) Chain-of-thought (CoT, Reason Only), (c) Act-only, and (d) ReAct (Reason+Act), solving a HotspotQA (Yang et al., 2018) question. (2) Comparison of (a) Act-only and (b) ReAct prompting to solve an AlfWorld (Shridhar et al., 2020b) game. In both domains, we omit in-context examples in the prompt, and only show task solving trajectories generated by the model (Act, Thought) and the environment (Obs).

answers from questions in arithmetic, commonsense, and symbolic reasoning tasks (Wei et al., 2022). However, this “chain-of-thought” reasoning is a static black box, in that the model uses its own internal representations to generate thoughts and is not grounded in the external world, which limits its ability to reason reactively or update its knowledge. This can lead to issues like fact hallucination and error propagation over the reasoning process (Figure 1 (1b)). On the other hand, recent work has explored the use of pre-trained language models for planning and acting in interactive environments (Ahn et al., 2022; Nakano et al., 2021; Yao et al., 2020; Huang et al., 2022a), with a focus on predicting actions via language priors. These approaches usually convert multi-modal observations into text, use a language model to generate domain-specific actions or plans, and then use a controller to choose or execute them. However, they do not employ language models to reason abstractly about high-level goals or maintain a working memory to support acting, barring Huang et al. (2022b) who perform a limited form of verbal reasoning to reiterate spatial facts about the current state. Beyond such simple embodied tasks to interact with a few blocks, there have not been studies on how reasoning and acting can be combined in a synergistic manner for general task solving, and if such a combination can bring systematic benefits compared to reasoning or acting alone.

In this work, we present ReAct, a general paradigm to combine reasoning and acting with language models for solving diverse language reasoning and decision making tasks (Figure 1). ReAct prompts LLMs to generate both verbal reasoning traces and actions pertaining to a task in an interleaved manner, which allows the model to perform dynamic reasoning to create, maintain, and adjust high-level plans for acting (reason to act), while also interact with the external environments (e.g. Wikipedia) to incorporate additional information into reasoning (act to reason).

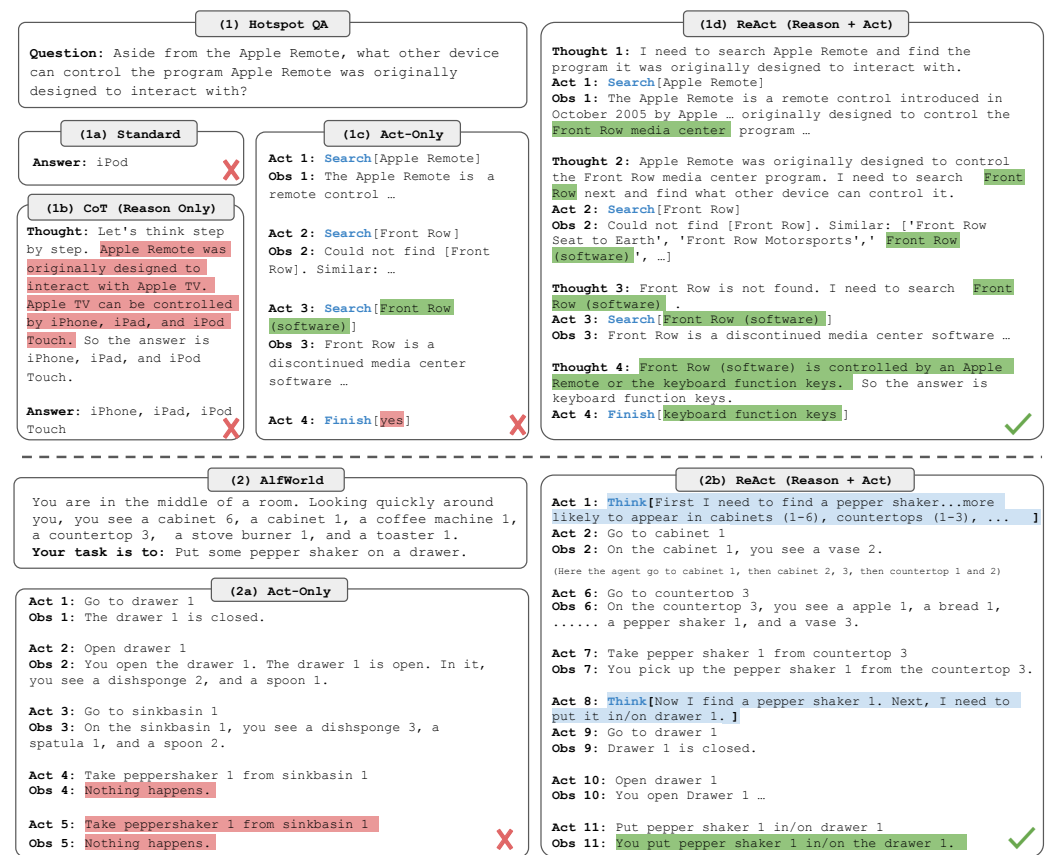


图1：（1）四种提示方法的比较，(a) 标准方法，(b) 连锁思维（CoT，纯推理），(c) 仅行动（Act-only），(d) ReAct (Reason+Act)，用于解决HotspotQA（Yang等，2018）问题。（2）(a) 仅行动提示与(b) ReAct提示在解决AlfWorld（Shridhar等，2020b）游戏中的比较。在两个领域中，我们省略了提示中的上下文示例，仅显示模型生成的任务解决轨迹（行动、思考）和环境（观察）。

在算术、常识和符号推理任务（Wei 等，2022）中，模型的回答通常基于“链式思维”推理。然而，这种“链式思维”推理是一种静态的黑箱，因为模型使用其内部表示生成思考，并未与外部世界建立联系，这限制了其进行反应性推理或更新知识的能力。这可能导致事实幻觉和推理过程中错误的传播（图 1 (1b)）。另一方面，近期的研究探索了预训练语言模型在交互环境中的规划与行动（Ahn 等，2022; Nakano 等，2021; Yao 等，2020; Huang 等，2022a）中的应用，重点在于通过语言先验预测行动。这些方法通常将多模态观察转化为文本，利用语言模型生成特定领域的行动或计划，然后由控制器选择或执行这些行动。然而，它们并未利用语言模型进行抽象的高层次目标推理或维护工作记忆以支持行动，Huang 等（2022b）除外，该研究采用有限形式的口头推理，重申关于当前状态的空间事实。除了与少量积木交互的简单体现任务外，目前尚无关于如何将推理与行动以协同方式结合以实现通用任务解决的研究，也尚不清楚这种结合是否能相较于单纯的推理或行动带来系统性的益处。

在本工作中，我们提出了 ReAct，一种结合推理与行动的通用范式，用于解决多样的语言推理与决策任务（图1）。ReAct 促使大型语言模型（LLMs）以交错的方式生成与任务相关的口头推理轨迹和行动，从而使模型能够进行动态推理，创建、维护并调整高层次的行动计划（推理以行动），同时与外部环境（如维基百科）交互，将额外信息融入推理过程（行动以推理）。

We conduct empirical evaluations of ReAct and state-of-the-art baselines on four diverse benchmarks: question answering (HotPotQA, Yang et al., 2018), fact verification (Fever, Thorne et al., 2018), text-based game (ALFWorld, Shridhar et al., 2020b), and webpage navigation (WebShop, Yao et al., 2022). For HotPotQA and Fever, with access to a Wikipedia API that the model can interact with, ReAct outperforms vanilla action generation models while being competitive with chain-of-thought reasoning (CoT) (Wei et al., 2022). The best approach overall is a combination of ReAct and CoT that allows for the use of both internal knowledge and externally obtained information during reasoning. On ALFWorld and WebShop, two or even one-shot ReAct prompting is able to outperform imitation or reinforcement learning methods trained with $10^3 \sim 10^5$ task instances, with an absolute improvement of 34% and 10% in success rates respectively. We also demonstrate the importance of sparse, versatile reasoning in decision making by showing consistent advantages over controlled baselines with actions only. Besides general applicability and performance boost, the combination of reasoning and acting also contributes to model interpretability, trustworthiness, and diagnosability across all domains, as humans can readily distinguish information from model’s internal knowledge versus external environments, as well as inspect reasoning traces to understand the decision basis of model actions.

To summarize, our key contributions are the following: (1) we introduce ReAct, a novel prompt-based paradigm to synergize reasoning and acting in language models for general task solving; (2) we perform extensive experiments across diverse benchmarks to showcase the advantage of ReAct in a few-shot learning setup over prior approaches that perform either reasoning or action generation in isolation; (3) we present systematic ablations and analysis to understand the importance of acting in reasoning tasks, and reasoning in interactive tasks; (4) we analyze the limitations of ReAct under the prompting setup (i.e. limited support of reasoning and acting behaviors), and perform initial finetuning experiments showing the potential of ReAct to improve with additional training data. Scaling up ReAct to train and operate on more tasks and combining it with complementary paradigms like reinforcement learning could further unlock the potential of large language models.

2 REACT: SYNERGIZING REASONING + ACTING

Consider a general setup of an agent interacting with an environment for task solving. At time step t , an agent receives an observation $o_t \in \mathcal{O}$ from the environment and takes an action $a_t \in \mathcal{A}$ following some policy $\pi(a_t|c_t)$, where $c_t = (o_1, a_1, \dots, o_{t-1}, a_{t-1}, o_t)$ is the *context* to the agent. Learning a policy is challenging when the mapping $c_t \mapsto a_t$ is highly implicit and requires extensive computation. For example, the agent shown in Figure 1(1c) is unable to generate the correct final action (Act 4) to finish the QA task as it requires complex reasoning over the trajectory context (Question, Act 1-3, Obs 1-3). Similarly, the agent shown in Figure 1(2a) fails to comprehend from the context that sinkbasin 1 does not contain peppershaker 1, thus keep producing hallucinating actions.

The idea of ReAct is simple: we augment the agent’s action space to $\hat{\mathcal{A}} = \mathcal{A} \cup \mathcal{L}$, where $\mathcal{L}$ is the space of language. An action $\hat{a}_t \in \mathcal{L}$ in the language space, which we will refer to as a *thought* or a *reasoning trace*, does not affect the external environment, thus leading to no observation feedback. Instead, a thought $\hat{a}_t$ aims to compose useful information by reasoning over the current context c_t , and update the context $c_{t+1} = (c_t, \hat{a}_t)$ to support future reasoning or acting. As shown in Figure 1, there could be various types of useful thoughts, e.g. decomposing task goals and create action plans (2b, Act 1; 1d, Thought 1), injecting commonsense knowledge relevant to task solving (2b, Act 1), extracting important parts from observations (1d, Thought2, 4), track progress and transit action plans (2b, Act 8), handle exceptions and adjust action plans (1d, Thought 3), and so on.

However, as the language space $\mathcal{L}$ is unlimited, learning in this augmented action space is difficult and requires strong language priors. In this paper, we mainly focus on the setup where a frozen large language model, PaLM-540B (Chowdhery et al., 2022)¹, is prompted with few-shot in-context examples to generate both domain-specific actions and free-form language thoughts for task solving (Figure 1 (1d), (2b)). Each in-context example is a human trajectory of actions, thoughts, and environment observations to solve a task instance (see Appendix C). For the tasks where reasoning is of primary importance (Figure 1(1)), we alternate the generation of thoughts and actions so that the task-solving trajectory consists of multiple thought-action-observation steps. In contrast, for decision making tasks that potentially involve a large number of actions (Figure 1(2)), thoughts only need to

¹We show some GPT-3 (Brown et al., 2020) results in Appendix A.1, which outperforms PaLM-540B.

我们在四个多样化的基准测试上对ReAct及其最先进的基线方法进行了实证评估：问答任务（HotPotQA, Yang等, 2018）、事实验证（Fever, Thorne等, 2018）、基于文本的游戏（ALFWorld, Shridhar等, 2020b）以及网页导航（WebShop, Yao等, 2022）。在HotPotQA和Fever任务中，借助模型可以交互的维基百科API，ReAct优于纯粹的行动生成模型，同时在链式思维（CoT, Wei等, 2022）方面具有竞争力。总体而言，最优方案是结合ReAct与CoT的方法，允许在推理过程中同时利用内部知识和外部获取的信息。在ALFWorld和WebShop任务中，单次或少量ReAct提示即可超越通过 $10^3 \sim 10^5$ 任务实例训练的模仿学习或强化学习方法，成功率分别实现34%和10%的绝对提升。我们还通过展示仅使用行动的受控基线，强调稀疏且多样化推理在决策中的重要性，显示出一致的优势。除了具有广泛适用性和性能提升外，推理与行动的结合还增强了模型的可解释性、可信度和可诊断性，因人类可以轻松区分模型内部知识与外部环境中的信息，并检查推理轨迹以理解模型行动的决策依据。

总之，我们的主要贡献包括：(1) 我们提出了 ReAct，一种新颖的基于提示的范式，用于在语言模型中协同推理与行动，以实现通用任务的解决；(2) 我们在多样化的基准测试中进行了广泛的实验，展示了 ReAct 在少样本学习设置下相较于仅进行推理或行动生成的先前方法的优势；(3) 我们进行了系统的消融和分析，以理解在推理任务中行动的重要性，以及在交互任务中推理的作用；(4) 我们分析了 ReAct 在提示设置下的局限性（即推理和行动行为支持有限），并进行了初步的微调实验，展示了通过增加训练数据，ReAct 有潜力得到改进。将 ReAct 扩展到更多任务的训练与操作，并结合强化学习等互补范式，可能进一步释放大语言模型的潜能。

2 REACT: SYNERGIZING REASONING + ACTING

考虑一个代理与环境交互以完成任务的通用设置。在时间步 t ，代理从环境接收观察 $o_t \in \mathcal{O}$ ，并根据某个策略 $\pi(a_t|c_t)$ 采取行动 $a_t \in \mathcal{A}$ ，其中 $c_t = (o_1, a_1, \dots, o_{t-1}, a_{t-1}, o_t)$ 是代理的上下文。当映射 $c_t \mapsto a_t$ 高度隐含且需要大量计算时，学习策略具有挑战性。例如，图1(1c)所示的代理无法生成正确的最终行动（Act 4）以完成问答任务，因为这需要对轨迹上下文（Question、Act 1-3、Obs 1-3）进行复杂推理。同样，图1(2a)所示的代理未能从上下文中理解到 sinkbasin 1 不包含 peppershaker 1，因此持续产生虚幻的行动。

ReAct 的想法很简单：我们将代理的行动空间扩展到 $\hat{\mathcal{A}} = \mathcal{A} \cup \mathcal{L}$ ，其中 $\mathcal{L}$ 是语言的空间。在语言空间中的一个行动 $\hat{a}_t \in \mathcal{L}$ ，我们将其称为 思考 或 推理轨迹，不会影响外部环境，因此不会产生观察反馈。相反，一个思考 $\hat{a}_t$ 旨在通过对当前上下文 c_t 进行推理，组合有用的信息，并更新上下文 $c_{t+1} = (c_t, \hat{a}_t)$ 以支持未来的推理或行动。如图 1 所示，可能存在各种类型的有用思考，例如：分解任务目标并制定行动计划（2b, Act 1; 1d, Thought 1）、注入与任务解决相关的常识知识（2b, Act 1）、提取观察中的重要部分（1d, Thought 2, 4）、跟踪进展并转变行动计划（2b, Act 8）、处理异常并调整行动计划（1d, Thought 3）等等。

然而，由于语言空间 $\mathcal{L}$ 是无限的，在这个增强的行动空间中学习具有一定的困难，并且需要强大的语言先验知识。在本文中，我们主要关注一种设置，即使用冻结的大型语言模型 PaLM-540B (Chowdhery等, 2022)¹，通过少样本上下文示例进行提示，以生成领域特定的行动和用于任务解决的自由形式语言思考（图1 (1d)，(2b)）。每个上下文示例都是一个人类轨迹，包括行动、思考和环境观察，用于解决一个任务实例（见附录C）。对于以推理为主要目标的任务（图1 (1)），我们交替生成思考和行动，使得任务解决轨迹由多个思考-行动-观察步骤组成。相反，对于可能涉及大量行动的决策任务（图1 (2)），思考仅需

¹我们在附录A.1中展示了一些GPT-3 (Brown等, 2020) 的结果，优于PaLM-540B。

appear sparsely in the most relevant positions of a trajectory, so we let the language model decide the asynchronous occurrence of thoughts and actions for itself.

Since decision making and reasoning capabilities are integrated into a large language model, ReAct enjoys several unique features: **A) Intuitive and easy to design:** Designing ReAct prompts is straightforward as human annotators just type down their thoughts in language on top of their actions taken. No ad-hoc format choice, thought design, or example selection is used in this paper. We detail prompt design for each task in Sections 3 and 4. **B) General and flexible:** Due to the flexible thought space and thought-action occurrence format, ReAct works for diverse tasks with distinct action spaces and reasoning needs, including but not limited to QA, fact verification, text game, and web navigation. **C) Performant and robust:** ReAct shows strong generalization to new task instances while learning solely from one to six in-context examples, consistently outperforming baselines with only reasoning or acting across different domains. We also show in Section 3 additional benefits when finetuning is enabled, and in Section 4 how ReAct performance is robust to prompt selections. **D) Human aligned and controllable:** ReAct promises an interpretable sequential decision making and reasoning process where humans can easily inspect reasoning and factual correctness. Moreover, humans can also control or correct the agent behavior on the go by thought editing, as shown in Figure 5 in Section 4.

3 KNOWLEDGE-INTENSIVE REASONING TASKS

We begin with knowledge-intensive reasoning tasks like multi-hop question answering and fact verification. As shown in Figure 1(1d), by interacting with a Wikipedia API, ReAct is able to retrieve information to support reasoning, while also use reasoning to target what to retrieve next, demonstrating a synergy of reasoning and acting.

3.1 SETUP

Domains We consider two datasets challenging knowledge retrieval and reasoning: (1) Hot-PotQA (Yang et al., 2018), a multi-hop question answering benchmark that requires reasoning over two or more Wikipedia passages, and (2) FEVER (Thorne et al., 2018), a fact verification benchmark where each claim is annotated SUPPORTS, REFUTES, or NOT ENOUGH INFO, based on if there exists a Wikipedia passage to verify the claim. In this work, we operate in a question-only setup for both tasks, where models only receive the question/claim as input without access to support paragraphs, and have to rely on their internal knowledge or retrieve knowledge via interacting with an external environment to support reasoning.

Action Space We design a simple Wikipedia web API with three types of actions to support interactive information retrieval: (1) **search**[entity], which returns the first 5 sentences from the corresponding entity wiki page if it exists, or else suggests top-5 similar entities from the Wikipedia search engine, (2) **lookup**[string], which would return the next sentence in the page containing string, simulating Ctrl+F functionality on the browser. (3) **finish**[answer], which would finish the current task with answer. We note that this action space mostly can only retrieve a small part of a passage based on exact passage name, which is significantly weaker than state-of-the-art lexical or neural retrievers. The purpose is to simulate how humans would interact with Wikipedia, and force models to retrieve via explicit reasoning in language.

3.2 METHODS

ReAct Prompting For HotpotQA and Fever, we randomly select 6 and 3 cases² from the training set and manually compose ReAct-format trajectories to use as few-shot exemplars in the prompts. Similar to Figure 1(d), each trajectory consists of multiple thought-action-observation steps (i.e. dense thought), where free-form thoughts are used for various purposes. Specifically, we use a combination of thoughts that decompose questions (“I need to search x, find y, then find z”), extract information from Wikipedia observations (“x was started in 1844”, “The paragraph does not tell x”), perform commonsense (“x is not y, so z must instead be...”) or arithmetic reasoning (“1844 < 1989”), guide

²We find more examples do not improve performance.

在轨迹的最相关位置稀疏出现，因此我们让语言模型自行决定思想与行动的异步发生。

由于决策能力和推理能力被整合到大型语言模型中，ReAct 具有若干独特特性：**A) 直观且易于设计：**设计ReAct提示非常简便，因为人工标注者只需在行动之上用语言表达他们的思考。本文未采用临时格式选择、思考设计或示例选择。我们在第3和第4节中详细介绍了每个任务的提示设计。**B) 通用且灵活：**由于思考空间和思考-行动发生格式的灵活性，ReAct适用于具有不同行动空间和推理需求的多样任务，包括但不限于问答、事实验证、文本游戏和网页导航。**C) 高效且稳健：**ReAct在仅通过一到六个上下文示例学习的情况下，展现出对新任务实例的强泛化能力，持续优于仅依赖推理或行动的基线模型，跨越不同领域。我们在第3节中还展示了启用微调时的额外优势，以及在第4节中ReAct性能对提示选择的鲁棒性。**D) 人类对齐且可控：**ReAct承诺实现可解释的序贯决策和推理过程，便于人类轻松检查推理和事实正确性。此外，人类还可以通过思考编辑，实时控制或纠正代理行为，如第4节第5图所示。

3 知识密集型推理任务

我们首先处理知识密集型推理任务，如多跳问答和事实验证。如图1(1d)所示，通过与维基百科API交互，ReAct能够检索信息以支持推理，同时也利用推理来指导下一步的检索，展示了推理与行动的协同作用。

3.1 设置

领域我们考虑两个具有挑战性的知识检索与推理数据集：(1) Hot- PotQA (Yang 等, 2018)，一个多跳问答基准，要求对两个或多个维基百科段落进行推理；(2) FEVER (Thorne 等, 2018)，一个事实验证基准，其中每个声明被标注为 SUPPORTS、REFUTES 或 NOT ENOUGH INFO，依据是否存在维基百科段落可以验证该声明。在本研究中，我们在两个任务中采用仅问句的设置，模型仅接收问题/声明作为输入，不访问支持段落，必须依赖其内部知识或通过与环境交互检索知识以支持推理。

行动空间我们设计了一个简易的维基百科网页API，包含三种类型的操作以支持交互式信息检索：(1) **search**[entity]，，如果对应实体的维基百科页面存在，则返回该实体页面的前五句，否则从维基百科搜索引擎中推荐前五个相似实体；(2) **lookup**[string]，，返回包含该字符串的页面中的下一句，模拟浏览器中的Ctrl+F功能；(3) **finish**[answer]，，用答案完成当前任务。我们注意到，这个行动空间主要只能基于精确的篇章名称检索篇章的一小部分内容，远不及最先进的词汇或神经检索器。其目的是模拟人类与维基百科的交互方式，促使模型通过显式推理在语言中进行检索。

3.2 方法

ReAct Prompting 对于 HotpotQA 和 Fever，我们从训练集中随机抽取 6 个和 3 个案例²，手动组合成 ReAct 格式的轨迹，作为少样本示例用于提示。类似于图 1(d)，每条轨迹由多个思考-行动-观察步骤组成（即密集思考），其中自由形式的思考用于多种目的。具体而言，我们使用结合的思考方式，包括分解问题（“我需要搜索 x，找到 y，然后找到 z”）、从维基百科观察中提取信息（“x 于 1844 年开始”，“该段落未提及 x”）、进行常识推理（“x 不是 y，因此 z 必须是...”）或算术推理（“1844 < 1989”），以引导

我们发现更多的示例并不能提升性能。

Prompt Method ^a	HotpotQA (EM)	Fever (Acc)
Standard	28.7	57.1
CoT (Wei et al., 2022)	29.4	56.3
CoT-SC (Wang et al., 2022a)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT-SC $\rightarrow$ ReAct	34.2	64.6
ReAct $\rightarrow$ CoT-SC	35.1	62.0
Supervised SoTA^b	67.5	89.5

Table 1: PaLM-540B prompting results on HotpotQA and Fever.

^aHotpotQA EM is 27.1, 28.9, 33.8 for Standard, CoT, CoT-SC in Wang et al. (2022b).

^b(Zhu et al., 2021; Lewis et al., 2020)

search reformulation (“maybe I can search/look up x instead”), and synthesize the final answer (“...so the answer is x”). See Appendix C for more details.

Baselines We systematically ablate ReAct trajectories to build prompts for multiple baselines (with formats as Figure 1(1a-1c)): (a) **Standard prompting** (Standard), which removes all thoughts, actions, observations in ReAct trajectories. (b) **Chain-of-thought prompting** (CoT) (Wei et al., 2022), which removes actions and observations and serve as a reasoning-only baseline. We also build a self-consistency baseline (CoT-SC) (Wang et al., 2022a;b) by sampling 21 CoT trajectories with decoding temperature 0.7 during inference and adopting the majority answer, which is found to consistently boost performance over CoT. (c) **Acting-only prompt** (Act), which removes thoughts in ReAct trajectories, loosely resembling how WebGPT (Nakano et al., 2021) interacts with the Internet to answer questions, though it operates on a different task and action space, and uses imitation and reinforcement learning instead of prompting.

Combining Internal and External Knowledge As will be detail in Section 3.3, we observe that the problem solving process demonstrated by ReAct is more factual and grounded, whereas CoT is more accurate in formulating reasoning structure but can easily suffer from hallucinated facts or thoughts. We therefore propose to incorporate ReAct and CoT-SC, and let the model decide when to switch to the other method based on the following heuristics: A) **ReAct $\rightarrow$ CoT-SC**: when ReAct fails to return an answer within given steps, back off to CoT-SC. We set 7 and 5 steps for HotpotQA and FEVER respectively as we find more steps will not improve ReAct performance³. B) **CoT-SC $\rightarrow$ ReAct**: when the majority answer among n CoT-SC samples occurs less than $n/2$ times (i.e. internal knowledge might not support the task confidently), back off to ReAct.

Finetuning Due to the challenge of manually annotating reasoning traces and actions at scale, we consider a bootstrapping approach similar to Zelikman et al. (2022), using 3,000 trajectories with correct answers generated by ReAct (also for other baselines) to finetune smaller language models (PaLM-8/62B) to decode trajectories (all thoughts, actions, observations) conditioned on input questions/claims. More details are in Appendix B.1.

3.3 RESULTS AND OBSERVATIONS

ReAct outperforms Act consistently Table 1 shows HotpotQA and Fever results using PaLM-540B as the base model with different prompting methods. We note that ReAct is better than Act on both tasks, demonstrating the value of reasoning to guide acting, especially for synthesizing the final answer, as shown in Figure 1 (1c-d). Fine-tuning results 3 also confirm the benefit of reasoning traces for more informed acting.

³Of all trajectories with correct final answers, those with 7 steps on HotpotQA and 5 steps on FEVER only take up 0.84% and 1.33% respectively.

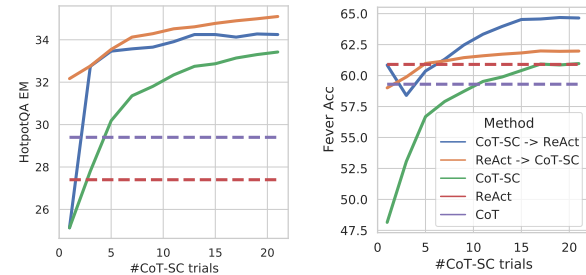


Figure 2: PaLM-540B prompting results with respect to number of CoT-SC samples used.

提示方法 ^a	HotpotQA (EM)	发热 (Acc)
ReAct: 利用搜索与推理子模型 模型中融合 3.2 方法	28.7	57.1
CoT (Wei 等人, 2022年)	29.4	56.3
CoT-SC (Wang 等人, 2022a)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT-SC $\rightarrow$ ReAct	34.2	64.6
ReAct $\rightarrow$ CoT-SC	35.1	62.0
监督学习最新技术^b	67.5	89.5

表 1: PaLM-540B 在 HotpotQA 和 Fever 上的提示结果

HotpotQA EM in Wang 等人 (2022b) 中的标准、CoT 和 CoT-SC 分别为 27.1、28.9 和 33.8。

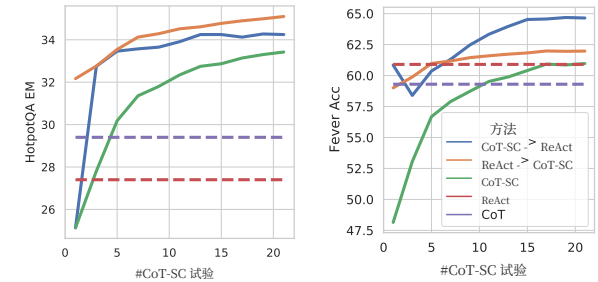


图2: PaLM-540B在不同CoT-SC样本数量下的提示结果。

搜索重述（“也许我可以搜索/查找x”），并合成最终答案（“……所以答案是x”）。详见附录C。

Baselines 我们系统地消融ReAct轨迹，以构建多种基线的提示（格式如图1(1a-1c)）：(a) **标准提示** (Standard)，移除ReAct轨迹中的所有思考、行动和观察。(b) **链式思维提示** (CoT) (Wei等，2022)，移除行动和观察，仅作为纯推理基线。我们还通过在推理过程中以解码温度0.7采样21条CoT轨迹并采用多数投票答案，构建了自一致性基线（CoT-SC）(Wang等，2022a;b)，发现其在性能上持续优于CoT。(c) **仅行动提示** (Act)，移除ReAct轨迹中的思考，类似WebGPT（Nakano等，2021）与互联网交互回答问题的方式，尽管其操作任务和行动空间不同，并采用模仿学习和强化学习而非提示。

结合内部与外部知识如第3.3节所述，我们观察到ReAct所展示的问题解决过程更为事实性和基础性，而CoT在构建推理结构方面更为准确，但容易出现虚构事实或思维。因此，我们提出将ReAct与CoT-SC结合，并让模型根据以下启发式规则决定何时切换到另一种方法：A)**ReAct $\rightarrow$ CoT-SC**：当ReAct在给定步骤内未能返回答案时，切换到CoT-SC。我们为HotpotQA和FEVER分别设定了7步和5步，因为我们发现更多步骤并不会提升ReAct的性能³。B)**CoT-SC $\rightarrow$ ReAct**：当 n 中大多数答案在 $n/2$ 次以下（即内部知识可能无法有信心地支持任务）时，切换回ReAct。

微调由于在大规模手动标注推理轨迹和行动方面存在挑战，我们考虑采用类似Zelikman等人（2022）的自举方法，使用由ReAct（以及其他基线方法）生成的3,000条正确答案轨迹，微调较小的语言模型（如PaLM-8/62B），以解码轨迹（所有思考、行动、观察），条件为输入的问题/主张。更多细节请参见附录B.1。

3.3 结果与观察

ReAct 优于 Act 持续地 表1显示了以PaLM-540B为基础模型，采用不同提示方法在HotpotQA和Fever任务上的结果。我们注意到，ReAct在两个任务中均优于Act，展示了推理引导行动的价值，尤其是在合成最终答案方面，如图1（1c-d）所示。微调结果3也证实了推理轨迹对于更有依据的行动的益处。

在所有具有正确最终答案的轨迹中，HotpotQA上需要7步的轨迹和FEVER上需要5步的轨迹分别仅占0.84%和1.33%。

	Type	Definition	ReAct	CoT
Success	True positive	Correct reasoning trace and facts	94%	86%
	False positive	Hallucinated reasoning trace or facts	6%	14%
Failure	Reasoning error	Wrong reasoning trace (including failing to recover from repetitive steps)	47%	16%
	Search result error	Search return empty or does not contain useful information	23%	-
	Hallucination	Hallucinated reasoning trace or facts	0%	56%
	Label ambiguity	Right prediction but did not match the label precisely	29%	28%

Table 2: Types of success and failure modes of ReAct and CoT on HotpotQA, as well as their percentages in randomly selected examples studied by human.

ReAct vs. CoT On the other hand, ReAct outperforms CoT on Fever (60.9 vs. 56.3) and slightly lags behind CoT on HotpotQA (27.4 vs. 29.4). Fever claims for SUPPORTS/REFUTES might only differ by a slight amount (see Appendix D.1), so acting to retrieve accurate and up-to-date knowledge is vital. To better understand the behavioral difference between ReAct and CoT on HotpotQA, we randomly sampled 50 trajectories with correct and incorrect answers (judged by EM) from ReAct and CoT respectively (thus 200 examples in total), and manually labeled their success and failure modes in Table 2. Some key observations are as follows:

A) **Hallucination is a serious problem for CoT**, resulting in much higher false positive rate than ReAct (14% vs. 6%) in success mode, and make up its major failure mode (56%). In contrast, the problem solving trajectory of ReAct is more grounded, fact-driven, and trustworthy, thanks to the access of an external knowledge base.

B) **While interleaving reasoning, action and observation steps improves ReAct’s groundedness and trustworthiness, such a structural constraint also reduces its flexibility in formulating reasoning steps**, leading to more reasoning error rate than CoT. we note that there is one frequent error pattern specific to ReAct, in which the model repetitively generates the previous thoughts and actions, and we categorize it as part of “reasoning error” as the model fails to reason about what the proper next action to take and jump out of the loop⁴.

C) **For ReAct, successfully retrieving informative knowledge via search is critical**. Non-informative search, which counts for 23% of the error cases, derails the model reasoning and gives it a hard time to recover and reformulate thoughts. This is perhaps an expected trade-off between factuality and flexibility, which motivates our proposed strategies of combining two methods.

We provide examples for each success and failure modes in Appendix E.1. We also find some HotpotQA questions may contain outdated answer labels, see Figure 4 for example.

ReAct + CoT-SC perform best for prompting LLMs Also shown in Table 1, the best prompting method on HotpotQA and Fever are ReAct $\rightarrow$ CoT-SC and CoT-SC $\rightarrow$ ReAct respectively. Furthermore, Figure 2 shows how different methods perform with respect to the number of CoT-SC samples used. While two ReAct + CoT-SC methods are advantageous at one task each, they both significantly and consistently outperform CoT-SC across different number of samples, reaching CoT-SC performance with 21 samples using merely 3-5 samples. These results indicate the value of properly combining model internal knowledge and external knowledge for reasoning tasks.

ReAct performs best for fine-tuning Figure 3 shows the scaling effect of prompting/finetuning four methods (Standard, CoT, Act, ReAct) on HotpotQA. With PaLM-8/62B, prompting ReAct performs worst among four methods due to the difficulty to learn both reasoning and acting from in-context examples. However, when finetuned with just 3,000 examples, ReAct becomes the best method among the four, with PaLM-8B finetuned ReAct outperforming all PaLM-62B prompting methods, and PaLM-62B finetuned ReAct outperforming all 540B prompting methods. In contrast, finetuning Standard or CoT is significantly worse than finetuning ReAct or Act for both PaLM-8/62B, as the former essentially teaches models to memorize (potentially hallucinated) knowledge facts, and the latter teaches models how to (reason and) act to access information from Wikipedia, a more generalizable skill for knowledge reasoning. As all prompting methods are still significantly far from domain-specific state-of-the-art approaches (Table 1), we believe finetuning with more human-written data might be a better way to unleash the power of ReAct.

⁴We suspect that this could be due to the sub-optimal greedy decoding procedure, and future work using better decoding (e.g. beam search) might help address this issue.

	Type	定义	ReAct	CoT
ReAct: 推理与行动于前 CoT: 推理与行动于后	真正阳性	正确的推理轨迹与事实	94%	86%
	误报	虚构的推理轨迹或事实	6%	14%
失败	推理错误	错误推理轨迹（包括未能从重复步骤中恢复）	47%	16%
	搜索结果错误	搜索返回为空或不包含有用信息	23%	-
	幻觉	虚构的推理轨迹或事实	0%	56%
	标签歧义	正确预测但未与标签完全匹配	29%	28%

表2: ReAct 和 CoT 在 HotpotQA 上的成功与失败模式类型及其在人类随机抽样示例中的百分比。

ReAct vs. CoT 另一方面, ReAct 在 Fever (60.9 对比 56.3) 上的表现优于 CoT, 在 HotpotQA (27.4 对比 29.4) 上则略逊于 CoT。Fever 中关于 SUPPORTS/REFUTES 的声明可能仅相差微小 (见附录 D.1), 因此采取行动以检索准确且最新的知识至关重要。为了更好地理解 ReAct 和 CoT 在 HotpotQA 上的行为差异, 我们随机抽取了 50 条具有正确和错误答案 (由 EM 判断) 的轨迹, 分别来自 ReAct 和 CoT (共 200 个样本), 并在表 2 中手动标注了它们的成功与失败模式。一些关键观察如下:

幻觉是CoT的一个严重问题, 在成功模式下, 其假阳性率远高于ReAct (14%对6%), 并且成为其主要的失败模式 (56%)。相比之下, ReAct的问题解决轨迹更加扎实、以事实为导向且值得信赖, 这得益于其对外部知识库的访问。

B) **在交错推理、行动和观察步骤的过程中, 能够提升ReAct的基础性和可信度**, 例如结构约束也会降低其在制定推理步骤时的灵活性, 从而导致比CoT更高的推理错误率。我们注意到, ReAct特有的一种频繁错误模式, 即模型反复生成之前的思考和行动, 我们将其归类为“推理错误”的一部分, 因为模型未能正确推断出下一步应采取的行动, 从而陷入循环⁴。

C) **对于ReAct, 通过搜索成功检索到有用知识至关重要**。非信息性搜索, 占错误案例的23%, 会干扰模型的推理, 导致其难以恢复和重新构建思路。这或许是事实性与灵活性之间的预期权衡, 这也促使我们提出结合两种方法的策略。

我们在附录E.1中提供了每种成功和失败模式的示例。我们还发现一些HotpotQA问题可能包含过时的答案标签, 详见图4的示例。

ReAct + CoT-SC 在提示大型语言模型 (LLMs) 方面表现最佳如表1所示, 在HotpotQA和Fever任务中, 最佳的提示方法分别是ReAct $\rightarrow$ 和CoT-SC $\rightarrow$ 。此外, 图2展示了不同方法在使用不同数量的CoT-SC样本时的表现。虽然两个ReAct + CoT-SC方法在各自任务中具有优势, 但它们都显著且持续优于CoT-SC, 无论样本数量多少, 都能达到CoT-SC的性能, 使用仅3-5个样本即可在21个样本的情况下实现。以上结果表明, 合理结合模型内部知识与外部知识对于推理任务具有重要价值。

ReAct 在微调方面表现最佳图3展示了在HotpotQA上, 提示/微调四种方法 (Standard、CoT、Act、ReAct) 的扩展效果。使用PaLM-8/62B时, 提示ReAct在四种方法中表现最差, 因为难以从上下文示例中同时学习推理和行动。然而, 当仅用3,000个示例进行微调时, ReAct成为四种方法中表现最优的, 经过微调的PaLM-8B ReAct优于所有PaLM-62B提示方法, 而PaLM-62B微调的ReAct又优于所有540B提示方法。相反, 对于PaLM-8/62B, 微调Standard或CoT明显不如微调ReAct或Act, 因为前者本质上教模型记忆 (可能存在幻觉的) 知识事实, 而后者则教模型如何 (推理和) 行动以访问信息源自维基百科, 这是一项更具泛化能力的知识推理技能。由于所有提示方法仍远远落后于领域特定的最先进方法 (表1), 我们认为通过更多人工撰写的数据进行微调, 可能是释放ReAct潜力的更佳途径。

我们怀疑这可能是由于次优的贪婪解码过程所致, 未来采用更优的解码方法 (例如束搜索) 可能有助于解决这一问题。

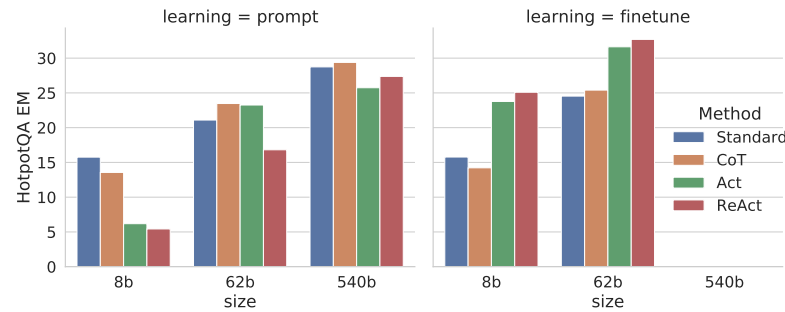


Figure 3: Scaling results for prompting and finetuning on HotPotQA with ReAct (ours) and baselines.

4 DECISION MAKING TASKS

We also test ReAct on two language-based interactive decision-making tasks, ALFWorld and WebShop, both of which feature complex environments that require agents to act over long horizons with sparse rewards, warranting the need for reasoning to act and explore effectively.

ALFWorld ALFWorld (Shridhar et al., 2020b) (Figure 1(2)) is a synthetic text-based game designed to align with the embodied ALFRED benchmark (Shridhar et al., 2020a). It includes 6 types of tasks in which an agent needs to achieve a high-level goal (e.g. examine paper under desk lamp) by navigating and interacting with a simulated household via text actions (e.g. go to coffeetable 1, take paper 2, use desk lamp 1). A task instance can have more than 50 locations and take an expert policy more than 50 steps to solve, thus challenging an agent to plan and track subgoals, as well as explore systematically (e.g. check all desks one by one for desk lamp). In particular, one challenge built into ALFWorld is the need to determine likely locations for common household items (e.g. desk lamps will likely be on desks, shelves, or dressers), making this environment a good fit for LLMs to exploit their pretrained commonsense knowledge. To prompt ReAct, we randomly annotate three trajectories from the training set for each task type, where each trajectory includes sparse thoughts that (1) decompose the goal, (2) track subgoal completion, (3) determine the next subgoal, and (4) reason via commonsense where to find an object and what to do with it. We show prompts used for ALFWorld in Appendix C.4. Following Shridhar et al. (2020b), we evaluate on 134 unseen evaluation games in a task-specific setup. For robustness, we construct 6 prompts for each task type through each permutation of 2 annotated trajectories from the 3 we annotate. Act prompts are constructed using the same trajectories, but without thoughts — since task instances are randomly chosen from the training set, it favors neither ReAct nor Act and provides a fair and controlled comparison to test the importance of sparse thoughts. For baselines, we use BUTLER (Shridhar et al., 2020b), an imitation learning agent trained on 10^5 expert trajectories for each task type⁵.

WebShop Can ReAct also interact with noisy real-world language environments for practical applications? We investigate WebShop (Yao et al., 2022), a recently proposed online shopping website environment with 1.18M real-world products and 12k human instructions. Unlike ALFWorld, Webshop contains a high variety of structured and unstructured texts (e.g. product titles, descriptions, and options crawled from Amazon), and requires an agent to purchase a product based on a user instruction (e.g. “I am looking for a nightstand with drawers. It should have a nickel finish, and priced lower than \$140”) through web interactions (e.g. search “nightstand drawers”, choose buttons such as “color: modern-nickel-white” or “back to search”). This task is evaluated by average score (percentage of desired attributes covered by the chosen product averaged across all episodes) and success rate (percentage of episodes where the chosen product satisfies all requirements) on 500 test instructions. We formulate Act prompts with actions to search, choose product, choose options, and buy, with ReAct prompts additionally reasoning to determine what to explore, when to buy, and what products options are relevant to the instruction. See Table 6 for an example prompt, and Table 10 for model predictions in the Appendix. We compare to an imitation learning (IL) method

⁵Micheli & Fleuret (2021) finetuned a GPT-2 model on 3553 task instances and achieved a much improved performance than BUTLER, but it is trained on all task types, thus not included as a baseline.

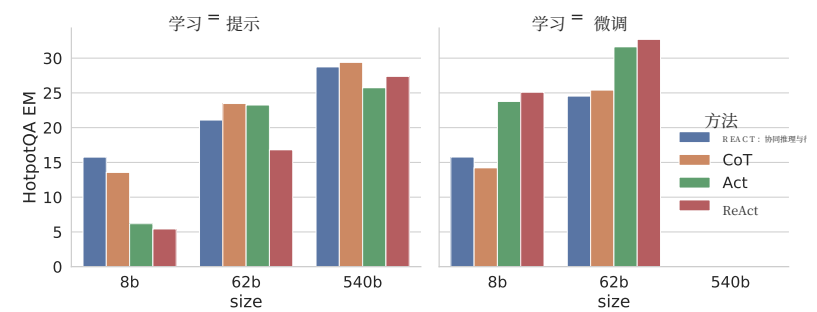


Figure 3: 在 HotPotQA 上使用 ReAct（我们的方法）和基线模型进行提示和微调的扩展结果 s.

4 决策任务

我们还在两个基于语言的交互式决策任务——ALFWorld 和 WebShop 上测试了 ReAct，这两个任务都具有复杂的环境，要求代理在长时间范围内采取行动，并应对稀疏的奖励，因此需要推理以有效地行动和探索。

ALFWorld ALFWorld (Shridhar 等, 2020b) (图1(2)) 是一款基于文本的合成游戏，旨在与具身的 ALFRED 基准 (Shridhar 等, 2020a) 保持一致。它包含六种任务类型，代理需要通过文本操作在模拟家庭环境中导航和交互，以实现高层次目标（例如，检查台灯下的纸张）。一个任务实例可能涉及超过50个地点，专家策略解决时间超过50步，因此对代理的规划和子目标追踪能力提出了挑战，同时需要系统性探索（例如，逐一检查所有书桌以寻找台灯）。特别地，ALFWorld 内置的一个挑战是需要判断常见家庭用品的可能位置（例如，台灯很可能在书桌、架子或衣柜上），使该环境非常适合大规模预训练语言模型（LLMs）利用其先验的常识知识。为了引导 ReAct，我们随机从每个任务类型的训练集中标注三个轨迹，每个轨迹包含稀疏的思考步骤，内容包括（1）分解目标，（2）追踪子目标完成情况，（3）确定下一个子目标，以及（4）通过常识推理找到物品的位置和处理方式。我们在附录C.4中展示了用于 ALFWorld 的提示。按照 Shridhar 等（2020b）的方法，我们在134个未见过的评估游戏中进行任务特定的评估。为了增强鲁棒性，我们为每个任务类型通过每种排列组合的两个标注轨迹构建6个提示。行动提示使用相同的轨迹，但不包含思考步骤——由于任务实例是随机从训练集中抽取的，这样的设计既不偏向 ReAct，也不偏向 Act，能够公平且受控地比较稀疏思考的重要性。作为基线，我们采用 BUTLER (Shridhar 等, 2020b)，这是一个在每个任务类型的 10^5 专家轨迹上训练的模仿学习代理⁵。

WebShop ReAct 是否也能在实际应用中与嘈杂的真实世界语言环境进行交互？我们研究了 WebShop (Yao 等, 2022)，这是一个最近提出的在线购物网站环境，包含118万件真实产品和1.2万条人工指令。与 ALFWorld 不同，WebShop 包含大量结构化和非结构化文本（例如从亚马逊爬取的产品标题、描述和选项），并要求代理根据用户指令（例如“我在找带抽屉的床头柜，镍色 finish，价格低于\$140”）通过网页交互进行购买（例如搜索 “nightstand drawers”，选择按钮如 “color: modern-nickel-white” 或 “back to search”）。该任务通过平均得分（所有任务中所选产品覆盖的期望属性百分比的平均值）和成功率（所选产品满足所有要求的任务百分比）在500个测试指令中进行评估。我们将行动提示设计为搜索、选择产品、选择选项和购买等操作，ReAct 提示还包括推理部分，用于判断何时探索、何时购买以及哪些产品选项与指令相关。见表6的示例提示，以及附录中的表10模型预测。我们将其与模仿学习 (IL) 方法进行了比较。

⁵Micheli & Fleuret (2021) 在3553个任务实例上对GPT-2模型进行了微调，取得了比BUTLER更显著的性能提升，但该模型是在所有任务类型上训练的，因此未被列为基线。

Method	Pick	Clean	Heat	Cool	Look	Pick 2	All	Method	Score	SR
Act (best of 6)	88	42	74	67	72	41	45	Act	62.3	30.1
ReAct (avg)	65	39	83	76	55	24	57	ReAct	66.6	40.0
ReAct (best of 6)	92	58	96	86	78	41	71	IL	59.9	29.1
ReAct-IM (avg)	55	59	60	55	23	24	48	IL+RL	62.4	28.7
ReAct-IM (best of 6)	62	68	87	57	39	33	53			
BUTLER _g (best of 8)	33	26	70	76	17	12	22	Human	82.1	59.6
BUTLER (best of 8)	46	39	74	100	22	24	37	Expert		

Table 3: AlfWorld task-specific success rates (%). BUTLER and BUTLER_g results are from Table 4 of Shridhar et al. (2020b). All methods use greedy decoding, except that BUTLER uses beam search.

trained with 1,012 human annotated trajectories, and a imitation + reinforcement learning (IL + RL) method additionally trained with 10,587 training instructions.

Results ReAct outperforms Act on both ALFWorld (Table 3) and Webshop (Table 4). On ALFWorld, the best ReAct trial achieves an average success rate of 71%, significantly outperforming the best Act (45%) and BUTLER (37%) trials. In fact, even the worse ReAct trial (48%) beats the best trial of both methods. Moreover, the advantage of ReAct over Act is consistent across six controlled trials, with relative performance gain ranging from 33% to 90% and averaging 62%. Qualitatively, we saw that, without any thoughts at all, Act fails to correctly decompose goals into smaller subgoals, or loses track of the current state of the environment. Example trajectories comparing ReAct and Act can be found in Appendix D.2.1 and Appendix D.2.2.

On Webshop, one-shot **Act** prompting already performs on par with IL and IL+RL methods. With additional sparse reasoning, **ReAct** achieves significantly better performance, with an absolute 10% improvement over the previous best success rate. By checking examples, we find that **ReAct** is more likely to identify instruction-relevant products and options by reasoning to bridge the gap between noisy observations and actions (e.g. “For ‘space-saving ottoman bench for living room’, the item has options ‘39x18x18inch’ and ‘blue’ and seems good to buy.”). However, existing methods are still far from the performance of expert humans (Table 4), who perform significantly more product explorations and query re-formulations that are still challenging for prompting-based methods.

On the value of internal reasoning vs. external feedback To our knowledge, ReAct is the first demonstration of combined reasoning and action using an LLM applied to an interactive environment within a closed-loop system. Perhaps the closest prior work is Inner Monologue (IM), from Huang et al. (2022b), in which actions from an embodied agent are motivated by an eponymous “inner monologue”. **However, IM’s “inner monologue” is limited to observations of the environment state and what needs to be completed by the agent for the goal to be satisfied.** In contrast, the reasoning traces in ReAct for decision making is flexible and sparse, allowing diverse reasoning types (see Section 2) to be induced for different tasks.

To demonstrate the differences between ReAct and IM, and to highlight the importance of internal reasoning vs. simple reactions to external feedback, we ran an ablation experiment using a thought pattern composed of IM-like dense external feedback. As can be seen in Table 3, ReAct substantially outperforms IM-style prompting (ReAct-IM) (71 vs. 53 overall success rate), with consistent advantages on five out of six tasks. Qualitatively, we observed that ReAct-IM often made mistakes in identifying when subgoals were finished, or what the next subgoal should be, due to a lack of high-level goal decomposition. Additionally, many ReAct-IM trajectories struggled to determine where an item would likely be within the ALFWorld environment, due to a lack of commonsense reasoning. Both shortcomings can be addressed in the ReAct paradigm. More details about ReAct-IM is in Appendix B.2. An example prompt for ReAct-IM can be found in Appendix C.4, and an example trajectory in Appendix D.2.3.

方法	Pick	ReAct (最佳6轮)	Heat	Cool	Look	选择 2	All
Act (最佳6轮)	88	42	74	67	72	41	45
ReAct (平均)	65	39	83	76	55	24	57
ReAct	92	58	96	86	78	41	71
ReAct-IM (平均)	55	59	60	55	23	24	48
ReAct-IM (最佳6)	62	68	87	57	39	33	53
BUTLER ^g (最佳的8个)	33	26	70	76	17	12	22
BUTLER (最佳8)	46	39	74	100	22	24	37

方法	ReAct (最佳6轮)	SR
Act	62.3	30.1
ReAct	66.6	40.0
IL	59.9	29.1
IL+RL	62.4	28.7
专家	82.1	59.6

表 3: AlfWorld 任务特定成功率 (%)。BUTLER 和 BUTLER_g 的结果来自 Shridhar 等人 (2020b) 表 4。所有方法均采用贪婪解码, 除 BUTLER 使用束搜索。

使用1,012条人工标注轨迹进行训练，并结合模仿学习 + 强化学习（IL + RL）方法，额外使用10,587条训练指令进行训练。

Results ReAct在ALFWorld（表3）和Webshop（表4）中的表现均优于Act。在ALFWorld上，最佳ReAct试验的平均成功率达到71%，显著高于最佳Act（45%）和BUTLER（37%）试验。实际上，即使是表现较差的ReAct试验（48%）也优于两种方法的最佳试验。此外，ReAct相对于Act的优势在六个对照试验中保持一致，性能提升的相对比例范围为33%至90%，平均为62%。从定性角度来看，我们发现，在完全没有思考的情况下，Act无法正确将目标分解为更小的子目标，或无法跟踪环境的当前状态。ReAct与Act的示例轨迹可在附录D.2.1和附录D.2.2中找到。

在Webshop上，一次性Act提示已与IL和IL+RL方法表现相当。通过额外的稀疏推理，ReAct实现了显著的性能提升，成功率比之前的最佳水平提高了10个百分点。通过检查示例，我们发现ReAct更有可能通过推理识别与指令相关的产品和选项，从而弥合噪声观察与行动之间的差距（例如，“对于‘节省空间的客厅脚凳’商品，该商品有‘39x18x18英寸’和‘蓝色’两个选项，似乎值得购买。”）。然而，现有方法仍远不及专家人类的表现（表4），后者进行的产品探索和查询重构明显更多，而这些仍然是基于提示的方法所面临的挑战。

关于内部推理与外部反馈的价值据我们所知，ReAct 是首次在封闭回路系统中，将大规模语言模型（LLM）应用于交互环境中，结合推理与行动的示范。或许最接近的前期工作是 Huang 等人（2022b）提出的 Inner Monologue（IM），其中由具身代理的行动由同名的“内心独白”驱动。然而，IM 的“内心独白”仅限于对环境状态的观察以及代理为实现目标所需完成的任务。相比之下，ReAct 中用于决策的推理轨迹具有灵活性和稀疏性，能够引发多样的推理类型（见第2节），以适应不同的任务。

REACT: 协同推理与行动于语言模型中的融合 为了展示ReAct与IM之间的差异，并强调内部推理与对外部反馈的简单反应的重要性，我们进行了一个消融实验，采用由类似IM的密集外部反馈组成的思维模式。如表3所示，ReAct在整体成功率上显著优于IM式提示（ReAct-IM）（71对比53），在六个任务中有五个任务表现出持续优势。从定性角度来看，我们观察到ReAct-IM在识别子目标何时完成或下一个子目标应为何时，常因缺乏高层次目标分解而出现错误。此外，许多ReAct-IM轨迹在判断某一项在ALFWorld环境中的可能位置时也存在困难，原因是缺乏常识推理。这两方面的不足都可以在ReAct范式中得到改善。关于ReAct-IM的更多细节请参见附录B.2。ReAct-IM的示例提示可以在附录C.4中找到，示例轨迹则在附录D.2.3中展示。

表4: 在Web-shop上的得分与成功率 (SR)。
IL/IL+RL取自Yao等人 (2022年)。

5 RELATED WORK

Language model for reasoning Perhaps the most well-known work of using LLMs for reasoning is Chain-of-Thought (CoT) (Wei et al., 2022), which reveals the ability of LLMs to formulate their own “thinking procedure” for problem solving. Several follow-up works have since been performed, including least-to-most prompting for solving complicated tasks (Zhou et al., 2022), zero-shot-CoT (Kojima et al., 2022), and reasoning with self-consistency (Wang et al., 2022a). Recently, (Madaan & Yazdanbakhsh, 2022) systematically studied the formulation and structure of CoT, and observed that the presence of symbols, patterns and texts is crucial to the effectiveness of CoT. Other work has also been extended to more sophisticated reasoning architecture beyond simple prompting. For example Selection-Inference (Creswell et al., 2022) divides the reasoning process into two steps of “selection” and “inference”. STaR (Zelikman et al., 2022) bootstraps the reasoning process by finetuning the model on correct rationales generated by the model itself. Faithful reasoning (Creswell & Shanahan, 2022) decomposes multi-step reasoning into three steps, each performed by a dedicated LM respectively. Similar approaches like Scratchpad (Nye et al., 2021), which finetunes a LM on intermediate computation steps, also demonstrate improvement on multi-step computation problems. In contrast to these methods, ReAct performs more than just isolated, fixed reasoning, and integrates model actions and their corresponding observations into a coherent stream of inputs for the model to reason more accurately and tackle tasks beyond reasoning (e.g. interactive decision making).

Language model for decision making The strong capability of LLMs has enabled them to perform tasks beyond language generation, and it is becoming more popular to take advantage of LLMs as a policy model for decision making, especially in interactive environments. WebGPT (Nakano et al., 2021) uses an LM to interact with web browsers, navigate through web pages, and infer answers to complicated questions from ELI5 (Fan et al., 2019). In comparison to ReAct, WebGPT does not explicitly model the thinking and reasoning procedure, instead rely on expensive human feedback for reinforcement learning. In conversation modeling, chatbots like BlenderBot (Shuster et al., 2022b) and Sparrow (Glaese et al., 2022) and task-oriented dialogue systems like SimpleTOD (Hosseini-Asl et al., 2020) also train LMs to make decision about API calls. Unlike ReAct, they do not explicitly consider the reasoning procedure either, and also relies on expensive datasets and human feedback collections for policy learning. In contrast, ReAct learns a policy in a much cheaper way, since the decision making process only requires language description of the reasoning procedure.⁶

LLMs have also been increasingly employed in interactive and embodied environments for planning and decision making. Perhaps most relevant to ReAct in this respect are SayCan (Ahn et al., 2022) and Inner Monologue (Huang et al., 2022b), which use LLMs for robotic action planning and decision making. In SayCan, LLMs were prompted to directly predict possible actions a robot can take, which is then reranked by an affordance model grounded on the visual environments for final prediction. Inner Monologue made further improvements by adding the eponymous “inner monologue”, which is implemented as injected feedback from the environment. To our knowledge, Inner Monologue is the first work that demonstrates such a closed-loop system, which ReAct builds on. However, we argue that Inner Monologue does not truly comprise of inner thoughts — this is elaborated in Section 4. We also note that leveraging language as semantically-rich inputs in the process of interactive decision making has been shown to be successful under other settings (Abramson et al., 2020; Karamcheti et al., 2021; Huang et al., 2022a; Li et al., 2022). It is becoming more evident that with the help of LLMs, language as a fundamental cognitive mechanism will play a critical role in interaction and decision making. What is more, progress in LLMs has also inspired the development of versatile and generalist agents like Reed et al. (2022).

6 CONCLUSION

We have proposed ReAct – a simple yet effective method for synergizing reasoning and acting in large language models. Through a diverse set of experiments on multi-hop question-answering, fact checking, and interactive decision-making tasks, we show that ReAct leads to superior performance with interpretable decision traces. Despite the simplicity of our method, complex tasks with large action spaces require more demonstrations to learn well, which unfortunately can easily go beyond the input length limit of in-context learning. We explore the fine-tuning approach on HotpotQA

⁶Human feedback can also be incorporated in a complementary manner but we leave it for future work.

5 相关工作

Language model for reasoning 或许最著名的利用大规模语言模型进行推理的工作是链式思维（Chain-of-Thought, CoT）（Wei 等，2022），它揭示了大规模语言模型为问题解决制定“思维程序”的能力。此后，出现了一系列后续研究，包括用于解决复杂任务的“由少到多”提示（Zhou 等，2022）、零-shot-CoT（Kojima 等，2022）以及自一致性推理（Wang 等，2022a）。近期，Madaan 和 Yazdanbakhsh（2022）系统性研究了链式思维的形成与结构，观察到符号、模式和文本的存在对链式思维的有效性至关重要。其他研究也扩展到超越简单提示的更复杂推理架构。例如，选择-推理（Selection-Inference）（Creswell 等，2022）将推理过程划分为“选择”和“推理”两个步骤。STaR（Zelikman 等，2022）通过在模型生成的正确推理依据上进行微调，启动推理过程。忠实推理（Creswell 和 Shanahan，2022）将多步骤推理分解为三个步骤，分别由专门的语言模型执行。类似的方法如Scratchpad（Nye 等，2021），通过在中间计算步骤上微调语言模型，也在多步骤计算问题上表现出改进。与这些方法不同，ReAct不仅仅执行孤立、固定的推理，还将模型的行动及其对应的观察整合为一个连贯的输入流，从而使模型能够更准确地推理并应对超越推理的任务（例如交互式决策制定）。

用于决策的语言模型大型语言模型（LLMs）的强大能力使其能够执行超出语言生成的任务，越来越多地将LLMs作为决策策略模型，特别是在交互式环境中。WebGPT（Nakano 等，2021）利用语言模型与网页浏览器交互，导航网页，并从ELI5（Fan 等，2019）推断复杂问题的答案。与ReAct相比，WebGPT并未明确建模思考和推理过程，而是依赖昂贵的人类反馈进行强化学习。在对话建模方面，像BlenderBot（Shuster 等，2022b）和 Sparrow（Glaese 等，2022）这样的聊天机器人，以及像SimpleTOD（Hosseini-Asl 等，2020）这样的任务导向对话系统，也训练语言模型对API调用做出决策。与ReAct不同，它们也未明确考虑推理过程，同样依赖昂贵的数据集和人类反馈收集进行策略学习。相比之下，ReAct以更低成本学习策略，因为决策过程仅需对推理过程的语言描述。⁶

LLMs 也越来越多地被应用于交互式 and 具象环境中的规划与决策。或许与 ReAct 最相关的是 SayCan (Ahn et al., 2022) 和 Inner Monologue (Huang et al., 2022b)，它们使用 LLMs 进行机器人行动规划和决策。在 SayCan 中，LLMs 被提示直接预测机器人可能采取的行动，然后由基于视觉环境的可行性模型进行重新排序，以获得最终预测。Inner Monologue 通过加入同名的“inner monologue”实现了进一步的改进，该“inner monologue”被实现为来自环境的注入反馈。据我们所知，Inner Monologue 是第一个展示此类闭环系统的工作，ReAct 正是在此基础上构建的。然而，我们认为 Inner Monologue 并不真正包含内在思想——这将在第 4 节中详细阐述。我们还注意到，在交互式决策过程中利用语言作为语义丰富的输入在其他设置中已被证明是成功的 (Abramson et al., 2020; Karamcheti et al., 2021; Huang et al., 2022a; Li et al., 2022)。越来越明显的是，在 LLMs 的帮助下，作为一种基本认知机制的语言将在交互和决策中发挥关键作用。更重要的是，LLMs 的进展也激发了像 Reed et al. (2022) 这样多功能和通用代理的开发。

6 结论

ReAct：在语言模型中协同推理与行动我们提出了一种名为ReAct的方法——一种简单而有效的在大型语言模型中协同推理与行动的策略。通过多跳问答、事实核查和交互式决策任务上的多样化实验，我们展示了ReAct在具有可解释决策轨迹的情况下，能够实现优越的性能。尽管我们的方法简单，但对于具有大量行动空间的复杂任务，学习效果通常需要更多示范样本，不幸的是，这很容易超出上下文学习的输入长度限制。我们还探索了在HotpotQA上的微调方法。

⁶人类反馈也可以以互补的方式加以整合，但我们将其留待未来的研究。

with initial promising results, but learning from more high-quality human annotations will be the desiderata to further improve the performance. Scaling up ReAct with multi-task training and combining it with complementary paradigms like reinforcement learning could result in stronger agents that further unlock the potential of LLMs for more applications.

ACKNOWLEDGMENTS

We thank the support and feedback of many people from Google Brain team and Princeton NLP Group. This work was supported in part by the National Science Foundation under Grant No. 2107048. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

REPRODUCIBILITY STATEMENT

Our main experiments are done on PaLM (Chowdhery et al., 2022), which is not an openly accessible model yet. To increase reproducibility, we have included all used prompts in Appendix C, additional experiments using GPT-3 (Brown et al., 2020) in Appendix A.1, and associated GPT-3 ReAct prompting code at <https://anonymous.4open.science/r/ReAct-2268/>.

ETHICS STATEMENT

ReAct prompts large language models to generate more human interpretable, diagnosable, and controllable task-solving trajectories than previous methods. However, hooking up a large language model with an action space to interact with external environments (e.g. the web, physical environments) has potential dangers, e.g. looking up inappropriate or private information, or taking harmful actions in an environment. Our experiments minimize such risks by limiting the interactions to specific websites (Wikipedia or WebShop) that are free of private information, without any dangerous actions in the action space design (i.e. models cannot really buy products on WebShop the research benchmark, or edit Wikipedia). We believe researchers should be aware of such risks before designing more extensive experiments in the future.

REFERENCES

Josh Abramson, Arun Ahuja, Iain Barr, Arthur Brussee, Federico Carnevale, Mary Cassin, Rachita Chhaparia, Stephen Clark, Bogdan Damoc, Andrew Dudzik, Petko Georgiev, Aurelia Guy, Tim Harley, Felix Hill, Alden Hung, Zachary Kenton, Jessica Landon, Timothy Lillicrap, Kory Mathewson, Soňa Mokrá, Alistair Muldal, Adam Santoro, Nikolay Savinov, Vikrant Varma, Greg Wayne, Duncan Williams, Nathaniel Wong, Chen Yan, and Rui Zhu. Imitating interactive intelligence, 2020. URL <https://arxiv.org/abs/2012.05672>.

Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Chuyuan Fu, Keerthana Gopalakrishnan, Karol Hausman, Alex Herzog, Daniel Ho, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Eric Jang, Rosario Jauregui Ruano, Kyle Jeffrey, Sally Jesmonth, Nikhil J Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Kuang-Huei Lee, Sergey Levine, Yao Lu, Linda Luu, Carolina Parada, Peter Pastor, Jornell Quiambao, Kanishka Rao, Jarek Rettinghouse, Diego Reyes, Pierre Sermanet, Nicolas Sievers, Clayton Tan, Alexander Toshev, Vincent Vanhoucke, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Mengyuan Yan, and Andy Zeng. Do as i can, not as i say: Grounding language in robotic affordances, 2022. URL <https://arxiv.org/abs/2204.01691>.

Ben Alderson-Day and Charles Fernyhough. Inner speech: development, cognitive functions, phenomenology, and neurobiology. *Psychological bulletin*, 141(5):931, 2015.

Alan Baddeley. Working memory. *Science*, 255(5044):556–559, 1992.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.

尽管初步取得了有希望的结果，但通过更多高质量的人类标注进行学习仍是进一步提升性能的理想途径。通过多任务训练扩大ReAct的规模，并将其与强化学习等互补范式相结合，可能会产生更强大的智能体，进一步释放大型语言模型在更多应用中的潜力。

致谢

我们感谢Google Brain团队和普林斯顿自然语言处理组的众多人员的支持与反馈。本工作部分得到了国家自然科学基金（Grant No. 2107048）的资助。本文所表达的任何观点、发现、结论或建议均为作者（们）的观点，不一定代表国家自然科学基金的立场。

可复现性声明

我们的主要实验是在 PaLM (Chowdhery et al., 2022) 上进行的，该模型尚未公开提供。为提高可复现性，我们在附录 C 中包含了所有使用的提示，在附录 A.1 中展示了使用 GPT-3 (Brown et al., 2020) 的额外实验，以及相关的 GPT-3 ReAct 提示代码，链接为 <https://anonymous.4open.science/r/ReAct-2268/>。

伦理声明

ReAct 促使大型语言模型生成比以往方法更具人类可解释性、可诊断性和可控性的任务解决轨迹。然而，将大型语言模型与行动空间结合以与外部环境（如网页、物理环境）交互存在潜在风险，例如查阅不当或私人信息，或在环境中采取有害行动。我们的实验通过限制交互仅限于特定网站（维基百科或WebShop）且这些网站不包含私人信息，避免了在行动空间设计中出现任何危险行为（例如，模型无法在WebShop上真正购买商品，或编辑维基百科），以最大程度降低此类风险。我们认为，研究人员在未来设计更大规模的实验之前，应充分意识到这些潜在风险。

参考文献

Josh Abramson, Arun Ahuja, Iain Barr, Arthur Brussee, Federico Carnevale, Mary Cassin, Rachita Chhaparia, Stephen Clark, Bogdan Damoc, Andrew Dudzik, Petko Georgiev, Aurelia Guy, Tim Harley, Felix Hill, Alden Hung, Zachary Kenton, Jessica Landon, Timothy Lillicrap, Kory Mathewson, So’na Mokrá, Alistair Muldal, Adam Santoro, Nikolay Savinov, Vikrant Varma, Greg Wayne, Duncan Williams, Nathaniel Wong, Chen Yan, and Rui Zhu. 模仿交互智能，2020年。URL <https://arxiv.org/abs/2012.05672>

Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Chuyuan Fu, Keerthana Gopalakrishnan, Karol Hausman, Alex Herzog, Daniel Ho, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Eric Jang, Rosario Jauregui Ruano, Kyle Jeffrey, Sally Jesmonth, Nikhil J Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Kuang-Huei Lee, Sergey Levine, Yao Lu, Linda Luu, Carolina Parada, Peter Pastor, Jornell Quiambao, Kanishka Rao, Jarek Rettinghouse, Diego Reyes, Pierre Sermanet, Nicolas Sievers, Clayton Tan, Alexander Toshev, Vincent Vanhoucke, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Mengyuan Yan, 以及 Andy Zeng。尽我所能，不是我说的：将语言与机器人可供性相结合，2022。网址 <https://arxiv.org/abs/2204.01691>。

Ben Alderson-Day 和 Charles Fernyhough。内在言语：发展、认知功能、现象学与神经生物学。*Psychological bulletin*, 141(5):931, 2015年。

艾伦·巴德利。工作记忆。*Science*, 255(5044):556–559, 1992年。

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell 等。语言模型是少样本学习者。神经信息处理系统的进展，第33卷：1877–1901，2020年。

Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, et al. Palm: Scaling language modeling with pathways. *arXiv preprint arXiv:2204.02311*, 2022.

Antonia Creswell and Murray Shanahan. Faithful reasoning using large language models, 2022. URL <https://arxiv.org/abs/2208.14271>.

Antonia Creswell, Murray Shanahan, and Irina Higgins. Selection-inference: Exploiting large language models for interpretable logical reasoning, 2022. URL <https://arxiv.org/abs/2205.09712>.

Angela Fan, Yacine Jernite, Ethan Perez, David Grangier, Jason Weston, and Michael Auli. ELI5: Long form question answering. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 3558–3567, Florence, Italy, July 2019. Association for Computational Linguistics. doi: 10.18653/v1/P19-1346. URL <https://aclanthology.org/P19-1346>.

Charles Fernyhough. Vygotsky, luria, and the social brain. *Self and social regulation: Social interaction and the development of social understanding and executive functions*, pp. 56–79, 2010.

Amelia Glaese, Nat McAleese, Maja Trebacz, John Aslanides, Vlad Firoiu, Timo Ewalds, Maribeth Rauh, Laura Weidinger, Martin Chadwick, Phoebe Thacker, Lucy Campbell-Gillingham, Jonathan Uesato, Po-Sen Huang, Ramona Comanescu, Fan Yang, Abigail See, Sumanth Dathathri, Rory Greig, Charlie Chen, Doug Fritz, Jaume Sanchez Elias, Richard Green, Soňa Mokrá, Nicholas Fernando, Boxi Wu, Rachel Foley, Susannah Young, Iason Gabriel, William Isaac, John Mellor, Demis Hassabis, Koray Kavukcuoglu, Lisa Anne Hendricks, and Geoffrey Irving. Improving alignment of dialogue agents via targeted human judgements, 2022. URL <https://storage.googleapis.com/deepmind-media/DeepMind.com/Authors-Notes/sparrow/sparrow-final.pdf>.

Ehsan Hosseini-Asl, Bryan McCann, Chien-Sheng Wu, Semih Yavuz, and Richard Socher. A simple language model for task-oriented dialogue. *Advances in Neural Information Processing Systems*, 33:20179–20191, 2020.

Wenlong Huang, Pieter Abbeel, Deepak Pathak, and Igor Mordatch. Language models as zero-shot planners: Extracting actionable knowledge for embodied agents. *arXiv preprint arXiv:2201.07207*, 2022a.

Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, et al. Inner monologue: Embodied reasoning through planning with language models. *arXiv preprint arXiv:2207.05608*, 2022b.

Siddharth Karamcheti, Megha Srivastava, Percy Liang, and Dorsa Sadigh. Lila: Language-informed latent actions. In *CoRL*, pp. 1379–1390, 2021. URL <https://proceedings.mlr.press/v164/karamcheti22a.html>.

Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. Large language models are zero-shot reasoners. *arXiv preprint arXiv:2205.11916*, 2022.

Angeliki Lazaridou, Elena Gribovskaya, Wojciech Stokowiec, and Nikolai Grigorev. Internet-augmented language models through few-shot prompting for open-domain question answering. *arXiv preprint arXiv:2203.05115*, 2022.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33: 9459–9474, 2020.

Shuang Li, Xavier Puig, Chris Paxton, Yilun Du, Clinton Wang, Linxi Fan, Tao Chen, De-An Huang, Ekin Akyürek, Anima Anandkumar, Jacob Andreas, Igor Mordatch, Antonio Torralba, and Yuke Zhu. Pre-trained language models for interactive decision-making, 2022. URL <https://arxiv.org/abs/2202.01771>.

Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann 等. Palm: 通过路径扩展语言模型的规模. *arXivpreprint arXiv:2204.02311*, 2022年。

Antonia Creswell 和 Murray Shanahan. 使用大型语言模型进行可靠推理, 2022年。网址 <https://arxiv.org/abs/2208.14271>

Antonia Creswell, Murray Shanahan, 和 Irina Higgins. Selection-inference: 利用大型语言模型实现可解释的逻辑推理, 2022。网址 <https://arxiv.org/abs/2205.09712>。

Angela Fan, Yacine Jernite, Ethan Perez, David Grangier, Jason Weston 和 Michael Auli. ELI5: 长形式问题回答。在第57届计算语言学协会年会论文集, 第3558–3567页, 意大利佛罗伦萨, 2019年7月。计算语言学协会。doi: 10.18653/v1/P19-1346。网址 <https://aclanthology.org/P19-1346>。

Charles Fernyhough. 维果茨基、卢里亚与社会大脑。自我与社会调节: 社会交互作用与社会理解及执行功能的发展, 第56–79页, 2010. *REACT: 协同推理与行动于语言模型*. Amelia Glaese, Nat McAleese, MajaTrebacz, JohnAslanides, Vlad Firoiu, Timo Ewalds, Maribeth Rauh, Laura Weidinger, Martin Chadwick, Phoebe Thacker, Lucy Campbell-Gillingham, Jonathan Uesato, Po-Sen Huang, Ramona Comanescu, Fan Yang, Abigail See, Sumanth Dathathri, Rory Greig, Charlie Chen, Doug Fritz, Jaume Sanchez Elias, Richard Green, Soňa Mokrá, Nicholas Fernando, Boxi Wu, Rachel Foley, Susannah Young, Iason Gabriel, William Isaac, John Mellor, Demis Hassabis, Koray Kavukcuoglu, Lisa Anne Hendricks 以及 Geoffrey Irving. 通过有针对性的人工判断改善对话代理的对齐, 2022年。网址 <https://storage.googleapis.com/deepmind-media/DeepMind.com/Authors-Notes/sparrow/sparrow-final.pdf>

Ehsan Hosseini-Asl, Bryan McCann, Chien-Sheng Wu, Semih Yavuz, 和 Richard Socher. 面向任务的对话的简易语言模型。神经信息处理系统进展, 第33卷: 20179–20191, 2020年。

黄文龙、Pieter Abbeel、Deepak Pathak 和 Igor Mordatch. 语言模型作为零样本规划器: 提取具行动性的知识用于具身代理. *arXiv* 预印本 *arXiv:2201.07207*, 2022a。

黄文龙, 夏飞, 肖泰德, 陈哈里斯, 梁杰奇, 弗洛伦斯·皮特, 曾安迪, 汤普森乔纳森, 莫尔达奇伊戈尔, 切博塔尔叶夫根, 等。内心独白: 通过语言模型进行规划的体现式推理. *arXiv preprint arXiv:2207.05608*, 2022b。

Siddharth Karamcheti, Megha Srivastava, Percy Liang, 和 Dorsa Sadigh. Lila: 基于语言的潜在动作。在 *CoRL*, 第1379–1390页, 2021年。网址 <https://proceedings.mlr.press/v164/karamcheti22a.html>

小島健、顾诗翔、Machel Reid、松尾丰、岩泽祐介。大型语言模型是零样本推理器. *arXiv*预印本 *arXiv:2205.11916*, 2022年。

Angeliki Lazaridou, Elena Gribovskaya, Wojciech Stokowiec, 以及 Nikolai Grigorev. 通过少样本提示实现互联网增强的语言模型用于开放域问答. *arXiv preprintarXiv:2203.05115*, 2022.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel 等. 增强检索的生成方法用于知识密集型自然语言处理任务. 神经信息处理系统进展, 33: 9459–9474, 2020.

李双、Xavier Puig、Chris Paxton、杜伊伦、王克林、西席凡、陈涛、黄德安、Ekin Akyürek、Anima Anandkumar、Jacob Andreas、Igor Mordatch、Antonio Torralba 及 Yuke Zhu。用于交互决策的预训练语言模型, 2022。网址 <https://arxiv.org/abs/2202.01771>

Aleksandr Romanovich Luria. Ls vygotsky and the problem of localization of functions. *Neuropsychologia*, 3(4):387–392, 1965.

Aman Madaan and Amir Yazdanbakhsh. Text and patterns: For effective chain of thought, it takes two to tango, 2022. URL <https://arxiv.org/abs/2209.07686>.

Vincent Micheli and François Fleuret. Language models are few-shot butlers. *arXiv preprint arXiv:2104.07972*, 2021.

Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, Xu Jiang, Karl Cobbe, Tyna Eloundou, Gretchen Krueger, Kevin Button, Matthew Knight, Benjamin Chess, and John Schulman. Webgpt: Browser-assisted question-answering with human feedback, 2021. URL <https://arxiv.org/abs/2112.09332>.

Maxwell Nye, Anders Johan Andreassen, Guy Gur-Ari, Henryk Michalewski, Jacob Austin, David Bieber, David Dohan, Aitor Lewkowycz, Maarten Bosma, David Luan, Charles Sutton, and Augustus Odena. Show your work: Scratchpads for intermediate computation with language models, 2021. URL <https://arxiv.org/abs/2112.00114>.

Scott Reed, Konrad Zolna, Emilio Parisotto, Sergio Gomez Colmenarejo, Alexander Novikov, Gabriel Barth-Maron, Mai Gimenez, Yury Sulsky, Jackie Kay, Jost Tobias Springenberg, Tom Eccles, Jake Bruce, Ali Razavi, Ashley Edwards, Nicolas Heess, Yutian Chen, Raia Hadsell, Oriol Vinyals, Mahyar Bordbar, and Nando de Freitas. A generalist agent, 2022. URL <https://arxiv.org/abs/2205.06175>.

Mohit Shridhar, Jesse Thomason, Daniel Gordon, Yonatan Bisk, Winson Han, Roozbeh Mottaghi, Luke Zettlemoyer, and Dieter Fox. Alfred: A benchmark for interpreting grounded instructions for everyday tasks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 10740–10749, 2020a.

Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. Alfworld: Aligning text and embodied environments for interactive learning. *arXiv preprint arXiv:2010.03768*, 2020b.

Kurt Shuster, Mojtaba Komeili, Leonard Adolphs, Stephen Roller, Arthur Szlam, and Jason Weston. Language models that seek for knowledge: Modular search & generation for dialogue and prompt completion. *arXiv preprint arXiv:2203.13224*, 2022a.

Kurt Shuster, Jing Xu, Mojtaba Komeili, Da Ju, Eric Michael Smith, Stephen Roller, Megan Ung, Moya Chen, Kushal Arora, Joshua Lane, Morteza Behrooz, William Ngan, Spencer Poff, Naman Goyal, Arthur Szlam, Y-Lan Boureau, Melanie Kambadur, and Jason Weston. Blenderbot 3: a deployed conversational agent that continually learns to responsibly engage, 2022b. URL <https://arxiv.org/abs/2208.03188>.

James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. Fever: a large-scale dataset for fact extraction and verification. *arXiv preprint arXiv:1803.05355*, 2018.

Lev S Vygotsky. Thinking and speech. *The collected works of LS Vygotsky*, 1:39–285, 1987.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models, 2022a. URL <https://arxiv.org/abs/2203.11171>.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, and Denny Zhou. Rationale-augmented ensembles in language models. *arXiv preprint arXiv:2207.00747*, 2022b.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Ed Chi, Quoc Le, and Denny Zhou. Chain of thought prompting elicits reasoning in large language models. *arXiv preprint arXiv:2201.11903*, 2022.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W Cohen, Ruslan Salakhutdinov, and Christopher D Manning. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. *arXiv preprint arXiv:1809.09600*, 2018.

Aleksandr Romanovich Luria. Ls vygotsky 与功能定位问题。 *Neuropsychologia*, 3(4): 387–392, 1965年。Aman Madaan 和 Amir Yazdanbakhsh. 文本与模式：为了实现有效的思维链，双人合作是必要的，2022年。网址 <https://arxiv.org/abs/2209.07686>。Vincent Micheli 和 François Fleuret. 语言模型是少样本管家。 *arXiv preprint arXiv:2104.07972*, 2021 年。Reiichiro Nakano、Jacob Hilton、Suchir Balaji、Jeff Wu、Long Ouyang、Christina Kim、Christopher Hesse、Shantanu Jain、Vineet Kosaraju、William Saunders、Xu Jiang、Karl Cobbe、Tyna Eloundou、Gretchen Krueger、Kevin Button、Matthew Knight、Benjamin Chess 和 John Schulman. Webgpt: 结合人类反馈的浏览器辅助问答，2021年。网址 <https://arxiv.org/abs/2112.09332>。

Maxwell Nye, Anders Johan Andreassen, Guy Gur-Ari, Henryk Michalewski, Jacob Austin, David Bieber, David Dohan, Aitor Lewkowycz, Maarten Bosma, David Luan, Charles Sutton, 和 Augustus Odena. Show your work: Scratchpads for intermediate computation with language models, 2021。URL <https://arxiv.org/abs/2112.00114>。

Scott Reed、Konrad Zolna、Emilio Parisotto、Sergio Gomez Colmenarejo、Alexander Novikov、Gabriel Barth-Maron、Mai Gimenez、Yury Sulsky、Jackie Kay、Jost Tobias Springenberg、Tom Eccles、Jake Bruce、Ali Razavi、Ashley Edwards、Nicolas Heess、Yutian Chen、Raia Hadsell、Oriol Vinyals、Mahyar Bordbar 以及 Nando de Freitas。通用智能体，2022。URL <https://arxiv.org/abs/2205.06175>。Mohit Shridhar、Jesse Thomason、Daniel Gordon、Yonatan Bisk、Winson Han、Roozbeh Mottaghi、Luke Zettlemoyer 以及 Dieter Fox。Alfred：用于解释日常任务中的 grounded 指令的基准测试。发表于 *IEEE/CVF 计算机视觉与模式识别会议论文集*，第10740–10749页，2020a。Mohit Shridhar、Xingdi Yuan、Marc-Alexandre Côté、Yonatan Bisk、Adam Trischler 以及 Matthew Hausknecht。Alfworld：对齐文本与具身环境以实现交互式学习。 *arXiv* 预印本 *arXiv:2010.03768*，2020b。Kurt Shuster、Mojtaba Komeili、Leonard Adolphs、Stephen Roller、Arthur Szlam 以及 Jason Weston。寻求知识的语言模型：对话与提示完成的模块化搜索与生成。 *arXiv* 预印本 *arXiv:2203.13224*，2022a。Kurt Shuster、Jing Xu、Mojtaba Komeili、Da Ju、Eric Michael Smith、Stephen Roller、Megan Ung、Moya Chen、Kushal Arora、Joshua Lane、Morteza Behrooz、William Ngan、Spencer Poff、Naman Goyal、Arthur Szlam、Y-Lan Boureau、Melanie Kambadur 以及 Jason Weston。Blenderbot 3：一个持续学习以负责任地交互的部署对话智能体，2022b。URL <https://arxiv.org/abs/2208.03188>。James Thorne、Andreas Vlachos、Christos Christodoulopoulos 以及 Arpit Mittal。Fever：一个大规模的事实提取与验证数据集。 *arXiv* 预印本 *arXiv:1803.05355*，2018。Lev S Vygotsky。思维与语言。 *LS Vygotsky* 文集，第1卷，第39–285页，1987。Xuezhi Wang、Jason Wei、Dale Schuurmans、Quoc Le、Ed Chi、Sharan Narang、Aakanksha Chowdhery 以及 Denny Zhou。自洽性提升语言模型中的链式推理能力，2022a。URL <https://arxiv.org/abs/2203.11171>。

Xuezhi Wang、Jason Wei、Dale Schuurmans、Quoc Le、Ed Chi 和 Denny Zhou。基于推理增强的集成方法在语言模型中的应用。 *arXiv* 预印本 *arXiv:2207.00747*，2022b。

Jason Wei、Xuezhi Wang、Dale Schuurmans、Maarten Bosma、Ed Chi、Quoc Le、Denny Zhou。Chain of thought prompting elicits reasoning in large language models. *arXiv preprint arXiv:2201.11903*, 2022. Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W Cohen, Ruslan Salakhutdinov, Christopher D Manning. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. *arXiv preprint arXiv:1809.09600*, 2018.

Shunyu Yao, Rohan Rao, Matthew Hausknecht, and Karthik Narasimhan. Keep CALM and explore: Language models for action generation in text-based games. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 8736–8754, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.704. URL <https://aclanthology.org/2020.emnlp-main.704>.

Shunyu Yao, Howard Chen, John Yang, and Karthik Narasimhan. Webshop: Towards scalable real-world web interaction with grounded language agents. *arXiv preprint arXiv:2207.01206*, 2022.

Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah D. Goodman. Star: Bootstrapping reasoning with reasoning, 2022. URL <https://arxiv.org/abs/2203.14465>.

Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Olivier Bousquet, Quoc Le, and Ed Chi. Least-to-most prompting enables complex reasoning in large language models, 2022. URL <https://arxiv.org/abs/2205.10625>.

Yunchang Zhu, Liang Pang, Yanyan Lan, Huawei Shen, and Xueqi Cheng. Adaptive information seeking for open-domain question answering. *arXiv preprint arXiv:2109.06747*, 2021.

Shunyu Yao, Rohan Rao, Matthew Hausknecht, 和 Karthik Narasimhan. 保持冷静并探索：基于文本游戏的行动生成语言模型。在2020年自然语言处理实证方法会议（*EMNLP*）论文集，第 8736–8754页，线上，2020年11月。计算语言学协会。doi: 10.18653/v1/2020.emnlp-main.704。网址 <https://aclanthology.org/2020.emnlp-main.704>

姚顺宇，陈浩然，杨约翰，纳拉西姆汉·卡尔蒂克。Webshop：迈向具有基础语言代理的可扩展真实世界网页交互。*arXiv*预印本 *arXiv:2207.01206*，2022年。

Eric Zelikman, Yuhuai Wu, Jesse Mu, 和 Noah D. Goodman. Star：引导推理与语言模型中的行动协同推理，2022。网址 <https://arxiv.org/abs/2203.14465>。

Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Olivier Bousquet, Quoc Le, 和 Ed Chi. 最少到最多提示（Least-to-most prompting）实现大规模语言模型中的复杂推理，2022年。URL <https://arxiv.org/abs/2205.10625>。

朱云长，庞亮，蓝艳艳，沈华为，程学旗。面向开放域问答的自适应信息检索。*arXivpreprint arXiv:2109.06747*，2021年。

A ADDITIONAL RESULTS

A.1 GPT-3 EXPERIMENTS

	PaLM-540B	GPT-3
HotpotQA (exact match)	29.4	30.8
ALFWorld (success rate %)	70.9	78.4

Table 5: ReAct prompting results using PaLM-540B vs. GPT-3 (text-davinci-002, greedy decoding). On HotpotQA, we randomly sample a subset of 500 validation questions. On ALFWorld, we use all 134 unseen validation task instances, and use the best prompt set according to PaLM-540B.

We run additional GPT-3 (Brown et al., 2020) experiments to confirm ReAct prompting performance is general across different large language models. As shown in Table 5, GPT-3 (text-davinci-002, greedy decoding) consistently outperforms PaLM-540B on HotpotQA and ALFWorld, possibly because it is finetuned with human instruction following. This indicates ReAct prompting is effective across different large language models on different tasks. The code for these experiments are at <https://react-lm.github.io/>.

A.2 REACT OBTAINS UP-TO-DATE KNOWLEDGE ON HOTPOTQA

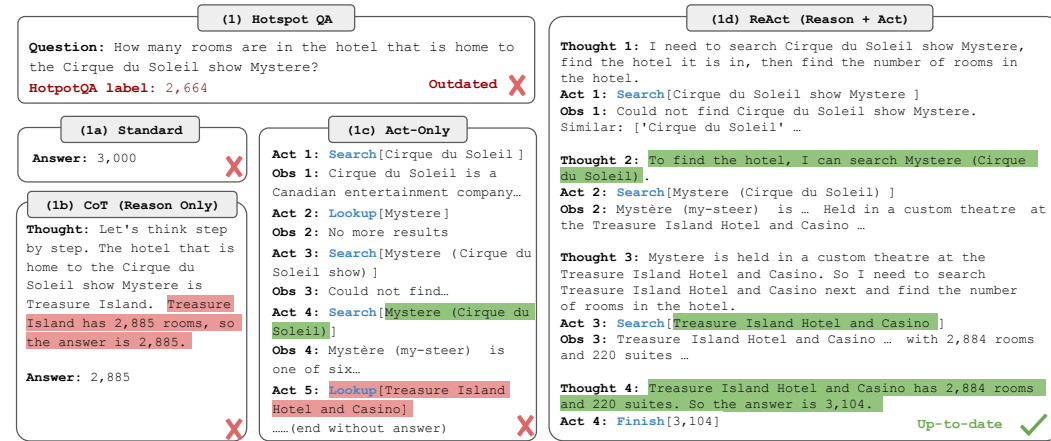


Figure 4: Another example HotpotQA question, where the original label is outdated. Only ReAct is able to obtain the up-to-date answer thanks to real-world web interaction plus reasoning.

During trajectory inspection, we also find that sometimes ReAct does not agree with dataset labels as the labels themselves could be outdated. For example, as shown in Figure 4, the question asks about the size of a hotel, which increased from the HotpotQA construction time. While Standard and CoT give wrong answers due to hallucination, Act fails despite the access of real-world web interaction, due to a lack of reasoning to guide how to interact with the Internet for QA. Only ReAct is able to retrieve up-to-date information from the Internet and provide a reasonable answer. Therefore, better incorporation of reasoning abilities might benefit recent Internet-augmented language models (Nakano et al., 2021; Lazaridou et al., 2022; Shuster et al., 2022a) for up-to-date task solving.

A.3 HUMAN-IN-THE-LOOP BEHAVIOR CORRECTION ON ALFWORLD

We also explore human-in-the-loop interaction with ReAct, to allow a human to inspect and edit ReAct’s reasoning traces. Figure 5 shows that by simply removing a hallucinating sentence in Act 17 and adding some hints in Act 23, ReAct can be made to change its behavior drastically to align with these human thought edits and succeed in the task. From a human perspective, solving such a task becomes significantly easier, from typing tens of actions to only editing a couple of thoughts, which enables new forms of human-machine collaboration. We note that such a policy edit on-the-go

A 额外结果

A.1 GPT-3 实验

	PaLM-540B	GPT-3
HotpotQA（精确匹配）	29.4	30.8
ALFWorld（成功率 %）	70.9	78.4

表5：使用PaLM-540B与GPT-3（text-davinci-002，贪婪解码）进行ReAct提示的结果。在HotpotQA上，我们随机抽取了500个验证问题的子集。在ALFWorld上，我们使用所有未见过的134个验证任务实例，并根据PaLM-540B选择最佳的提示集。

我们进行了额外的GPT-3（Brown等，2020）实验，以确认ReAct提示在不同大型语言模型中的表现具有普遍性。如表5所示，GPT-3（text-davinci-002，贪婪解码）在HotpotQA和ALFWorld任务中始终优于PaLM-540B，可能是由于其经过了人类指令遵循的微调。这表明ReAct提示在不同任务和不同大型语言模型中都具有有效性。这些实验的代码可在 <https://react-lm.github.io/>获取。

A.2 REACT 获取关于HotpotQA的最新知识

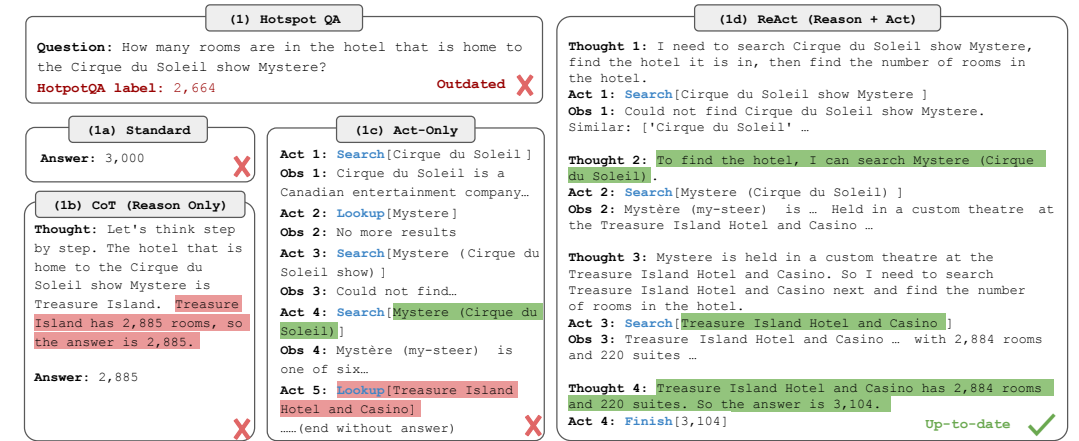


图4：另一个 HotpotQA 问题示例，其中原始标签已过时。只有 ReAct 能够通过结合真实世界的网页交互和推理，获得最新的答案。在轨迹检查过程中，我们还发现有时ReAct与数据集标签不一致，因为标签本身可能已经过时。例如，如图4所示，问题询问酒店的规模，而该信息自HotpotQA构建时间以来已发生变化。虽然Standard和CoT由于幻觉而给出错误答案，但Act在访问真实世界网页交互的情况下仍然失败，原因在于缺乏引导如何与互联网交互以进行问答的推理能力。只有ReAct能够从互联网检索最新信息并提供合理的答案。因此，更好地结合推理能力可能有助于近年来的互联网增强型语言模型（Nakano等，2021；Lazaridou等，2022；Shuster等，2022a）实现最新任务的解决。

A.3 人类在环行为校正 ONALFWORLD

我们还探索了与ReAct的人机交互方式，允许人为检查和编辑ReAct的推理轨迹。图5显示，通过简单地删除Act 17中的一段幻觉句子并在Act 23中添加一些提示，ReAct的行为可以被大幅度调整，以符合这些人为的思考修改并成功完成任务。从人类的角度来看，解决此类任务变得显著更为容易，从输入数十个动作到仅需编辑几条思考，这为人机协作开辟了新的形式。我们注意到，这种策略的现场编辑具有潜在的应用价值。

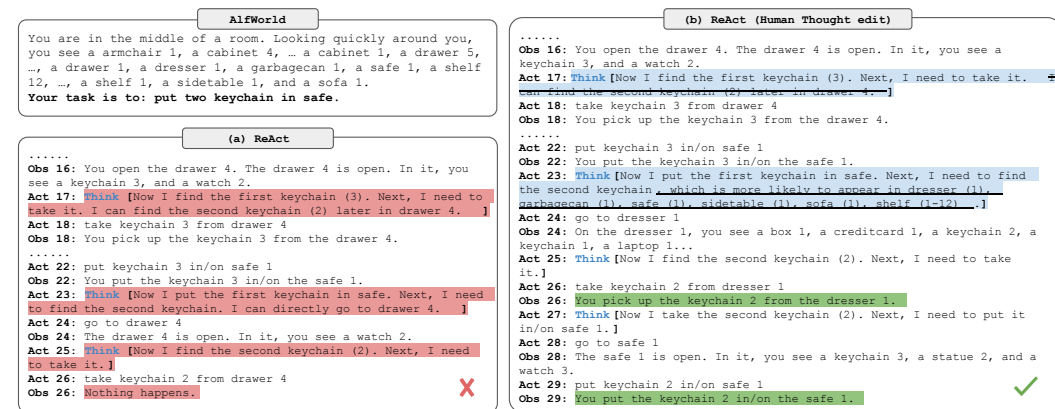


Figure 5: A human-in-the-loop behavior correction example with ReAct in AlfWorld. (a) ReAct trajectory fails due to a hallucinating thought (Act 17). (b) By a human simply editing two thoughts (Act 17, 23), the ReAct trajectory produces desirable reasoning traces and actions and succeeds.

is difficult for Act and previous RL methods, as a human cannot change the model parameters, and changing a few actions might not edit the rest of the model behavior. This paradigm is also more than human dialogue to update the goal or subgoal as in Huang et al. (2022b) — while editing ReAct thoughts can do these, it can also modify the model’s internal belief, reasoning styles, or anything the flexible thought space supports, for better task solving. We believe this is an exciting direction for human alignment and leave more systematic study as future work.

B EXPERIMENT DETAILS

B.1 HOTPOTQA FINETUNING DETAILS

For all finetuning we use a batch size of 64. On PaLM-8B, we finetune ReAct and Act methods for 4,000 steps and Standard and CoT methods for 2,000 steps. On PaLM-62B, we finetune ReAct and Act methods for 4,000 steps and Standard and CoT methods for 1,000 steps. We find ReAct and Act methods generally benefit from more training steps (and more training data), while Standard and CoT methods degrade soon after finetuning.

B.2 ALFWORLD IM-STYLE DETAILS

For the IM-style ablation, the same expert trajectories used in ReAct are reannotated with dense external feedback thoughts within these trajectories, that limit ReAct-IM to only think about (1) decomposing the current goal and (2) the current subgoal that needs to be completed. In particular, ReAct-IM lacks thoughts that (1) determine when a subgoal is completed (2) determine what the next subgoal should be (3) inducing the LLM to refer to its internal pretraining knowledge to identify where items can be within the environment.

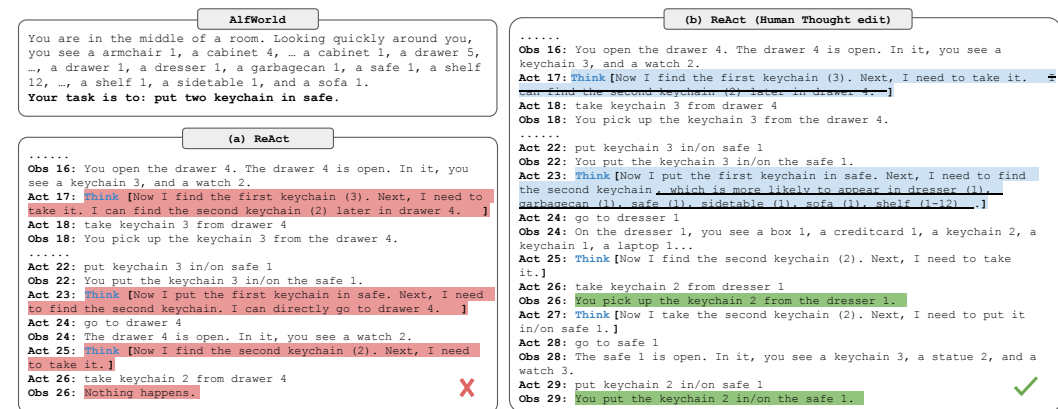


图5：在AlfWorld中使用ReAct的人机交互行为纠正示例。（a）由于出现幻觉性思维（Act 17），ReAct轨迹失败。（b）通过人为简单编辑两个思维（Act 17, 23），ReAct轨迹产生了理想的推理轨迹和行动，并取得了成功。

对于 Act 和之前的 RL 方法来说，这是具有挑战性的，因为人类无法直接改变模型参数，改变少数几个动作可能无法修改模型的其余行为。这一范式不仅限于人类对话中更新目标或子目标，如 Huang 等人（2022b）所示——虽然编辑 ReAct 思考可以实现这些，还可以修改模型的内部信念、推理风格或任何由灵活思考空间支持的内容，以实现更优的任务解决。我们相信这是一个令人振奋的人类对齐研究方向，并将更多系统性研究留待未来工作。

B 实验细节

B.1 HOTPOTQA 微调细节

对于所有微调，我们采用批次大小为64。在PaLM-8B上，我们对ReAct和Act方法进行4,000步的微调，对Standard和CoT方法进行2,000步的微调。在PaLM-62B上，我们对ReAct和Act方法进行4,000步的微调，对Standard和CoT方法进行1,000步的微调。我们发现ReAct和Act方法通常受益于更多的训练步骤（以及更多的训练数据），而Standard和CoT方法在微调后不久性能会下降。

B.2 ALFWORLD IM-风格细节

对于IM风格的消融实验，使用与ReAct相同的专家轨迹，在这些轨迹中重新注释了密集的外部反馈思路，限制ReAct-IM仅考虑（1）分解当前目标和（2）需要完成的当前子目标。特别地，ReAct-IM缺乏以下思考：（1）判断何时完成子目标；（2）确定下一个子目标应是什么；（3）引导大规模语言模型（LLM）参考其内部预训练知识，以识别环境中物品可能所在的位置。

C PROMPTS

C.1 HOTPOTQA

Hotpot QA Prompts	
Original	
Question	What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into?
Answer	1,800 to 7,000 ft
Question	Musician and satirist Allie Goertz wrote a song about the "The Simpsons" character Milhouse, who Matt Groening named after who?
Answer	Richard Nixon
Question	Which documentary is about Finnish rock groups, Adam Clayton Powell or The Saimaa Gesture?
Answer	The Saimaa Gesture
Question	What profession does Nicholas Ray and Elia Kazan have in common?
Answer	director, screenwriter, actor
Question	Which magazine was started first Arthur’s Magazine or First for Women?
Answer	Arthur’s Magazine
Question	Were Pavel Urysohn and Leonid Levin known for the same type of work?
Answer	Yes
Act	
Question	What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into?
Action 1	Search[Colorado orogeny]
Observation 1	The Colorado orogeny was an episode of mountain building (an orogeny) in Colorado and surrounding areas.
Action 2	Lookup[eastern sector]
Observation 2	(Result 1 / 1) The eastern sector extends into the High Plains and is called the Central Plains orogeny.
Action 3	Search[High Plains]
Observation 3	High Plains refers to one of two distinct land regions:
Action 4	Search[High Plains (United States)]
Observation 4	The High Plains are a subregion of the Great Plains. From east to west, the High Plains rise in elevation from around 1,800 to 7,000 ft (550 to 2,130 m).[3]
Action 5	Finish[1,800 to 7,000 ft]
Question	Musician and satirist Allie Goertz wrote a song about the "The Simpsons" character Milhouse, who Matt Groening named after who?
Action 1	Search[Milhouse]
Observation 1	Milhouse Mussolini Van Houten is a recurring character in the Fox animated television series The Simpsons voiced by Pamela Hayden and created by Matt Groening.
Action 2	Lookup[named after]
Observation 2	(Result 1 / 1) Milhouse was named after U.S. president Richard Nixon, whose middle name was Milhous.
Action 3	Finish[Richard Nixon]
Question	Which documentary is about Finnish rock groups, Adam Clayton Powell or The Saimaa Gesture?
Action 1	Search[Adam Clayton Powell]
Observation 1	Could not find [Adam Clayton Powell]. Similar: ['Adam Clayton Powell III', 'Seventh Avenue (Manhattan)', 'Adam Clayton Powell Jr. State Office Building', 'Isabel Washington Powell', 'Adam Powell', 'Adam Clayton Powell (film)', 'Giancarlo Esposito'].
Action 2	Search[Adam Clayton Powell (film)]
Observation 2	Adam Clayton Powell is a 1989 American documentary film directed by Richard Kilberg. The film is about the rise and fall of influential African-American politician Adam Clayton Powell Jr.[3][4] It was later aired as part of the PBS series The American Experience.
Action 3	Finish[The Saimaa Gesture]
Question	What profession does Nicholas Ray and Elia Kazan have in common?
Action 1	Search[Nicholas Ray]
Observation 1	Nicholas Ray (born Raymond Nicholas Kienzle Jr., August 7, 1911 – June 16, 1979) was an American film director, screenwriter, and actor best known for the 1955 film Rebel Without a Cause.
Action 2	Search[Elia Kazan]
Observation 2	Elia Kazan was an American film and theatre director, producer, screenwriter and actor.

Continued on next page

C 提示

C.1 HOTPOTQA

火锅问答提示	
SELECT NAME FROM FACT	
问题	东部区域的海拔范围是多少？ 科罗拉多造山运动延伸至？ 1,800 到 7,000 英尺
REA CT：协同推理与行动于语言模型	
问题	音乐家兼讽刺作家Allie Goertz创作了一首关于《辛普森一家》的歌曲角色米尔豪斯，Matt Groening 将其命名为谁的名字？ 理查德·尼克松
REA CT：协同推理与行动于语言模型	
问题	关于芬兰摇滚乐队的纪录片是《Adam Clayton Powell》还是《The》Saimaa Gesture? 赛马伊手势
REA CT：协同推理与行动于语言模型	
问题	尼古拉斯·雷和伊利亚·卡赞的共同职业是什么？ 导演，编剧，演员
REA CT：协同推理与行动于语言模型	
问题	Arthur’ s Magazine 先于 First for Women 创刊吗？ Arthur’ s Magazine
REA CT：协同推理与行动于语言模型	
问题	帕维尔·乌里松和列昂尼德·列文是否以相同类型的工作而闻名？ Yes
REA CT：协同推理与行动于语言模型	
Act	
问题	东部区域的海拔范围是多少？ 科罗拉多造山运动延伸至？ 搜索 [科罗拉多造山运动]
行动 1	科罗拉多造山运动是一次山脉形成（造山运动）事件，发生在科罗拉多及周边地区
观察 1	查找 [东部地区]
行动 2	东部区域延伸至高原地区，称为
观察 2	中原造山运动
行动 3	搜索 [高原]
观察 3	高原指的是两个不同土地区域之一：
行动 4	搜索 [高原地区（美国）]
观察 4	高原是大平原的一个子区域。从东向西， 高原的海拔从大约1,800英尺（550米）升高至7,000英尺（2,130米）
行动 5	m).[3] Finish[1,800 到7,000英尺]
问题	音乐家兼讽刺作家Allie Goertz创作了一首关于《辛普森一家》的歌曲角色米尔豪斯，Matt Groening 将其命名为谁的名字？ 搜索 [米尔豪斯]
行动 1	Milhouse Mussolini Van Houten 是《辛普森一家》动画中的常驻角色
观察 1	电视剧《辛普森一家》由帕梅拉·海登配音，由马特创作
行动 2	格鲁宁
观察 2	查找 [{name}]命名
行动 3	Milhouse 的命名源自美国总统理查德·尼克松，whose 中间名是Milhous。 Finish [理查德·尼克松]
问题	关于芬兰摇滚乐队的纪录片是《Adam Clayton Powell》还是《The》Saimaa Gesture? 搜索 [亚当·克莱顿·鲍威尔]
行动 1	未能找到 [Adam Clayton Powell]。
观察 1	类似：[' 亚当·克莱顿·鲍威尔 III' 第七大道（曼哈顿）， 亚当·克莱顿·鲍威尔 Jr. 州政府 “Building” 伊莎贝尔·华盛顿·鲍威尔 ‘Adam Powell’ 亚当·克莱顿·鲍威尔 (电影)， ‘Giancarlo Esposito’]。
行动 2	搜索 [亚当·克莱顿·鲍威尔（电影）]
观察 2	Adam Clayton Powell 是一部由1989年美国纪录片导演的影片 理查德·基尔伯格 该电影讲述了具有影响力的兴衰历程 非裔美国政治家亚当·克莱顿·鲍威尔 Jr. [3] [4] 后来被播出 作为 PBS 系列《美国体验》的一部分。
行动 3	完成 [塞马伊手势]
问题	尼古拉斯·雷和伊利亚·卡赞的共同职业是什么？ 搜索 [尼古拉斯·雷]
行动 1	Nicholas Ray（出生名为 Raymond Nicholas Kienzle Jr., 1911年8月7日—） 1979年）是一位美国电影导演， 编剧， 以及因其在……方面的杰出表现而闻名的演员
观察 1	1955年电影《青少年怒吼》
行动 2	搜索 [Elia Kazan]
观察 2	Elia Kazan 是一位美国电影和戏剧导演、制片人、编剧和演员。

续见下一页

Hotpot QA Prompts – Continued from previous page	
Action 3	Finish[director, screenwriter, actor]
Question	Which magazine was started first Arthur’s Magazine or First for Women?
Action 1	Search[Arthur’s Magazine]
Observation 1	Arthur’s Magazine (1844-1846) was an American literary periodical published in Philadelphia in the 19th century.
Action 2	Search[First for Women]
Observation 2	First for Women is a woman’s magazine published by Bauer Media Group in the USA.[1] The magazine was started in 1989.
Action 3	Finish[Arthur’s Magazine]
Question	Were Pavel Urysohn and Leonid Levin known for the same type of work?
Action 1	Search[Pavel Urysohn]
Observation 1	Pavel Samuilovich Urysohn (February 3, 1898 - August 17, 1924) was a Soviet mathematician who is best known for his contributions in dimension theory.
Action 2	Search[Leonid Levin]
Observation 2	Leonid Anatolievich Levin is a Soviet-American mathematician and computer scientist.
Action 3	Finish[yes]

CoT	
Question	What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into?
Thought	Let’s think step by step. The eastern sector of Colorado orogeny extends into the High Plains. High Plains rise in elevation from around 1,800 to 7,000 ft, so the answer is 1,800 to 7,000 ft.
Answer	1,800 to 7,000 ft
Question	Musician and satirist Allie Goertz wrote a song about the "The Simpsons" character Milhouse, who Matt Groening named after who?
Thought	Let’s think step by step. Milhouse was named after U.S. president Richard Nixon, so the answer is Richard Nixon.
Answer	Richard Nixon
Question	Which documentary is about Finnish rock groups, Adam Clayton Powell or The Saimaa Gesture?
Thought	Let’s think step by step. Adam Clayton Powell (film) is a documentary about an African-American politician, not Finnish rock groups. So the documentary about Finnish rock groups must instead be The Saimaa Gesture.
Answer	The Saimaa Gesture
Question	What profession does Nicholas Ray and Elia Kazan have in common?
Thought	Let’s think step by step. Professions of Nicholas Ray are director, screenwriter, and actor. Professions of Elia Kazan are director, producer, screenwriter, and actor. So profession Nicholas Ray and Elia Kazan have in common is director, screenwriter, and actor.
Answer	director, screenwriter, actor
Question	Which magazine was started first Arthur’s Magazine or First for Women?
Thought	Let’s think step by step. Arthur’s Magazine was started in 1844. First for Women was started in 1989. 1844 (Arthur’s Magazine) < 1989 (First for Women), so Arthur’s Magazine was started first.
Answer	Arthur’s Magazine
Question	Were Pavel Urysohn and Leonid Levin known for the same type of work?
Thought	Let’s think step by step. Pavel Urysohn is a mathematician. Leonid Levin is a mathematician and computer scientist. So Pavel Urysohn and Leonid Levin have the same type of work.
Answer	Yes

ReAct	
Question	What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into?
Thought 1	I need to search Colorado orogeny, find the area that the eastern sector of the Colorado orogeny extends into, then find the elevation range of the area.
Action 1	Search[Colorado orogeny]
Observation 1	The Colorado orogeny was an episode of mountain building (an orogeny) in Colorado and surrounding areas.
Thought 2	It does not mention the eastern sector. So I need to look up eastern sector.
Action 2	Lookup[eastern sector]
Observation 2	(Result 1 / 1) The eastern sector extends into the High Plains and is called the Central Plains orogeny.
Thought 3	The eastern sector of Colorado orogeny extends into the High Plains. So I need to search High Plains and find its elevation range.
Action 3	Search[High Plains]
Observation 3	High Plains refers to one of two distinct land regions
Thought 4	I need to instead search High Plains (United States).
Action 4	Search[High Plains (United States)]

Continued on next page

火锅问答提示——续前页	
行动 3	Finish[导演, 编剧, actor]
问题	Arthur’s Magazine 先于 First for Women 创刊吗?
行动 1	搜索[Arthur的杂志]
观察 1	Arthur’s Magazine (1844-1846) 是一份美国文学期刊, 出版于在19世纪的费城。
行动 2	搜索[First for Women]
观察 2	First for Women 是由 Bauer Media Group 出版的一本女性杂志 USA.[1] 该杂志于1989年创刊。
行动 3	Finish[Arthur的杂志]
问题	帕维尔·乌里松和列昂尼德·列文是否以相同类型的工作而闻名?
行动 1	搜索[帕维尔·乌里松]
观察 1	帕维尔·萨穆伊洛维奇·乌里苏恩 (February 3, 1898年8月17日 1924) 是一位苏联在维度理论方面贡献卓著的数学家
行动 2	搜索[列奥尼德·列文]
观察 2	Leonid Anatolievich Levin 是一位苏联裔美国数学家和计算机科学家
行动 3	科学家
完成[是]	

CoT	
问题	东部区域的海拔范围是多少?
思考	科罗拉多造山运动延伸至? 让我们一步步思考。科罗拉多造山带东部区域延伸进入高原地区。高原的海拔从大约1800米升高至 7,000 英尺, 所以答案是 1,800 到 7,000 英尺
REA CT: 协同推理与行动于语言模型	
问题	音乐家兼讽刺作家Allie Goertz创作了一首关于《辛普森一家》的歌曲角色米尔豪斯, Matt Groening 将其命名为谁的名字?
思考	让我们一步步思考。Milhouse 的命名源自美国总统理查德·尼克松, 所以答案是理查德·尼克松。
REA CT: 协同推理与行动于语言模型	
问题	关于芬兰摇滚乐队的纪录片是《Adam Clayton Powell》还是《The Saimaa Gesture?
思考	让我们一步步思考。亚当·克莱顿·鲍威尔 (电影) 是一部关于一位非裔美国政治家, 而非芬兰摇滚乐队。所以纪录片关于芬兰摇滚乐队, 必须是 The Saimaa Gesture。赛马伊手势
REA CT: 协同推理与行动于语言模型	
问题	尼古拉斯·雷和伊利亚·卡赞的共同职业是什么?
思考	让我们一步步思考。尼古拉斯·雷的职业是导演 伊利亚·卡赞的职业包括导演、制片人、编剧和演员。因此, 专业的尼古拉斯·雷和伊利亚·卡赞在 common 是 director, 编剧, 和演员。
REA CT: 协同推理与行动于语言模型	
问题	导演, 编剧, 演员
问题	Arthur’s Magazine 先于 First for Women 创刊吗?
思考	让我们一步步思考。Arthur’s Magazine 于 1844 年创刊。为女性而设的项目于1989年启动。1844 (Arthur’s Magazine) < 1989 (首次为妇女), 因此亚瑟杂志首先创刊。
REA CT: 协同推理与行动于语言模型	
问题	Arthur’s Magazine
问题	帕维尔·乌里松和列昂尼德·列文是否以相同类型的工作而闻名?
思考	让我们一步步思考。Pavel Urysohn 是一位数学家。莱昂尼德·列文是一位数学家和计算机科学家。所以帕维尔·乌里松和列昂尼德·列文具有相同类型的工作。
REA CT: 协同推理与行动于语言模型	
Yes	

ReAct	
问题	东部区域的海拔范围是多少?
思考 1	科罗拉多造山运动延伸至? 我需要搜索科罗拉多造山事件, 找到东部区域的相关区域 科罗拉多造山运动的范围延伸至, 然后找出其海拔范围。
行动 1	搜索[科罗拉多造山运动]
观察 1	科罗拉多造山运动是一次山脉形成 (造山运动) 事件, 发生在科罗拉多及周边地区
思考 2	它没有提及东部地区。所以我需要查找东方行业。
行动 2	查找[东部地区]
观察 2	东部区域延伸至高原地区, 称为中原造山运动
思考 3	科罗拉多造山带的东部区域延伸至高原平原。需要搜索High Plains并查找其海拔范围。 So I
行动 3	搜索[高原]
观察 3	High Plains 指的是两个不同土地区域之一
思考 4	我需要改为搜索 High Plains (美国)。
行动 4	搜索[高原地区 (美国)]

续页

Hotpot QA Prompts – Continued from previous page	
Observation 4	The High Plains are a subregion of the Great Plains. From east to west, the High Plains rise in elevation from around 1,800 to 7,000 ft (550 to 2,130 m).[3]
Thought 5	High Plains rise in elevation from around 1,800 to 7,000 ft, so the answer is 1,800 to 7,000 ft.
Action 5	Finish[1,800 to 7,000 ft]
Question	Musician and satirist Allie Goertz wrote a song about the "The Simpsons" character Milhouse, who Matt Groening named after who?
Thought 1	The question simplifies to "The Simpsons" character Milhouse is named after who. I only need to search Milhouse and find who it is named after.
Action 1	Search[Milhouse]
Observation 1	Milhouse Mussolini Van Houten is a recurring character in the Fox animated television series The Simpsons voiced by Pamela Hayden and created by Matt Groening.
Thought 2	The paragraph does not tell who Milhouse is named after, maybe I can look up "named after".
Action 2	Lookup[named after]
Observation 2	(Result 1 / 1) Milhouse was named after U.S. president Richard Nixon, whose middle name was Milhous.
Thought 3	Milhouse was named after U.S. president Richard Nixon, so the answer is Richard Nixon.
Action 3	Finish[Richard Nixon]
Question	Which documentary is about Finnish rock groups, Adam Clayton Powell or The Saimaa Gesture?
Thought 1	I need to search Adam Clayton Powell and The Saimaa Gesture, and find which documentary is about Finnish rock groups.
Action 1	Search[Adam Clayton Powell]
Observation 1	Could not find [Adam Clayton Powell]. Similar: ['Adam Clayton Powell III', 'Seventh Avenue (Manhattan)', 'Adam Clayton Powell Jr. State Office Building', 'Isabel Washington Powell', 'Adam Powell', 'Adam Clayton Powell (film)', 'Giancarlo Esposito'].
Thought 2	To find the documentary, I can search Adam Clayton Powell (film).
Action 2	Search[Adam Clayton Powell (film)]
Observation 2	Adam Clayton Powell is a 1989 American documentary film directed by Richard Kilberg. The film is about the rise and fall of influential African-American politician Adam Clayton Powell Jr.[3][4] It was later aired as part of the PBS series The American Experience.
Thought 3	Adam Clayton Powell (film) is a documentary about an African-American politician, not Finnish rock groups. So the documentary about Finnish rock groups must instead be The Saimaa Gesture.
Action 3	Finish[The Saimaa Gesture]
Question	What profession does Nicholas Ray and Elia Kazan have in common?
Thought 1	I need to search Nicholas Ray and Elia Kazan, find their professions, then find the profession they have in common.
Action 1	Search[Nicholas Ray]
Observation 1	Nicholas Ray (born Raymond Nicholas Kienzle Jr., August 7, 1911 – June 16, 1979) was an American film director, screenwriter, and actor best known for the 1955 film Rebel Without a Cause.
Thought 2	Professions of Nicholas Ray are director, screenwriter, and actor. I need to search Elia Kazan next and find his professions.
Action 2	Search[Elia Kazan]
Observation 2	Elia Kazan was an American film and theatre director, producer, screenwriter and actor.
Thought 3	Professions of Elia Kazan are director, producer, screenwriter, and actor. So profession Nicholas Ray and Elia Kazan have in common is director, screenwriter, and actor.
Action 3	Finish[director, screenwriter, actor]
Question	Which magazine was started first Arthur’s Magazine or First for Women?
Thought 1	I need to search Arthur’s Magazine and First for Women, and find which was started first.
Action 1	Search[Arthur’s Magazine]
Observation 1	Arthur’s Magazine (1844-1846) was an American literary periodical published in Philadelphia in the 19th century.
Thought 2	Arthur’s Magazine was started in 1844. I need to search First for Women next.
Action 2	Search[First for Women]
Observation 2	First for Women is a woman’s magazine published by Bauer Media Group in the USA.[1] The magazine was started in 1989.
Thought 3	First for Women was started in 1989. 1844 (Arthur’s Magazine) < 1989 (First for Women), so Arthur’s Magazine was started first.
Action 3	Finish[Arthur’s Magazine]
Question	Were Pavel Urysohn and Leonid Levin known for the same type of work?
Thought 1	I need to search Pavel Urysohn and Leonid Levin, find their types of work, then find if they are the same.
Action 1	Search[Pavel Urysohn]

火锅问答提示——续前页	
观察 4	高原是大平原的一个子区域。从东向西，高原的海拔从大约1,800英尺（550米）升高至7,000英尺（2,130米）m).[3]
思考 5	高原的海拔从大约1800英尺升高到7000英尺，因此答案是 1,800 到 7,000 英尺
行动 5	Finish [1,800 到7,000英尺]
问题	音乐家兼讽刺作家Allie Goertz创作了一首关于《辛普森一家》的歌曲角色米尔豪斯，Matt Groening 将其命名为谁的名字？
思考 1	这个问题简化为“辛普森一家”角色米尔豪斯的名字来源于哪里 who。我只需要搜索Milhouse, 看看它是以谁的名字命名的。
行动 1	搜索 [米尔豪斯]
观察 1	Milhouse Mussolini Van Houten 是《辛普森一家》动画中的常驻角色电视剧《辛普森一家》由帕梅拉·海登配音，由马特创作格鲁宁
思考 2	这段话没有说明 Milhouse 的名字来源，也许我可以查找“以……命名”
行动 2	查找 [以{name}命名]
观察 2	Milhouse 的命名源自美国总统理查德·尼克松，whose 中间名是Milhous。
思考 3	米尔豪斯的名字取自美国总统理查德·尼克松，因此答案是理查德·尼克松
行动 3	Finish [理查德·尼克松]
问题	关于芬兰摇滚乐队的纪录片是《Adam Clayton Powell》还是《The》Saimaa Gesture?
思考 1	我需要搜索 Adam Clayton Powell 和 The Saimaa Gesture, 并找出哪个纪录片关于芬兰摇滚乐队。
行动 1	搜索 [亚当·克莱顿·鲍威尔]
观察 1	未能找到 [Adam Clayton Powell]。类似: [' 亚当·克莱顿·鲍威尔 III' 第七大道 (曼哈顿) 亚当·克莱顿·鲍威尔 Jr. 州政府 “Building” 伊莎贝尔·华盛顿·鲍威尔 ‘Adam Powell’ 亚当·克莱顿·鲍威尔 (电影), ‘Giancarlo Esposito’]。
思考 2	要查找该纪录片，我可以搜索 Adam Clayton Powell (film)。
行动 2	搜索 [亚当·克莱顿·鲍威尔 (电影)]
观察 2	Adam Clayton Powell 是一部由1989年美国纪录片导演的影片理查德·基尔伯格 该电影讲述了具有影响力的兴衰历程非裔美国政治家亚当·克莱顿·鲍威尔 Jr.[3][4] 后来被播出作为 PBS 系列《美国体验》的一部分。
思考 3	亚当·克莱顿·鲍威尔 (电影) 是一部关于非裔美国人的纪录片政治人物，而非芬兰摇滚乐队。所以关于芬兰摇滚的纪录片组别必须是 The Saimaa Gesture。
行动 3	完成 [塞马伊手势]
问题	尼古拉斯·雷和伊利亚·卡赞的共同职业是什么？
思考 1	我需要搜索 Nicholas Ray 和 Elia Kazan, 查找他们的职业，然后找出他们共同的职业。
行动 1	搜索 [尼古拉斯·雷]
观察 1	Nicholas Ray (出生名为 Raymond Nicholas Kienzle Jr., 1911年8月7日—1979年) 是一位美国电影导演，编剧，以及因其在……方面的杰出表现而闻名的演员1955年电影《青少年怒吼》
思考 2	尼古拉斯·雷的职业是导演、编剧和演员。我需要
行动 2	搜索 Elia Kazan 并查找他的职业。
观察 2	搜索 [Elia Kazan] Elia Kazan 是一位美国电影和戏剧导演、制片人、编剧和演员。
思考 3	Elia Kazan 的职业包括导演、制片人、编剧和演员。所以尼古拉斯·雷和伊利亚·卡赞共同的职业是导演、编剧，和演员。
行动 3	Finish [导演，编剧，actor]
问题	Arthur’s Magazine 先于 First for Women 创刊吗？
思考 1	我需要搜索Arthur’s Magazine和First for Women, 找出哪个是首先开始。
行动 1	搜索 [Arthur的杂志]
观察 1	Arthur’s Magazine (1844-1846) 是一份美国文学期刊，出版于在19世纪的费城。
思考 2	Arthur’s Magazine 于 1844 年创刊。我需要先为女性进行搜索下一步。
行动 2	搜索 [First for Women]
观察 2	First for Women 是由 Bauer Media Group 出版的一本女性杂志 USA.[1] 该杂志于1989年创刊。
思考 3	First for Women 于 1989 年成立。1844 (Arthur's Magazine) < 1989 (First (为女性而设)，因此Arthur的杂志首先创刊。
行动 3	Finish [Arthur的杂志]
问题	帕维尔·乌里松和列昂尼德·列文是否以相同类型的工作而闻名？
思考 1	我需要搜索 Pavel Urysohn 和 Leonid Levin, 了解他们的工作类型。然后判断它们是否相同。
行动 1	搜索 [帕维尔·乌里松]

Hotpot QA Prompts – Continued from previous page	
Observation 1	Pavel Samuilovich Urysohn (February 3, 1898 – August 17, 1924) was a Soviet mathematician who is best known for his contributions in dimension theory.
Thought 2	Pavel Urysohn is a mathematician. I need to search Leonid Levin next and find its type of work.
Action 2	Search[Leonid Levin]
Observation 2	Leonid Anatolievich Levin is a Soviet-American mathematician and computer scientist.
Thought 3	Leonid Levin is a mathematician and computer scientist. So Pavel Urysohn and Leonid Levin have the same type of work.
Action 3	Finish[yes]

火锅问答提示——续前页			
观察 1	帕维尔·萨穆伊洛维奇·乌里苏恩 (February 3, 1898年8月17日	1924)	是一位苏联
思考 2	在维度理论方面贡献卓越的数学家 Pavel Urysohn 是一位数学家。	我需要接下来搜索Leonid Levin。	
行动 2	找出其工作类型。		
观察 2	搜索 [列奥尼德·列文] Leonid Anatolievich Levin 是一位苏联裔美国数学家和计算机科学家		
思考 3	科学家 Leonid Levin 是一位数学家和计算机科学家。	所以帕维尔·乌里松	
行动 3	和列奥尼德·列文具有相同类型的工作。		
	完成 [是]		

C.2 FEVER

FEVER Prompts	
Original	Determine if there is Observation that SUPPORTS or REFUTES a Claim, or if there is NOT ENOUGH INFORMATION.
Claim Answer	Nikolaj Coster-Waldau worked with the Fox Broadcasting Company. SUPPORTS
Claim Answer	Stranger Things is set in Bloomington, Indiana. REFUTES
Claim Answer	Beautiful reached number two on the Billboard Hot 100 in 2003. NOT ENOUGH INFO
Ask	Determine if there is Observation that SUPPORTS or REFUTES a Claim, or if there is NOT ENOUGH INFORMATION.
Claim Action2 Observation 1	Nikolaj Coster-Waldau worked with the Fox Broadcasting Company. Search[Nikolaj Coster-Waldau] Nikolaj William Coster-Waldau (born 27 July 1970) is a Danish actor and producer. He graduated from the Danish National School of Performing Arts in Copenhagen in 1993,[1] and had his breakthrough role in Denmark with the film Nightwatch (1994). He played Jaime Lannister in the HBO fantasy drama series Game of Thrones, for which he received two Primetime Emmy Award nominations for Outstanding Supporting Actor in a Drama Series.. Coster-Waldau has appeared in numerous films in his native Denmark and Scandinavia, including Headhunters (2011) and A Thousand Times Good Night (2013). In the U.S, his debut film role was in the war film Black Hawk Down (2001), playing Medal of Honor recipient Gary Gordon.[2] He then played a detective in the short-lived Fox television series New Amsterdam (2008), and appeared in the 2009 Fox television film Virtuality, originally intended as a pilot.
Action 2	Finish[SUPPORTS]
Claim Action 1 Observation 1	Stranger Things is set in Bloomington, Indiana. Search[Stranger Things] Stranger Things is an American science fiction horror drama television series created by the Duffer Brothers. Set in the 1980s, primarily in the fictional town of Hawkins, Indiana, the series centers on a number of mysteries and supernatural events occurring around the town and their impact on an ensemble of child and adult characters.
Action 2	Finish[REFUTES]
Claim Action 1 Observation 1	Beautiful reached number two on the Billboard Hot 100 in 2003. Search[Beautiful] Could not find [Beautiful]. Similar: ['Beautiful', 'Beautiful, Beautiful', 'A Beautiful Mind (film)', 'Beautiful (Christina Aguilera song)', 'Life Is Beautiful'].
Action 2 Observation 2	Search[Beautiful (Christina Aguilera song)] "Beautiful" is a song recorded by American singer Christina Aguilera for her fourth studio album, Stripped (2002).
Action 3 Observation 3	Lookup[Billboard Hot 100] (Result 1 / 3) The song peaked at number two on the Billboard Hot 100 in the United States, where it was certified Gold for 500,000 units shipped.
Action 4	Finish[NOT ENOUGH INFO]
Ask	Determine if there is Observation that SUPPORTS or REFUTES a Claim, or if there is NOT ENOUGH INFORMATION.
Claim Thought Answer	Nikolaj Coster-Waldau worked with the Fox Broadcasting Company. Nikolaj William Coster-Waldau appeared in the 2009 Fox television film Virtuality, so he has worked with the Fox Broadcasting Company. SUPPORTS
Claim Thought Answer	Stranger Things is set in Bloomington, Indiana. Stranger Things is in the fictional town of Hawkins, Indiana, not in Bloomington, Indiana. REFUTES

Continued on next page

C.2 发热

FEVER 提示	
REACT: 推理与行动于语言模型中的融合 C PROMPTS	判断是否存在支持或反驳某一主张的观察，或是否信息不足。
声明 REACT: 协同推理与行动于语言模型	尼古拉·科斯特-瓦尔道与福克斯广播公司合作。支持
声明 REACT: 协同推理与行动于语言模型	《怪奇物语》设定于印第安纳州布卢明顿。反驳
声明 REACT: 协同推理与行动于语言模型	Beautiful 在2003年登顶 Billboard Hot 100 排行榜第二名。 REACT: 协同推理与行动于语言模型
Ask	判断是否存在支持或反驳主张的观察结果信息不足。 , 或者如果
声明 REACT: 协同推理与行动于语言模型 观察 1	尼古拉·科斯特-瓦尔道与福克斯广播公司合作。搜索 [尼古拉·科斯特-瓦尔道] 尼古拉·威廉·科斯特-瓦尔道（出生于1970年7月27日）是丹麦演员和制作者。他毕业于丹麦国家表演艺术学院在哥本哈根的 1993,[1] , 并在丹麦取得了他的突破性角色电影《夜巡》(1994) 他在HBO奇幻剧中饰演詹姆·兰尼斯特电视剧《权力的游戏》, 因此他获得了两项黄金时段艾美奖电视剧最佳男配角提名奖 Coster-Waldau 在其祖国丹麦出演了众多电影, 并斯堪的纳维亚, 包括《猎头》(2011) 和《一千次好夜》(2013). 在美国 , 他的处女作是在战争电影《黑鹰坠落》中担任角色 (2001), 扮演荣获荣誉勋章的加里·戈登。[2] 然后他扮演了一个角色。侦探在短暂播出的福克斯电视系列剧《新阿姆斯特丹》(2008) 中出现, 在2009年福克斯电视电影《虚拟世界》中, 最初的设想是试点。 , and
行动 2	Finish [支持]
声明 行动 1 观察 1	《怪奇物语》设定于印第安纳州布卢明顿。搜索 [怪奇物语] 《怪奇物语》是一部美国科幻恐怖剧情电视节目由Duffer Brothers创作的系列作品。 设定于1980年代, 主要在虚构城镇霍金斯 , 印第安纳 , 该系列围绕一系列小镇周围发生的神秘事件和超自然现象及其影响在儿童和成人角色的合奏中。
行动 2	Finish [反驳]
声明 行动 1 观察 1	Beautiful 在2003年登顶 Billboard Hot 100 排行榜第二名。搜索 [美丽] 未能找到 [美丽]。 类似: 'Beautiful' , 美丽 , “Beautiful” , 《美丽心灵》(电影) 《Beautiful (Christina Aguilera歌曲)》 , 生命是 “Beautiful” 。
行动 2 观察 2	搜索 [Beautiful (Christina Aguilera歌曲)] “Beautiful” 是美国歌手克里斯蒂娜·阿奎莱拉为她的第四张录音室专辑 Stripped (2002)
行动 3 观察 3	查找 [公告牌热榜 100] (Result 1 / 3) 这首歌在 Billboard Hot 100 排行榜上最高排名为第二名。美国, 获得了50万台出货量的金牌认证。
行动 4	Finish [信息不足]
Ask	判断是否存在支持或反驳主张的观察结果信息不足。 , 或者如果
声明 思考 REACT: 协同推理与行动于语言模型	尼古拉·科斯特-瓦尔道与福克斯广播公司合作。尼古拉·威廉·科斯特-瓦尔道出现在2009年福克斯电视电影中虚拟性, 因此他曾与福克斯广播公司合作。支持
声明 思考 REACT: 协同推理与行动于语言模型	《怪奇物语》设定于印第安纳州布卢明顿。《怪奇物语》发生在虚构的印第安纳州霍金斯市, 而不是在印第安纳州布卢明顿 反驳

续页

Fever Prompts – Continued from previous page	
Claim	Beautiful reached number two on the Billboard Hot 100 in 2003.
Thought	The song peaked at number two on the Billboard Hot 100 in the United States, but not sure if it was in 2003.
Answer	NOT ENOUGH INFO
ReAct	Determine if there is Observation that SUPPORTS or REFUTES a Claim, or if there is NOT ENOUGH INFORMATION.
Claim	Nikolaj Coster-Waldau worked with the Fox Broadcasting Company.
Thought 1	I need to search Nikolaj Coster-Waldau and find if he has worked with the Fox Broadcasting Company.
Action 1	Search[Nikolaj Coster-Waldau]
Observation 1	Nikolaj William Coster-Waldau (born 27 July 1970) is a Danish actor and producer. He graduated from the Danish National School of Performing Arts in Copenhagen in 1993,[1] and had his breakthrough role in Denmark with the film Nightwatch (1994). He played Jaime Lannister in the HBO fantasy drama series Game of Thrones, for which he received two Primetime Emmy Award nominations for Outstanding Supporting Actor in a Drama Series.. Coster-Waldau has appeared in numerous films in his native Denmark and Scandinavia, including Headhunters (2011) and A Thousand Times Good Night (2013). In the U.S, his debut film role was in the war film Black Hawk Down (2001), playing Medal of Honor recipient Gary Gordon.[2] He then played a detective in the short-lived Fox television series New Amsterdam (2008), and appeared in the 2009 Fox television film Virtuality, originally intended as a pilot.
Thought 2	Because he "appeared in the 2009 Fox television film Virtuality", he should have worked with the Fox Broadcasting Company.
Action 2	Finish[SUPPORTS]
Claim	Stranger Things is set in Bloomington, Indiana.
Thought 1	I should search for Stranger Things, and see if it is set in Bloomington, Indiana.
Action 1	Search[Stranger Things]
Observation 1	Stranger Things is an American science fiction horror drama television series created by the Duffer Brothers. Set in the 1980s, primarily in the fictional town of Hawkins, Indiana, the series centers on a number of mysteries and supernatural events occurring around the town and their impact on an ensemble of child and adult characters.
Thought 2	The observation says that it is set in a "fictional town of Hawkins, Indiana", so it is not set in Bloomington.
Action 2	Finish[REFUTES]
Claim	Beautiful reached number two on the Billboard Hot 100 in 2003.
Thought 1	I need to search the song Beautiful and find if it reached number two on the Billboard Hot 100 in 2003.
Action 1	Search[Beautiful]
Observation 1	Could not find [Beautiful]. Similar: [‘Beautiful’, ‘Beautiful, Beautiful’, ‘A Beautiful Mind (film)’, ‘Beautiful (Christina Aguilera song)’, ‘Life Is Beautiful’].
Thought 2	From suggestions, I should search "Beautiful (Christina Aguilera song)" to find the song.
Action 2	Search[Beautiful (Christina Aguilera song)]
Observation 2	"Beautiful" is a song recorded by American singer Christina Aguilera for her fourth studio album, Stripped (2002).
Thought 3	It does not mention Billboard, so I need to look up "Billboard Hot 100" to find if it reached number two on it in 2003.
Action 3	Lookup[Billboard Hot 100]
Observation 3	(Result 1 / 3) The song peaked at number two on the Billboard Hot 100 in the United States, where it was certified Gold for 500,000 units shipped.
Thought 4	It only says the song peaked at number two on the Billboard Hot 100, but not if it was in 2003. I am not sure if this claim is true or not.
Action 4	Finish[NOT ENOUGH INFO]

Fever Prompts – 续前页	
声明 思考	Beautiful 在2003年登顶 Billboard Hot 100 排行榜第二名。 这首歌在美国Billboard Hot 100排行榜上最高排名第二。 但不确定是否是在2003年。
REACT: 协同推理与行动于语言模型 言模型	REACT: 协同推理与行动手册
ReAct	判断是否存在支持或反驳主张的观察结果 信息不足。 , 或者如果
声明 思考 1	尼古拉·科斯特-瓦尔道与福克斯广播公司合作。 我需要搜索Nikolaj Coster-Waldau, 查找他是否曾与<style id='1'>xxx</style>合作过。
行动 1 观察 1	搜索[尼古拉·科斯特-瓦尔道] 尼古拉·威廉·科斯特-瓦尔道（出生于1970年7月27日）是丹麦演员和制作者。他毕业于丹麦国家表演艺术学院 在哥本哈根的 1993,[1] , 并在丹麦取得了他的突破性角色 电影《夜巡》（1994）他在HBO奇幻剧中饰演詹姆·兰尼斯特 电视剧《权力的游戏》, 因此他获得了两项黄金时段艾美奖 电视剧最佳男配角提名奖 Coster-Waldau 在其祖国丹麦出演了众多电影, 并 斯堪的纳维亚, 包括《猎头》（2011）和《一千次良夜》 （2013）。在美国 , 他的处女作是在战争电影《黑鹰坠落》中担任角色 （2001）, 扮演荣获荣誉勋章的加里·戈登。[2] 然后他扮演了一个角色。 侦探在短暂播出的福克斯电视系列剧《新阿姆斯特丹》（2008）中 出现在2009年福克斯电视电影《虚拟世界》中, 最初的设想是 试点。 因为他 “出现在2009年福克斯电视电影《虚拟世界》中” , 他应该 曾与福克斯广播公司合作过。
思考 2	
行动 2	Finish[支持]
声明 思考 1	《怪奇物语》设定于印第安纳州布卢明顿。 我应该搜索《怪奇物语》, 看看是否设定在布卢明顿, 印第安纳。
行动 1 观察 1	搜索[怪奇物语] 《怪奇物语》是一部美国科幻恐怖剧情电视节目 由Duffer Brothers创作的系列作品。 设定于1980年代, 主要在 虚构城镇霍金斯 , 印第安纳, 该系列围绕一系列 小镇周围发生的神秘事件和超自然现象及其影响 在儿童和成人角色的合奏中。 观察指出, 它设定在一个“虚构的霍金斯镇” 印第安纳”, 因此并未设定在布卢明顿。
思考 2	
行动 2	Finish[反驳]
声明 思考 1	Beautiful 在2003年登顶 Billboard Hot 100 排行榜第二名。 我需要搜索歌曲《Beautiful》并查找它是否登顶至第二名。
行动 1 观察 1	2003年公告牌百强单曲榜 搜索[美丽] 未能找到[美丽]。 类似: ‘Beautiful’ , 美丽 , “Beautiful” , 《美丽心灵》（电影） 《Beautiful (Christina Aguilera歌曲)》 生命是 “Beautiful” 。
思考 2	根据建议, 我应搜索 “Beautiful (Christina Aguilera song)” 以 找歌曲。
行动 2 观察 2	搜索[Beautiful (Christina Aguilera歌曲)] “Beautiful” 是美国歌手克里斯蒂娜·阿奎莱拉为她的 第四张录音室专辑 Stripped (2002)
思考 3	它没有提及 Billboard, 因此我需要查找 “Billboard Hot 100” 以 查找它是否在2003年达到第二名。
行动 3 观察 3	查找[公告牌热榜 100] (Result 1/ 3) 这首歌在 Billboard Hot 100 排行榜上最高排名为第二名。 美国, 获得了50万台出货量的金牌认证。
思考 4	它只提到这首歌在Billboard Hot 100排行榜上排名第二, 但没有提及其他内容 如果是在2003年。 我不确定这个说法是否属实。
行动 4	Finish[信息不足]

C.3 WEBSHOP

Table 6: Prompts used for the Webshop experiments in Section 4. We use the prompt on the left for Act, and use the prompt on the right (add reasoning) for ReAct.

Instruction: i would like a 3 ounce bottle of bright citrus deodorant for sensitive skin, and price lower than 50.00 dollars	
Act	ReAct
Action: search [3 ounce bright citrus deodorant sensitive skin] Observation: [Back to Search] Page 1 (Total results: 50) [Next] [B078GWRC1J] Bright Citrus Deodorant by Earth Mama Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce \$10.99 [B078GTKVXY] Ginger Fresh Deodorant by Earth Mama Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce \$10.99 [B08KBVJ4XN] Barrel and Oak - Aluminum-Free Deodorant, Deodorant for Men , Essential Oil-Based Scent, 24-Hour Odor Protection, Cedar & Patchouli Blend, Gentle on Sensitive Skin (Mountain Sage, 2.7 oz, 2-Pack) \$15.95 Action: click [B078GWRC1J] Observation: [Back to Search] [Prev] scent [assorted scents][bright citrus][calming lavender][ginger [fresh] simply non-scents] size [travel set (4-pack)][3 ounce (pack of 1)][3-ounce (2 -pack)] Bright Citrus Deodorant by Earth Mama Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce Price: \$10.99 Rating: N.A. [Description] [Features] [Reviews] [Buy Now] Action: click[bright citrus] Observation: You have clicked bright citrus. Action: click[3 ounce (pack of 1)] Observation: You have clicked 3 ounce (pack of 1). Action: click[Buy Now]	Action: search [3 ounce bright citrus deodorant sensitive skin] Observation: [Back to Search] Page 1 (Total results: 50) [Next] [B078GWRC1J] Bright Citrus Deodorant by Earth Mama Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce \$10.99 [B078GTKVXY] Ginger Fresh Deodorant by Earth Mama Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce \$10.99 [B08KBVJ4XN] Barrel and Oak - Aluminum-Free Deodorant, Deodorant for Men , Essential Oil-Based Scent, 24-Hour Odor Protection, Cedar & Patchouli Blend, Gentle on Sensitive Skin (Mountain Sage, 2.7 oz, 2-Pack) \$15.95 Action: think[B078GWRC1J and B078GTKVXY are bright citrus deodorant less then 50 dollars. I can check B078GWRC1J first. Observation: OK. Action: click [B078GWRC1J] Observation: [Back to Search] [Prev] scent [assorted scents][bright citrus][calming lavender][ginger [fresh] simply non-scents] size [travel set (4-pack)][3 ounce (pack of 1)][3-ounce (2 -pack)] Bright Citrus Deodorant by Earth Mama Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce Price: \$10.99 Rating: N.A. [Description] [Features] [Reviews] [Buy Now] Action: think[For 3 ounce bottle of bright citrus deodorant for sensitive skin, the item has options 'bright citrus' and '3 ounce (pack of 1)' and seems good to buy.] Observation: OK. Action: click[bright citrus] Observation: You have clicked bright citrus. Action: click[3 ounce (pack of 1)] Observation: You have clicked 3 ounce (pack of 1). Action: click[Buy Now]

C.3 网店

表6：第4节中用于网络商店实验的提示。我们使用左侧的提示进行Act，并使用右侧的提示（添加推理）进行ReAct。

Instruction: 我想要一瓶3盎司的亮丽柑橘香型敏感肌肤用除臭剂，价格低于50.00美元	
Act	ReAct
Action: search [3 ounce bright citrus deodorant sensitive skin] 观察: 返回搜索 R E A C T：协同推理与行动于语言模型中的融合 C PROMPTS 下一步 B078GWRC1J Earth Mama 的亮丽柑橘除臭剂 天然安全敏感肌肤、孕期与哺乳期，含有有机成分金盏花 3盎司 \$10.99 B078GTKVXY Earth Mama 天然清新除臭剂 敏感肌肤、孕期与哺乳期，含有有机成分金盏花 3盎司 \$10.99 B08KBVJ4XN Barrel and Oak - 无铝除臭剂，男士除臭剂，精油基香氛，24小时气味防护，雪松 & Patchouli 混合物，温和适用于敏感肌肤（山鼠尾草，2.7盎司，双包装 \$15.95 Action: click [B078GWRC1J] 观察: 返回搜索 R E A C T：协同推理与行动于语言模型中的结合 C PROMPTS 香氛 [混合香氛][明亮的柑橘][舒缓的薰衣草][生姜 新鲜的 简单的非感官体验 size [旅行套装（4件装）][3盎司（1包）][3-盎司（2 -包）] Earth Mama 的亮丽柑橘除臭剂 天然安全敏感肌肤、孕期与哺乳期，含有有机成分金盏花 3盎司 价格: \$10.99 评级: 不适用 [描述] 特征 评论 立即购买 Action: click[bright citrus] Observation: 您已点击明亮的柑橘。 Action: click[3 盎司（1包）] Observation: 您已点击 3 盎司（1 包）。 Action: click[立即购买]	Action: search [3 ounce bright citrus deodorant sensitive skin] 观察: 返回搜索 R E A C T：协同推理与行动于语言模型中的融合 C PROMPTS 下一步 B078GWRC1J Earth Mama亮橙香体露 天然安全敏感肌肤、孕期与哺乳期，含有有机成分金盏花 3盎司 \$10.99 B078GTKVXY Earth Mama 天然清新除臭剂 敏感肌肤、孕期与哺乳期，含有有机成分金盏花 3盎司 \$10.99 B08KBVJ4XN Barrel and Oak - 无铝除臭剂，男士除臭剂，精油基香氛，24小时气味防护，雪松 & Patchouli 混合物，温和适用于敏感肌肤（山鼠尾草，2.7盎司，双包装 \$15.95 Action: think[B078GWRC1J 和 B078GTKVXY 是明亮的柑橘不超过50美元的除臭剂。我可以先查看B078GWRC1J。 观察: OK。 Action: click [B078GWRC1J] 观察: 返回搜索 R E A C T：协同推理与行动于语言模型中的结合 C PROMPTS 香氛 [混合香氛][明亮的柑橘][舒缓的薰衣草][生姜 新鲜的 简单的非感官体验 size [旅行套装（4件装）][3盎司（1包）][3-盎司（2 -包）] Earth Mama 的亮丽柑橘除臭剂 天然安全敏感肌肤、孕期与哺乳期，含有有机成分金盏花 3盎司 价格: \$10.99 评级: 不适用 [描述] 特征 评论 立即购买 Action: think[对于3盎司亮丽柑橘香体露的瓶子对于敏感肌肤，该产品具有“明亮柑橘”选项和“3盎司（1包）”看起来不错，可以购买。] 观察: OK。 Action: click[bright citrus] Observation: 您已点击明亮的柑橘。 Action: click[3盎司（1包）] Observation: 您已点击 3 盎司（1 包）。 Action: click[立即购买]

C.4 ALFWORLD

Table 7: An Act prompt on the ALFWorld clean task. No thoughts are provided.

You are in the middle of a room. Looking quickly around you, you see a cabinet 13, a cabinet 12, a cabinet 11, a cabinet 10, a cabinet 9, a cabinet 8, a cabinet 7, a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 1, a diningtable 1, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean lettuce in diningtable. > go to fridge 1 The fridge 1 is closed. > open fridge 1 You open the fridge 1. The fridge 1 is open. In it, you see a cup 3, a egg 2, a potato 3, and a potato 2. > go to diningtable 1 On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a lettuce 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1. > take lettuce 1 from diningtable 1 You pick up the lettuce 1 from the diningtable 1. > go to sinkbasin 1 On the sinkbasin 1, you see a apple 2, a ladle 2, a spoon 1, and a tomato 3. > clean lettuce 1 with sinkbasin 1 You clean the lettuce 1 using the sinkbasin 1. > think: Now I clean a lettuce (1). Next, I need to put it in/on diningtable 1. OK. > go to diningtable 1 On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1. > put lettuce 1 in/on diningtable 1 You put the lettuce 1 in/on the diningtable 1.
--

C.4 ALFWORLD

表7：关于ALFWorld清洁任务的Act提示。未提供思路。

你位于房间中央。迅速环顾四周，你看到一个13号橱柜、一个12号橱柜、一个11号橱柜、一个10号橱柜、一个9号橱柜、一个8号橱柜、一个7号橱柜、一个6号橱柜、一个5号橱柜、一个4号橱柜、一个3号橱柜、一个2号橱柜、一个1号橱柜，一个1号咖啡机，一个1号台面，一个1号餐桌，一个1号抽屉，一个1号冰箱，一个1号垃圾桶，一个1号微波炉，一个3号架子，一个2号架子，一个1号架子，一个1号水槽盆，一个4号炉灶灶头，一个3号炉灶灶头，一个2号炉灶灶头，一个1号炉灶灶头，以及一个1号烤面包机。你的任务是：将一片干净的生菜放在餐桌上。>前往冰箱1。冰箱1是关闭的。>打开冰箱1。你打开了冰箱1。冰箱1是打开的。在里面，你看到一个3号杯子、两个鸡蛋、三个土豆、两个土豆。>前往餐桌1。在餐桌1上，你看到一个1号苹果、一个1号面包、两个1号黄油刀、两个2号杯子、两个2号叉子、两个2号刀子、一个1号刀、一个1号汤勺、一个1号生菜、两个2号马克杯、一个1号平底锅、一个1号胡椒粉罐、三个3号铲子、两个2号番茄、一个1号番茄。>从餐桌1取出生菜1。你从餐桌1拿起了生菜1。>前往水槽盆1。在水槽盆1上，你看到一个2号苹果、一个2号汤勺、一个1号勺子、三个3号番茄。>用水槽盆1清洗生菜1。你用水槽盆1清洗了生菜1。>思考：现在我在清洗一片生菜（1）。接下来，我需要将它放在餐桌1上。好的。>前往餐桌1。在餐桌1上，你看到一个1号苹果、一个1号面包、两个1号黄油刀、两个2号杯子、两个2号叉子、两个2号刀子、一个1号刀、一个1号汤勺、一个1号番茄、两个2号番茄、一个1号平底锅、一个1号胡椒粉罐、三个3号铲子。>将生菜1放在餐桌1上。你将生菜1放在餐桌1上。
--

Table 8: An ReAct prompt on the ALFWorld clean task.

You are in the middle of a room. Looking quickly around you, you see a cabinet 13, a cabinet 12, a cabinet 11, a cabinet 10, a cabinet 9, a cabinet 8, a cabinet 7, a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 1, a diningtable 1, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean lettuce in diningtable. > think: To solve the task, I need to find and take a lettuce, then clean it with sinkbasin, then put it in diningtable. OK. > think: First I need to find a lettuce. A lettuce is more likely to appear in fridge (1), diningtable (1), sinkbasin (1), stoveburner (1-3), cabinet (1-13). I can check one by one, starting with fridge 1. OK. > go to fridge 1 The fridge 1 is closed. > open fridge 1 You open the fridge 1. The fridge 1 is open. In it, you see a cup 3, a egg 2, a potato 3, and a potato 2. > go to diningtable 1 On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a lettuce 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1. > think: Now I find a lettuce (1). Next, I need to take it. OK. > take lettuce 1 from diningtable 1 You pick up the lettuce 1 from the diningtable 1. > think: Now I take a lettuce (1). Next, I need to go to sinkbasin (1) and clean it. OK. > go to sinkbasin 1 On the sinkbasin 1, you see a apple 2, a ladle 2, a spoon 1, and a tomato 3. > clean lettuce 1 with sinkbasin 1 You clean the lettuce 1 using the sinkbasin 1. > think: Now I clean a lettuce (1). Next, I need to put it in/on diningtable 1. OK. > go to diningtable 1 On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1. > put lettuce 1 in/on diningtable 1 You put the lettuce 1 in/on the diningtable 1.
--

表8：ALFWorld清洁任务上的ReAct提示。

你位于房间的中间。快速环顾四周，你看到一个柜子13、一个柜子12、一个柜子11、一个柜子10、一个柜子9、一个柜子8、一个柜子7、一个柜子6、一个柜子5、一个柜子4、一个柜子3、一个柜子2、一个柜子1、一个咖啡机1、一个台面1、一个餐桌1、一个抽屉1、一个冰箱1、一个垃圾桶1、一个微波炉1、一个架子3、一个架子2、一个架子1、一个水槽盆1、一个炉灶火炉4、一个炉灶火炉3、一个炉灶火炉2、一个炉灶火炉1，以及一个烤面包机1。你的任务是：将一片干净的生菜放在餐桌上。> 思考：为了解决这个任务，我需要找到并取出一片生菜，然后用水槽盆清洗它，最后放在餐桌上。好的。> 思考：首先我需要找到一片生菜。生菜更可能出现在冰箱（1）、餐桌（1）、水槽盆（1）、炉灶火炉（1-3）、柜子（1-13）中。我可以逐个检查，从冰箱1开始。好的。> 前往冰箱1 冰箱1是关闭的。> 打开冰箱1 你打开了冰箱1。冰箱1是打开的。在里面，你看到一个杯子3、两个鸡蛋2、三个土豆3、两个土豆2。> 前往餐桌1 在餐桌1上，你看到一个苹果1、一块面包1、两个黄油刀2、两个杯子2、两个叉子2、两个刀子2、一个刀子1、一个汤勺1、一个生菜1、两个马克杯2、一个平底锅2、一个胡椒粉瓶1、一个铲子3、两个番茄2、一个番茄1。> 思考：现在我找到了生菜（1）。接下来，我需要将它取出来。好的。> 从餐桌1取出生菜1 你从餐桌1上拿起了生菜1。> 思考：现在我拿到了一片生菜（1）。接下来，我需要去水槽盆（1）清洗它。好的。> 前往水槽盆1 在水槽盆1上，你看到两个苹果2、两个汤勺2、一个勺子1、三个番茄3。> 用水槽盆1清洗生菜1 你用水槽盆1清洗了生菜1。> 思考：现在我清洗了一片生菜（1）。接下来，我需要将它放在餐桌上。好的。> 前往餐桌1 在餐桌1上，你看到一个苹果1、一块面包1、两个黄油刀2、两个杯子2、两个叉子2、两个刀子2、一个刀子1、一个汤勺1、两个马克杯2、一个平底锅2、一个胡椒粉瓶1、一个铲子3、两个番茄2、一个番茄1。> 将生菜1放在餐桌上 你将生菜1放在餐桌上。

Table 9: An ReAct-IM prompt on the ALFWorld clean task.

You are in the middle of a room. Looking quickly around you, you see a cabinet 13, a cabinet 12, a cabinet 11, a cabinet 10, a cabinet 9, a cabinet 8, a cabinet 7, a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 1, a diningtable 1, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean lettuce in diningtable. > think: To solve the task, I need to find and take a lettuce, then clean it with sinkbasin, then put it in diningtable. OK. > think: First I need to find a lettuce. OK. > go to fridge 1 The fridge 1 is closed. > think: First I need to find a lettuce. OK. > open fridge 1 You open the fridge 1. The fridge 1 is open. In it, you see a cup 3, a egg 2, a potato 3, and a potato 2. > think: First I need to find a lettuce. OK. > go to diningtable 1 On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a lettuce 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1. > think: First I need to find a lettuce. OK. > take lettuce 1 from diningtable 1 You pick up the lettuce 1 from the diningtable 1. > think: I need to clean this lettuce (1) using sinkbasin 1. OK. > go to sinkbasin 1 On the sinkbasin 1, you see a apple 2, a ladle 2, a spoon 1, and a tomato 3. > think: I need to clean this lettuce (1) using sinkbasin 1. OK. > clean lettuce 1 with sinkbasin 1 You clean the lettuce 1 using the sinkbasin 1. > think: I need to put this cleaned lettuce (1) in/on diningtable 1. OK. > go to diningtable 1 On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1. > think: I need to put this cleaned lettuce (1) in/on diningtable 1. OK. > put lettuce 1 in/on diningtable 1 You put the lettuce 1 in/on the diningtable 1.

D TRAJECTORIES

D.1 FEVER TRAJECTORIES

We provide some trajectories produced by ReAct, CoTand Acton several randomly selected examples from the FEVER dev set below. Search results are omitted to space.

表9: ALFWorld清洁任务上的ReAct-IM提示。

你位于房间的中间。快速环顾四周，你看到一个橱柜 13, 一个柜子 12, 一个柜子 11, 一个柜子 10, 一个柜子 9, 一个柜子 8, 一个柜子 7, 一个柜子 6, 一个柜子 5, 一个柜子 4, 一个柜子 3, 一个柜子 2, 一个柜子 1, 咖啡机 1、操作台 1、餐桌 1、抽屉 1、冰箱 1、垃圾桶 1、微波炉 1, 架子 3, 架子 2, 架子 1, 水槽盆 1, a 炉灶燃烧器 4, 炉灶灶头 3, 炉灶灶头 2, 炉灶灶头 1, 和一个烤面包机 1. 您的任务是: 将一片干净的生菜放在餐桌上。 思考: 为了解决该任务, 我需要找到并拿一片生菜, 然后清理它, 使用水槽, 然后放在餐桌上。 OK. 思考: 首先我需要找到一片生菜。 OK. 前往冰箱 1 冰箱 1 关闭。 思考: 首先我需要找到一片生菜。 OK. 打开冰箱 1 你打开冰箱 1. 冰箱 1 已打开。在其中, 你看到一个杯子 3, 一个鸡蛋 2, 一个土豆3, 以及土豆2。 思考: 首先我需要找到一片生菜。 OK. 前往餐桌 1 在餐桌1上, 你看到一个苹果1, 一块面包1, 一把黄油刀2, 一个杯子2, 一个 fork 2, 刀 2, 刀 1, 汤勺 1, 生菜 1, 马克杯 2, 马克杯 1, 平底锅 2, a peppershaker 1、刮刀 3、番茄 2 和番茄 1。 思考: 首先我需要找到一片生菜。 OK. 取餐桌 1 上的生菜 1 你从餐桌 1 上拿起生菜 1。 思考: 我需要用水槽1清洗这片生菜 (1) 。 OK. 前往水槽盆1 在水槽盆1上, 你看到一个苹果2, 一个汤勺2, 一个勺子1, 以及一个番茄3。 思考: 我需要用水槽1清洗这片生菜 (1) 。 OK. 用水槽盆清洗生菜1 你用水槽盆1清洗生菜1。 思考: 我需要将这份清洗过的生菜 (1) 放在餐桌1上。 OK. 前往餐桌 1 在餐桌1上, 你看到一个苹果1, 一个面包1, 一把黄油刀2, 一个杯子2, 一把叉子 2, 把刀 2, 把刀 1, 汤勺 1, 马克杯 2, 马克杯 1, 平底锅 2, 胡椒粉罐 1, a spatula 3, 一个番茄 2, 以及一个番茄 1。 思考: 我需要将这份清洗过的生菜 (1) 放在餐桌1上。 OK. 将生菜 1 放在/放在餐桌 1 上 你将生菜1放在餐桌1上。
--

D 轨迹

D.1 发热轨迹

我们提供了由ReAct、CoT和Act在以下从FEVER开发集随机选择的几个示例中生成的部分轨迹。搜索结果为节省空间而省略。

Example 2491 (gt: REFUTES)	
ReAct	
Claim:	Bermuda Triangle is a loosely-defined region in the Pacific Ocean.
Thought 1:	I need to search Bermuda Triangle and find if it is in the Pacific Ocean.
Action 1:	Search[Bermuda Triangle]
Observation 1:	Reportedly haunted locations...
Thought 2:	The observation says that it is in the western part of the North Atlantic Ocean, so it is not in the Pacific Ocean.
Action 2:	Finish[REFUTES]
Observation 2:	Episode finished
Act	
Claim:	Bermuda Triangle is a loosely-defined region in the Pacific Ocean.
Action 1	: Search[Bermuda Triangle]
Observation 1	: Reportedly haunted locations:...
Action 2:	Finish[REFUTES]
Observation 2:	Episode finished
Cot	
Claim:	Bermuda Triangle is a loosely-defined region in the Pacific Ocean.
Thought:	Let’s think step by step. The Bermuda Triangle is a loosely-defined region in the Atlantic Ocean. The Pacific Ocean is not the Atlantic Ocean.
Answer:	REFUTES

Example 5908 (gt: SUPPORTS)	
ReAct	
Claim:	Princess Mononoke is a film.
Thought 1:	I need to search Princess Mononoke and find if it is a film.
Action 1:	Search[Princess Mononoke]
Observation 1:	Princess Mononoke ...
Thought 2:	From the observation, it says that Princess Mononoke is a film.
Action 2:	Finish[SUPPORTS]
Observation 2:	Episode finished
Act	
Claim:	Princess Mononoke is a film.
Action 1:	Search[Princess Mononoke]
Observation 1:	Princess Mononoke ...
Action 2:	Finish[SUPPORTS]
Observation 2:	Episode finished
Cot	
Claim:	Princess Mononoke is a film.
Thought:	Let’s think step by step. Princess Mononoke is a movie, and movies are films.
Answer:	SUPPORTS

Example 1951 (gt: REFUTES)	
ReAct	
Claim:	Soyuz was part of the American space program.
Thought 1:	I need to search Soyuz and find if it was part of the American space program.
Action 1:	Search[Soyuz]
Observation 1:	Soyuz is a transliteration...
Thought 2:	It does not mention the American space program, so I need to search "American space program" to find if it was part of it.
Action 2:	Search[American space program]
Observation 2:	The space policy...
Thought 3:	It does not mention Soyuz, so I am not sure if this claim is true or not.
Action 3:	Finish[NOT ENOUGH INFO]
Observation 3:	Episode finished
Act	
Claim:	Soyuz was part of the American space program.
Action 1:	Search[Soyuz]
Observation 1:	Soyuz is a transliteration...
Action 2:	Finish[NOT ENOUGH INFO]
Observation 2:	Episode finished
Cot	
Claim:	Soyuz was part of the American space program.
Thought:	Let’s think step by step. Soyuz is a Russian spacecraft. The American space program is NASA. NASA and Russia have worked together on the International Space Station.
Answer:	SUPPORTS

示例 2491（原文：REFUTES）	
ReAct	
声明:	百慕大三角是太平洋中一个定义较为宽松的区域。
思考 1:	我需要搜索百慕大三角，看看它是否位于太平洋。
行动 1:	搜索[百慕大三角]
观察 1:	据报道的闹鬼地点……
思考 2:	观察表明，它位于北大西洋的西部地区海洋，所以它不在太平洋。
行动 2:	Finish(反驳)
观察 2:	第九集：在 ALFWorld 清洁任务中的 ReAct-IM 提示
Act	
声明:	百慕大三角是太平洋中一个定义较为宽松的区域。
行动 1	: 搜索[百慕大三角]
观察 1	: 据报道的闹鬼地点: ...
行动 2:	Finish(反驳)
观察 2:	第九集：在 ALFWorld 清洁任务中的 ReAct-IM 提示
Cot	
声明:	百慕大三角是太平洋中一个定义较为宽松的区域。
思考:	让我们一步步思考。百慕大三角是一个定义较为宽松的区域太平洋不是大西洋。
REA CT：在语言模型中协同推理与行动	反驳
Table 9: ALFWorld清洁任务上的 ReAct-IM提示	

示例 5908（gt：支持）	
ReAct	
声明:	《幽灵公主》是一部电影。
思考 1:	我需要搜索《幽灵公主》，并确认它是否是一部电影。
行动 1:	搜索[《幽灵公主》]
观察 1:	《幽灵公主》...
思考 2:	根据观察，指出《幽灵公主》是一部电影。
行动 2:	Finish(支持)
观察 2:	第九集：在 ALFWorld 清洁任务中的 ReAct-IM 提示
Act	
声明:	《幽灵公主》是一部电影。
行动 1:	搜索[《幽灵公主》]
观察 1:	《幽灵公主》...
行动 2:	Finish(支持)
观察 2:	第九集：在 ALFWorld 清洁任务中的 ReAct-IM 提示
Cot	
声明:	《幽灵公主》是一部电影。
思考:	让我们一步步思考。《幽灵公主》是一部电影，而电影是影片。支持
REA CT：在语言模型中协同推理与行动	支持
Table 9: ALFWorld清洁任务上的 ReAct-IM提示	

示例 1951（原文：REFUTES）	
ReAct	
声明:	苏联号曾是美国太空计划的一部分。
思考 1:	我需要搜索Soyuz，查明它是否曾是美国太空任务的一部分
行动 1:	程序
观察 1:	Search[Soyuz]
思考 2:	Soyuz 是一个音译... 它没有提及美国太空计划，因此我需要进行搜索“美国太空计划”以查明其是否为其一部分。
行动 2:	Search[美国太空计划]
观察 2:	空间政策...
思考 3:	它没有提及联盟号，因此我不确定这个说法是否属实。
行动 3:	Finish(信息不足)
观察 3:	第九集：在 ALFWorld 清洁任务中的 ReAct-IM 提示
Act	
声明:	苏联号曾是美国太空计划的一部分。
行动 1:	Search[Soyuz]
观察 1:	Soyuz 是一个音译...
行动 2:	Finish(信息不足)
观察 2:	第九集：在 ALFWorld 清洁任务中的 ReAct-IM 提示
Cot	
声明:	苏联号曾是美国太空计划的一部分。
思考:	让我们一步步思考。Soyuz 是一款俄罗斯航天器。The American 空间项目由美国国家航空航天局（NASA）负责。NASA与俄罗斯在此方面曾合作过。国际空间站
REA CT：在语言模型中协同推理与行动	支持
Table 9: ALFWorld清洁任务上的 ReAct-IM提示	

Example 3208 (gt: REFUTES)	
ReAct	
Claim:	Reign Over Me is an American film made in 2010.
Thought 1:	I need to search for Reign Over Me and find if it is an American film made in 2010.
Action 1:	Search[Reign Over Me]
Observation 1:	Reign Over Me ...
Thought 2:	The observation says that it is an American film made in 2007, so it is not made in 2010.
Action 2:	Finish[REFUTES]
Observation 2:	Episode finished
Act	
Claim:	Reign Over Me is an American film made in 2010.
Action 1:	Search[Reign Over Me]
Action 2:	Finish[SUPPORTS]
Observation 2:	Episode finished
CoT	
Claim:	Reign Over Me is an American film made in 2010.
Thought:	Let’s think step by step. First, is Reign Over Me an American film? Yes, it is. Second, was it made in 2010? Yes, it was.
Answer:	SUPPORTS

D.2 ALFWORLD TRAJECTORIES

We provide an example trajectory of ReAct, Act, and ReAct-IM on the same ALFWorld game, pick_clean_then_place_in_recep-Knife-None-CounterTop-10/trial_T20190909_110445_675754.

D.2.1 REACT TRAJECTORY

ReAct is able to correctly find the knife, clean it, and then place it on a countertop.

示例 3208 (原文: REFUTES)	
ReAct	
声明:	Reign Over Me 是一部于2010年制作的美国电影。
思考 1:	我需要搜索《Reign Over Me》，并查明它是否是一部美国电影。在2010年。
行动 1:	Search [Reign Over Me]
观察 1:	统治我……
思考 2:	观察显示，这是一部于2007年拍摄的美国电影，因此不是于2010年制造。
行动 2:	Finish [反驳]
观察 2:	第九集：在 ALFWorld 清洁任务中的 ReAct-IM 提示
Act	
声明:	Reign Over Me 是一部于2010年制作的美国电影。
行动 1:	Search [Reign Over Me]
行动 2:	Finish [支持]
观察 2:	第九集：在 ALFWorld 清洁任务中的 ReAct-IM 提示
CoT	
声明:	Reign Over Me 是一部于2010年制作的美国电影。
思考:	让我们一步步思考。First, is Reign Over Me an American film? Yes, 它是。第二，它是在2010年制作的吗？Yes，它曾是。
REACT:	支持

在语言模型中
协同推理与行
动 Table 9:
ALFWorld清
洁任务上的
ReAct-IM提
示

D.2 ALFWORLD 轨迹

我们提供了在相同ALFWorld游戏中ReAct、Act和ReAct-IM的示例轨迹，pick_clean_then_place_in_recep-Knife-None-CounterTop-10/ 试验 T20190909 110445 675754. _ _

D.2.1 REACT 轨迹

ReAct 能够正确找到刀具，清洗它，然后将其放置在台面上。

ReAct ALFWorld trajectory
You are in the middle of a room. Looking quickly around you, you see a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 3, a countertop 2, a countertop 1, a drawer 3, a drawer 2, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean knife in countertop. > think: To solve the task, I need to find and take a knife, then clean it with sinkbasin, then put it in countertop. OK. > think: First I need to find a knife. A knife is more likely to appear in cabinet (1-6), drawer (1-3), countertop (1-3), fridge (1), garbagecan (1), shelf (1-3), sinkbasin (1), stoveburner (1-4), toaster (1). I can check one by one, starting with cabinet 1. OK. > go to cabinet 1 On the cabinet 1, you see a bowl 1. > go to cabinet 2 The cabinet 2 is closed. > go to cabinet 3 On the cabinet 3, you see a glassbottle 1. > go to cabinet 4 On the cabinet 4, you see a mug 1. > go to cabinet 5 The cabinet 5 is closed. > go to cabinet 6 The cabinet 6 is closed. > go to drawer 1 The drawer 1 is closed. > go to drawer 2 The drawer 2 is closed. > go to drawer 3 The drawer 3 is closed. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a knife 1, a plate 2, a potato 3, and a statue 1. > think: Now I find a knife (1). Next, I need to take it. OK. > take knife 1 from countertop 2 You pick up the knife 1 from the countertop 2. > think: Now I take a knife (1). Next, I need to go to sinkbasin (1) and clean it. OK. > go to sinkbasin 1 On the sinkbasin 1, you see a fork 3, a lettuce 3, and a spatula 2. > clean knife 1 with sinkbasin 1 You clean the knife 1 using the sinkbasin 1. > think: Now I clean a knife (1). Next, I need to put it in/on countertop 1. OK. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > put knife 1 in/on countertop 1 You put the knife 1 in/on the countertop 1.

D.2.2 ACT TRAJECTORY

Act is able to correctly find the knife, but tries to clean it using a sinkbasin before actually going to the sinkbasin. From there on, the model is unable to recover, and is stuck repeating the same sequence of commands. With ReAct, an existing thought would have concretely identified that the knife had been taken, and that the next subgoal was to go to a sinkbasin to clean it.

ReAct ALFWorld 轨迹
你位于房间的中间。快速环顾四周，你看到一个橱柜6、一个橱柜5、一个橱柜4、一个橱柜3、一个橱柜2、一个橱柜1、一台咖啡机1、一个台面3、一个台面2、一个台面1、一个抽屉3、一个抽屉2、一个抽屉1、一个冰箱1、一个垃圾桶1、一个微波炉1、一个架子3、一个架子2、一个架子1、一个水槽盆1、一个炉灶灶头4、一个炉灶灶头3、一个炉灶灶头2、一个炉灶灶头1，以及一个烤面包机1。你的任务是：将一把干净的刀放在台面上。> 思考：为了解决这个任务，我需要找到并取出一把刀，然后用水槽盆清洗它，最后放在台面上。好的。> 思考：首先我需要找到一把刀。刀更可能出现在橱柜（1-6）、抽屉（1-3）、台面（1-3）、冰箱（1）、垃圾桶（1）、架子（1-3）、水槽盆（1）、炉灶灶头（1-4）或烤面包机（1）中。我可以逐个检查，从橱柜1开始。好的。> 前往橱柜1 在橱柜1上，你看到一个碗1。 > 前往橱柜2 橱柜2是关闭的。> 前往橱柜3 在橱柜3上，你看到一个玻璃瓶1。> 前往橱柜4 在橱柜4上，你看到一个马克杯1。> 前往橱柜5 橱柜5是关闭的。> 前往橱柜6 橱柜6是关闭的。> 前往抽屉1 抽屉1是关闭的。> 前往抽屉2 抽屉2是关闭的。> 前往抽屉3 抽屉3是关闭的。> 前往台面1 在台面1上，你看到生菜2、马克杯2、胡椒粉瓶1和汤匙2。> 前往台面2 在台面2上，你看到碗1、洗碗海绵1、玻璃瓶3、刀1、盘子2、土豆3和雕像1。> 思考：现在我找到了刀（1）。接下来，我需要取出它。好的。> 从台面2取出刀1 你从台面2拿起了刀1。> 思考：现在我拿到了一把刀（1）。接下来，我需要前往水槽盆（1）并清洗它。好的。> 前往水槽盆1 在水槽盆1上，你看到叉子3、生菜3和铲子2。> 用水槽盆1清洗刀1 你用水槽盆1清洗了刀1。> 思考：现在我清洗了一把刀（1）。接下来，我需要将它放在台面1上。好的。> 前往台面1 在台面1上，你看到生菜2、马克杯2、胡椒粉瓶1和汤匙2。> 将刀1放在台面1上 你将刀1放在台面1上。

D.2.2 ACT 轨迹

Act 能够正确找到刀，但在实际前往水槽之前，试图用水槽盆清洗它。从此之后，模型无法恢复，陷入重复执行相同指令的循环。采用 ReAct 时，已有的思考会具体识别出刀已被取走，下一步子目标是前往水槽盆进行清洗。

Act ALFWorld trajectory
You are in the middle of a room. Looking quickly around you, you see a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 3, a countertop 2, a countertop 1, a drawer 3, a drawer 2, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean knife in countertop. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a knife 1, a plate 2, a potato 3, and a statue 1. > take knife 1 from countertop 2 You pick up the knife 1 from the countertop 2. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > clean knife 1 with sinkbasin 1 Nothing happens. > go to countertop 3 On the countertop 3, you see a bread 3, a butterknife 2, a cellphone 1, a creditcard 1, a fork 2, a houseplant 1, a knife 2, a spatula 1, a statue 3, a tomato 3, a tomato 2, a tomato 1, and a vase 2. > take knife 2 from countertop 3 Nothing happens. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a plate 2, a potato 3, and a statue 1. > take knife 1 from countertop 2 Nothing happens. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > clean knife 1 with sinkbasin 1 Nothing happens. > go to countertop 3 On the countertop 3, you see a bread 3, a butterknife 2, a cellphone 1, a creditcard 1, a fork 2, a houseplant 1, a knife 2, a spatula 1, a statue 3, a tomato 3, a tomato 2, a tomato 1, and a vase 2. > take knife 2 from countertop 3 Nothing happens. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a plate 2, a potato 3, and a statue 1. > take knife 1 from countertop 2 Nothing happens. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > clean knife 1 with sinkbasin 1 Nothing happens. > go to countertop 3 On the countertop 3, you see a bread 3, a butterknife 2, a cellphone 1, a creditcard 1, a fork 2, a houseplant 1, a knife 2, a spatula 1, a statue 3, a tomato 3, a tomato 2, a tomato 1, and a vase 2. > take knife 2 from countertop 3 Nothing happens. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a plate 2, a potato 3, and a statue 1. > take knife 1 from countertop 2 Nothing happens. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > clean knife 1 with sinkbasin 1 Nothing happens.

D.2.3 REACT-IM TRAJECTORY

ReAct-IM is able to correctly find the knife, but is unable to clean it as well. The LLM then gets stuck trying to place this knife on a countertop. One possible issue is the incorrect thought think: I need to find a clean knife., which seems to trick the LLM in believing that the knife is already cleaned. In ReAct, this sequence of subgoals would be concretely captured by each thought.

Act ALFWorld 轨迹
你位于房间的中部。快速环顾四周，你看到一个柜子6, a 柜子 5, 一个柜子 4, 一个柜子 3, 一个柜子 2, 一个柜子 1, 一台咖啡机 1, a 台面 3, 台面 2, 台面 1, 抽屉 3, 抽屉 2, 抽屉 1, 抽屉 冰箱 1, 垃圾桶 1, 微波炉 1, 架子 3, 架子 2, 架子 1, 水槽盆 1, 炉灶燃烧器 4, 炉灶燃烧器 3, 灶具炉头 2, 炉灶燃烧器 1, 和一个烤面包机 1. 您的任务是: 将一把干净的刀放在台面上。 前往操作台1 在台面1上, 你看到生菜2、马克杯2、胡椒粉罐1和汤匙2。 前往台面2 在台面2上, 你看到一个杯子1, 一个洗碗海绵1, 一个玻璃瓶3, 一把刀1, 一个 plate 2, a potato 3 and a statue 1。 从台面2取刀1 你从台面 2 上拿起刀具 1。 前往操作台1 在台面1上, 你看到生菜2、马克杯2、胡椒粉罐1和汤匙2。 用水槽清洗刀具1 没有任何反应。 前往操作台3 在操作台3上, 你看到一个面包3, 一把黄油刀2, 一部手机1, 一张信用卡 1、一个叉子 2、一盆室内植物 1、一把刀 2、一把铲子 1、一尊雕像 3、一个番茄 2, 一个番茄 1, 和一个花瓶 2. 从台面 3 取刀 2 没有任何反应。 前往台面2 在台面2上, 你看到一个杯子1, 一个洗碗海绵1, 一个玻璃瓶3, 一个盘子2, 一个土豆 3 和一尊雕像 1。 从台面2取刀1 没有任何反应。 前往操作台1 在台面1上, 你看到生菜2、马克杯2、胡椒粉罐1和汤匙2。 用水槽清洗刀具1 没有任何反应。 前往操作台3 在操作台3上, 你看到一个面包3, 一把黄油刀2, 一部手机1, 一张信用卡 1、一个叉子 2、一盆室内植物 1、一把刀 2、一把铲子 1、一尊雕像 3、一个番茄 3、一个番茄 2, 一个番茄 1, 和一个花瓶 2. 从台面 3 取刀 2 没有任何反应。 前往台面2 在台面2上, 你看到一个杯子1, 一个洗碗海绵1, 一个玻璃瓶3, 一个盘子2, 一个土豆 3 和一尊雕像 1。 从台面2取刀1 没有任何反应。 前往操作台1 在台面1上, 你看到生菜2、马克杯2、胡椒粉罐1和汤匙2。 用水槽清洗刀具1 没有任何反应。 前往操作台3 在操作台3上, 你看到一个面包3, 一把黄油刀2, 一部手机1, 一张信用卡 1、一个叉子 2、一盆室内植物 1、一把刀 2、一把铲子 1、一尊雕像 3、一个番茄 3、一个番茄 2, 一个番茄 1, 和一个花瓶 2. 从台面 3 取刀 2 没有任何反应。 前往台面2 在台面2上, 你看到一个杯子1, 一个洗碗海绵1, 一个玻璃瓶3, 一个盘子2, 一个土豆 3 和一尊雕像 1。 从台面2取刀1 没有任何反应。 前往操作台1 在台面1上, 你看到生菜2、马克杯2、胡椒粉罐1和汤匙2。 用水槽清洗刀具1 没有任何反应。

D.2.3 REACT-IM 轨迹

ReAct-IM 能够正确找到刀具，但在清洁方面表现不佳。随后，LLM 在尝试将这把刀放置在台面上时陷入困境。一个可能的问题是错误的思考：我需要找到一把干净的刀具，这似乎误导了 LLM，使其相信刀已经被清洁。在 ReAct 中，这一系列子目标会被每个思考具体地捕捉。

ReAct-IM ALFWorld trajectory
You are in the middle of a room. Looking quickly around you, you see a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 3, a countertop 2, a countertop 1, a drawer 3, a drawer 2, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean knife in countertop. > think: To solve the task, I need to find and take a clean knife, then put it in countertop. OK. > think: I need to find a clean knife. OK. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > think: I need to find a clean knife. OK. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a knife 1, a plate 2, a potato 3, and a statue 1. > think: I need to find a clean knife. OK. > take knife 1 from countertop 2 You pick up the knife 1 from the countertop 2. > think: I need to put this knife (1) in/on countertop 1. OK. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > think: I need to put this knife (1) in/on countertop 1. OK. > put knife 1 in/on countertop 1 You put the knife 1 in/on the countertop 1. > think: I need to put this knife (1) in/on countertop 1. OK. > put knife 1 in/on countertop 1 Nothing happens. > think: I need to put this knife (1) in/on countertop 1. OK. > put knife 1 in/on countertop 1 Nothing happens. > think: I need to put this knife (1) in/on countertop 1. OK. > put knife 1 in/on countertop 1 Nothing happens. > think: I need to put this knife (1) in/on countertop 1. OK. > put knife 1 in/on countertop 1 Nothing happens. > think: I need to put this knife (1) in/on countertop 1. OK.

ReAct-IM ALFWorld 轨迹
你位于房间的中间。快速环顾四周，你看到一个橱柜6、一个橱柜5、一个橱柜4、一个橱柜3、一个橱柜2、一个橱柜1、一台咖啡机1、一个台面3、一个台面2、一个台面1、一个抽屉3、一个抽屉2、一个抽屉1、一个冰箱1、一个垃圾桶1、一个微波炉1、一个架子3、一个架子2、一个架子1、一个水槽盆1、一个炉灶灶头4、一个炉灶灶头3、一个炉灶灶头2、一个炉灶灶头1，以及一个烤面包机1。你的任务是：将一把干净的刀放在台面上。> 思考：为完成任务，我需要找到并取出一把干净的刀，然后将其放在台面上。好的。>思考：我需要找到一把干净的刀。好的。> 前往台面1 在台面1上，你看到生菜2、杯子2、胡椒粉瓶1和勺子2。> 思考：我需要找到一把干净的刀。好的。> 前往台面2 在台面2上，你看到杯子1、洗碗海绵1、玻璃瓶3、刀子1、盘子2、土豆3和雕像1。> 思考：我需要找到一把干净的刀。好的。> 从台面2取出刀子1 你从台面2拿起刀子1。> 思考：我需要将这把刀（1）放在台面1上。好的。> 前往台面1 在台面1上，你看到生菜2、杯子2、胡椒粉瓶1和勺子2。> 思考：我需要将这把刀（1）放在台面1上。好的。> 将刀子1放在台面1上 你将刀子1放在台面1上。> 思考：我需要将这把刀（1）放在台面1上。好的。> 将刀子1放在台面1上 无任何反应。> 思考：我需要将这把刀（1）放在台面1上。好的。> 将刀子1放在台面1上 无任何反应。> 思考：我需要将这把刀（1）放在台面1上。好的。> 将刀子1放在台面1上 无任何反应。> 思考：我需要将这把刀（1）放在台面1上。好的。> 将刀子1放在台面1上 无任何反应。> 思考：我需要将这把刀（1）放在台面1上。好的。> 将刀子1放在台面1上 无任何反应。> 思考：我需要将这把刀（1）放在台面1上。好的。> 将刀子1放在台面1上 无任何反应。> 思考：我需要将这把刀（1）放在台面1上。好的。

D.3 WEBSHOP TRAJECTORIES

Table 10: Example trajectories for Webshop predicted by `Act` and `ReAct`. Compared to `Act`, `ReAct` uses reasoning to find products that satisfy all target attributes.

Instruction: get me a sixteen pack of apple cinnamon freeze dried banana chips, and price lower than 50.00 dollars	
Act	ReAct
Action: search[sixteen pack apple cinnamon freeze dried banana chips] Observation: [Back to Search] Page 1 (Total results: 50) [Next] [B00611VFZE] Brothers-ALL-Natural Fruit Crisps, Strawberry Banana, 0.42 - Ounce (Pack of 100) \$85.0 [B096H2P6G2] Moon Fruit Freeze Dried Fruit Snacks. Fruit Snacks for Kids - (Variety Pack) \$18.99 [B092JLLYK6] Nature's Turn Freeze-Dried Fruit Snacks - Banana Crisps - Perfect For School Lunches or an On-The-Go Snack - No Sugar Added, Non GMO, Gluten Free, Nothing Artificial (0.53oz) 6-Pack \$12.99 Action: click[B00611VFZE] Observation: [Back to Search] [Prev] flavor name [asian pear][banana][fuji apple & cinnamon][strawberry banana] Brothers-ALL-Natural Fruit Crisps, Strawberry Banana, 0.42 Ounce (Pack of 100) Price: \$85.0 Rating: N.A. [Description] [Features] [Reviews] [Buy Now] Action: click[Buy Now]	Action: search[sixteen pack apple cinnamon freeze dried banana chips] Observation: [Back to Search] Page 1 (Total results: 50) [Next] [B00611VFZE] Brothers-ALL-Natural Fruit Crisps, Strawberry Banana, 0.42 - Ounce (Pack of 100) \$85.0 [B096H2P6G2] Moon Fruit Freeze Dried Fruit Snacks. Fruit Snacks for Kids - (Variety Pack) \$18.99 [B092JLLYK6] Nature's Turn Freeze-Dried Fruit Snacks - Banana Crisps - Perfect For School Lunches or an On-The-Go Snack - No Sugar Added, Non GMO, Gluten Free, Nothing Artificial (0.53oz) 6-Pack \$12.99 Action: think[B00611VFZE is strawberry banana, not apple cinnamon. B096H2P6G2 is fruit snacks, not freeze dried banana chips. B092JLLYK6 is banana crisps, not apple cinnamon. I can check B092JLLYK6 first. Observation: OK. Action: click[B092JLLYK6] Observation: [Back to Search] [Prev] flavor name [apple][apple cinnamon][banana][cantaloupe][peach][pear][strawberry][strawberry banana][sampler variety pack][mega variety pack][orchard variety pack] size [0.53 ounce (pack of 6)][0.53 ounce (pack of 8)][0.53 ounce (pack of 16)] Nature's Turn Freeze-Dried Fruit Snacks - Banana Crisps - Perfect For School Lunches or an On-The-Go Snack - No Sugar Added, Non GMO, Gluten Free, Nothing Artificial (0.53oz) 6-Pack Price: \$12.99 Rating: N.A. [Description] [Features] [Reviews] [Buy Now] Action: think[For sixteen pack of apple cinnamon freeze dried banana chips, the item has options 'apple cinnamon' and '0.53 ounce (pack of 16)' and seems good to buy.] Observation: OK. Action: click[apple cinnamon] Observation: You have clicked apple cinnamon. Action: click[0.53 ounce (pack of 16)] Observation: You have clicked 0.53 ounce (pack of 16). Action: click[Buy Now]
Score: 0.125	Score: 1.0

D.3 网店轨迹

表10: `Act`和`ReAct`预测的Webshop示例轨迹。与`Act`相比，`ReAct`通过推理找到满足所有目标属性的产品。

Instruction: 获取一箱十六包苹果肉桂风干香蕉片，价格低于50.00美元	
Act	ReAct
Action: search[十六包苹果肉桂冻干香蕉 chips] 观察: 返回搜索 第1页（共50个结果） 下一步 REA CT：协同推理与行动于语言模型 Brothers-ALL-Natural 水果脆片，草莓香蕉，0.42 盎司（100包装） REA CT：协同推理与行动于语言模型 [B096H2P6G2] Moon Fruit 冻干水果零食。儿童水果零食 - （多样包） \$18.99 B092JLLYK 自然之转冻干水果零食 - 香蕉脆片 - 完美用于学校午餐或随身零食——无添加糖 非转基因，无麸质，无人工添加（0.53盎司）6包装 \$12.99 Action: click[B00611VFZE] 观察: 返回搜索 Prev 口味名称 [亚洲梨][香蕉][富士苹果与肉桂][草莓 banana] Brothers-ALL-Natural 水果脆片，草莓香蕉，0.42 盎司（100包装） 价格: \$85.0 评分: 不适用 [描述] 特征 评论 立即购买 Action: click[立即购买]	Action: search[十六包苹果肉桂冻干香蕉 chips] 观察: 返回搜索 第1页（共50个结果） 下一步 REA CT：协同推理与行动于语言模型 Brothers-ALL-Natural 水果脆片，草莓香蕉，0.42 盎司（100包装） REA CT：协同推理与行动于语言模型 [B096H2P6G2] Moon Fruit 冻干水果零食。儿童水果零食 - （多样包） \$18.99 B092JLLYK 自然之转冻干水果零食 - 香蕉脆片 - 完美用于学校午餐或随身零食——无添加糖 非转基因，无麸质，无人工添加（0.53盎司）6包装 \$12.99 Action: think[B00611VFZE 是草莓香蕉，不是苹果肉桂。B096H2P6G2 是水果零食，不是冻干香蕉片。B092JLLYK6 是香蕉脆片，不是苹果肉桂。我可以确认。B092JLLYK6 观察: OK。 Action: click[B092JLLYK6] 观察: 返回搜索 Prev 风味名称 [苹果][苹果肉桂][香蕉][哈密瓜][桃子][梨] 草莓[草莓香蕉][采样品种包][超级品种 pack][果园品种包] size [0.53 盎司（6包装）][0.53 盎司（8包装）][0.53 盎司（包装第16页）] 自然之转冻干水果零食 - 香蕉脆片 - 完美用于学校午餐或随身零食——无添加糖，非 GMO，gluten free，无人工添加（0.53盎司）6包装 价格: \$12.99 评分: 不适用 [描述] 特征 评论 立即购买 Action: think[对于十六包苹果肉桂冻干香蕉芯片，该项目有“苹果肉桂”和“0.53盎司（包装）”两个选项16)，看起来适合购买。] 观察: OK。 Action: click[apple cinnamon] Observation: 您已点击苹果肉桂。 Action: click[0.53 盎司（16包装）] Observation: 您已点击 0.53 盎司（16 包）。 Action: click[立即购买]
Score: 0.125	Score: 1.0

E MORE ANALYSIS

E.1 SUCCESS AND FAILURE MODES ANALYSIS

We provide some examples corresponding to the success and error mode analysis given in Sec. 3.3. Search results and non-representative steps are omitted to reduce space.

Success: True positive	
ReAct	
Question:	Author David Chanoff has collaborated with a U.S. Navy admiral who served as the ambassador to the United Kingdom under which President?
Thought 1:	I need to search David Chanoff and find the U.S. Navy admiral he collaborated with.
Action 1:	Search[David Chanoff]
Observation 1:	David Chanoff is a noted author...
Action 2:	Search[William J. Crowe]
Observation 2:	William J. Crowe is...
Thought 3:	William J. Crowe served as the ambassador to the United Kingdom under President Bill Clinton. So the answer is Bill Clinton.
Action 3:	Finish[Bill Clinton]
Observation 3:	Episode finished
CoT	
Question:	What was the name of the team that represented an Arizona university in the conference that covers a broad expanse of the western United States, including Arizona, California, and New Mexico?
Thought:	Let’s think step by step. The name of the team that represented an Arizona university in the conference that covers a broad expanse of the western United States, including Arizona, California, and New Mexico is Arizona State Sun Devils.
Answer:	Arizona State Sun Devils

Success: False positive	
ReAct	
Question:	Did the board game San Marco or About Time come out first?
Thought 1:	I need to search San Marco and About Time, and find which came out first.
Action 1:	Search[San Marco]
Observation 1:	San Marco is one of the ... The was built in venice.
Thought 2:	San Marco is a sestiere of Venice. I need to search About Time next.
Action 2:	Search[About Time]
Observation 2:	Could not find [About Time]. Similar: [...].
Thought 3:	About Time is a 2013 film. So San Marco came out first.
Action 3:	Finish[San Marco]
Observation 3:	Episode finished
CoT	
Question:	Was Vanderbilt University or Emory University founded first?
Thought:	Let’s think step by step. Vanderbilt University was founded in 1873. Emory University was founded in 1836. 1873 (Vanderbilt University) > 1836 (Emory University), so Vanderbilt University was founded first.
Answer:	Emory University

Failure: Reasoning error	
ReAct	
Question:	What was the middle name of the actress who was cast alongside Harvey Fierstein, Linda Hart, Dick Latessa, Clarke Thorell, Mary Bond Davis, Laura Bell Bundy, Matthew Morrison, Corey Reynolds, and Marissa Jaret Winokur in Hairspray?
Thought 1:	I need to search Harvey Fierstein, Linda Hart, Dick Latessa, Clarke Thorell, Mary Bond Davis, Laura Bell Bundy, Matthew Morrison, Corey Reynolds, and Marissa Jaret Winokur, find the actress that was cast alongside them in Hairspray, then find its middle name.
Action 1:	Search[Harvey Fierstein]
Observation 1:	Harvey Forbes Fierstein ...
Thought 2:	Harvey Fierstein is an actor, playwright, and screenwriter. I need to search Linda Hart next...
Action 2:	Search[Linda Hart]...
CoT	
Question:	Who is older Danny Green or James Worthy?
Thought:	Let’s think step by step. Danny Green was born in 1987. James Worthy was born in 1961. 1987 (Danny Green) > 1961 (James Worthy), so Danny Green is older.
Answer:	Danny Green

E 更多分析

E.1 成功与失败模式分析

我们提供一些与第3.3节中成功与失败模式分析相对应的示例。为节省空间，搜索结果和非代表性步骤省略。

成功：真正例	
ReAct	
REA CT：协同推理与行动于语言模型中的结合 E.1 成功与失败模式分析	作者David Chanoff曾与一位曾任职的美国海军上将合作在什么总统领导下担任英国大使？ 我需要搜索David Chanoff，找到他所关联的美国海军将军合作。 搜索 [David Chanoff] David Chanoff 是一位著名的作者…… 搜索 [William J. Crowe] 威廉·J·克罗是... 威廉·J·克劳担任英国大使期间 比尔·克林顿总统 所以答案是比尔·克林顿。 完成 [比尔·克林顿] REA CT：协同推理与行动于语言模型中的结合 E.1 成功与失败模式分析
CoT	
REA CT：协同推理与行动于语言模型中的结合 E.1 成功与失败模式分析	代表亚利桑那州立大学的团队的名称是什么？ 涵盖美国西部广泛区域的会议 包括亚利桑那州、加利福尼亚州和新墨西哥州吗？ 让我们一步步思考。代表亚利桑那州的团队名称 在涵盖西部广泛范围的会议中，大学 美国，包括亚利桑那州、加利福尼亚州和新墨西哥州是亚利桑那州 州太阳魔鬼 亚利桑那州立大学太阳神
REA CT：协同推理与行动于语言模型中的结合 E.1 成功与失败模式分析	

成功：误报	
ReAct	
REA CT：协同推理与行动于语言模型中的结合 E.1 成功与失败模式分析	桌面游戏《San Marco》还是《About Time》先推出？ 我需要搜索《圣马可》和《关于时间》，并找出哪一个先上映。 搜索 [圣马可] San Marco 是其中之一…… 该建筑位于威尼斯。 圣马可可是威尼斯的一个区。 我接下来需要搜索 About Time。 搜索 [关于时间] 无法找到 [关于时间]。 类似: [...]. 关于时间是一部2013年的电影。 所以圣马可第一名出场。 Finish [圣马可] REA CT：协同推理与行动于语言模型中的结合 E.1 成功与失败模式分析
CoT	
REA CT：协同推理与行动于语言模型中的结合 E.1 成功与失败模式分析	范德堡大学还是埃默里大学先成立？ 让我们一步步思考。 范德堡大学成立于1873年。 大学成立于1836年。 1873（范德比尔特大学） > 1836（埃默里大学） 范德比尔特大学（Vanderbilt University）因此首先成立。 埃默里大学 艾默里
REA CT：协同推理与行动于语言模型中的结合 E.1 成功与失败模式分析	

失败：推理错误	
ReAct	
REA CT：协同推理与行动于语言模型中的结合 E.1 成功与失败模式分析	与哈维共同出演的女演员的中间名是什么 Fierstein, Linda Hart, Dick Latessa, Clarke Thorell, Mary Bond Davis, Laura Bell Bundy, Matthew Morrison, Corey Reynolds, and Marissa Jaret Winokur 在 发胶？ 我需要搜索Harvey Fierstein, Linda Hart, Dick Latessa, Clarke Thorell, Mary Bond Davis, Laura Bell Bundy, Matthew Morrison, Corey Reynolds, 以及 Marissa Jaret Winokur，找到与他们一同出演的女演员 定型喷雾，然后找到它的中间名。 搜索 [Harvey Fierstein] 哈维·福布斯·费尔斯坦 ... 哈维·费尔斯坦是一位演员、剧作家和编剧。 我需要 搜索 Linda Hart 下一步…… 搜索 [Linda Hart]...
CoT	
REA CT：协同推理与行动于语言模型中的结合 E.1 成功与失败模式分析	丹尼·格林比詹姆斯·沃西年长吗？ 让我们一步步思考。 丹尼·格林出生于1987年。 詹姆斯·沃西曾是出生于1961年。 1987 (Danny Green) > 1961 (James Worthy)，因此Danny Green是较老。 丹尼·格林
REA CT：协同推理与行动于语言模型中的结合 E.1 成功与失败模式分析	

Failure: Search error	
ReAct	
Question:	What does the goddess associated with the goddess frigg consists of what tales?
Thought 1:	I need to search goddess frigg and find its associated goddess.
Action 1:	Search[goddess frigg]
Observation 1:	Could not find goddess frigg. Similar: ['Frigg', 'Frigg and Freyja common origin hypothesis', 'Freyja (section Relation to Frigg and other goddesses and figures)', 'Eir (redirect from Eir (goddess))', 'Baldr'].
Thought 2:	I need to search Frigg instead...

Failure: Hallucination	
Cot	
Question:	What was the first year a scientific journal published by an organization located in the Bronx was published?
Thought:	Let’s think step by step. The first year a scientific journal published by an organization located in the Bronx was published was 1916.
Answer:	1916 (Label: 1909)

Failure: Label ambiguity	
ReAct	
Question:	What nationality is the man who developed the dictator game?
Thought 1:	I need to search dictator game, find the man who developed it, then find his nationality.
Action 1:	Search[dictator game]
Observation 1:	The dictator game is a popular experimental instrument in social psychology and economics...
Action 3:	Search[Amos Tversky]
Observation 3:	Amos Nathan Tversky...
Thought 4:	Amos Tversky is an Israeli cognitive and mathematical psychologist. So the man who developed the dictator game is Israeli.
Action 4:	Finish[Israeli] (label: Israel-American)
Cot	
Question:	Kurt Fuller from Stockton, CA appeared in which American Detective Comedy Drama television series created by Steve Franks?
Thought:	Let’s think step by step. Kurt Fuller from Stockton, CA appeared in Psych. Psych is an American Detective Comedy Drama television series created by Steve Franks.
Answer:	Psych (label: Psych is an American detective comedy-drama)

失败：搜索错误	
ReAct	
REACT: 协同推理与行动 于语言模型中的融合 E.1 成功与失败模式分析	与女神弗丽嘉相关的女神由什么组成
故事吗？ 思考1: 我需要搜索女神弗丽嘉 (Frigg) 并找到与之相关的女神。	
观察1: 搜索[女神弗丽嘉] 未能找到女神弗丽嘉。类似: ‘Frigg’ Frigg 和 Freyja 的共性原始假设 Freyja (节: 与弗丽嘉及其他女神的关系) (包括图表)’ Eir (女神)’ ‘Baldr’].	
思考 2:	我需要搜索 Frigg 代替……

失败：幻觉	
Cot	
REACT: 协同推理与行动 于语言模型中的融合 E.1 成功与失败模式分析	由组织发表科学期刊的第一年是哪一年 位于布朗克斯的文章何时发表？ 思考: 让我们一步步思考。由科学期刊首次发表的位于布朗克斯的一个组织于1916年发表。
REACT: 协同推理与行动 于语言模型中的结合 E.1 成功与失败模式分析	1916 (标签: REACT: 协同推理与行动 于语言模型中的结合 E.1 成功与失败模式分析
失败：标签歧义	

ReAct	
REACT: 协同推理与行动 于语言模型中的融合 E.1 成功与失败模式分析	开发独裁者游戏的那个人是哪国人？ 思考1: 我需要搜索独裁者游戏，找到开发它的那个人，然后找到他的国籍。 观察1: 搜索[独裁者游戏] 独裁者游戏是社会心理学中一种常用的实验工具 和经济学…… 搜索[阿莫斯·特维尔斯基] Amos Nathan Tversky... 阿莫斯·特维尔斯基是以色列的认知与数学心理学家。因此， 开发独裁者游戏的研究人员是以色列人。 行动 4: 完成[以色列] (标签: 以色列-美国)
Cot	
REACT: 协同推理与行动 于语言模型中的融合 E.1 成功与失败模式分析	库尔特·富勒 (Kurt Fuller) 来自加利福尼亚州斯托克顿 (Stockton, CA)，出现在哪部美国侦探喜剧中 由Steve Franks创作的剧情电视系列剧 让我们一步步思考。来自加利福尼亚州斯托克顿的库尔特·富勒在《心理》剧中亮相。 Psych 是一部由美国侦探喜剧剧集，由创建者创作。 史蒂夫·弗兰克斯 心理 (标签: Psych 是一部美国侦探喜剧剧集)
REACT: 协同推理与行动 于语言模型中的结合 E.1 成功与失败模式分析	